



VIPER

Visual-Infrared Position Estimation for Reconnaissance

Samina Farheen, Noah Venegas, Andrew Hariman, Andrew Mills, Nicholas Carrano,
William Tu, Annie Do, Carter Brown, Eugene Antone, Wesley Zalvin

9 June 2026



TABLE OF CONTENTS

TABLE OF CONTENTS.....	2
I. Executive Summary.....	4
II. Background and Research Summary.....	5
III. Design.....	7
3.1 - Product Definition.....	7
3.2 - Updated Product Specifications.....	8
3.3 - Subsystems.....	12
3.3.1 - Sensors Subsystem.....	12
A. Carrier Board PCB.....	12
B. Sensor Board PCB.....	14
C. Sensor Trade Studies.....	16
D. PCB Manufacturing and Stackup.....	25
3.3.2 - Software Subsystem.....	27
A. STM Dataflow: Sensor Data and Timestamp Generation.....	27
B. Timestamp Generation.....	28
C. Interrupt-Driven Architecture.....	28
D. Per-Sensor Dataflow.....	28
E. Stream Path to the Raspberry Pi.....	29
F. Interrupt Driven Architecture Summary.....	29
G. Raspberry Pi Dataflow Overview: The IR and Visual Cameras.....	30
3.3.3 - Housing Subsystem.....	31
A. Enclosure.....	31
B. Heat Sink.....	34
C. Environmental Sealing.....	35
D. Vibration Isolator.....	36
E. Manufacturing.....	37
F. Assembly.....	39
IV. Analysis.....	40
4.1 - PCB.....	41
4.1.1 - PCB Simulations: Buck Converter.....	41
4.2 - Software.....	43
4.2.1 - Data throughput.....	43
4.2.2 - Cyclic Redundancy Check (CRC) for Data Transmission.....	44
4.2.3 - Buffers and Overflow Handling.....	44
4.2.5 - Data Post Processing.....	45
A. Kalibr Calibration.....	45

Calibration setup.....	46
Table X: IMU Noise Parameters.....	46
Camera intrinsics.....	46
Camera–IMU extrinsics and timing.....	47
B. Spectacular AI Position Estimation.....	48
C. Dataset Formats.....	48
4.3 - Thermal.....	49
4.3.1 - Material trade studies.....	49
4.3.2 - Benchmark test.....	51
4.3.3 - Heat Sink Optimization.....	52
A. Purpose.....	52
B. Methods.....	52
C. Results.....	54
4.3.4 - Full System Simulations.....	56
4.4 - Vibrational.....	60
4.5 - Weight Analysis.....	62
V. Risk Mitigation.....	67
VI. Testing.....	70
6.1 - Thermal Test.....	70
6.2 - Flight Test.....	71
6.3 - VIO Results and Metrics.....	72
VII. Task Distribution.....	75
VIII. Cost.....	76
8.1 - R&D Expenditure.....	76
8.2 - Production BOM.....	77
IX. Conclusion.....	79
X. Works Cited.....	80
XI. Appendix.....	85
11.1 - Carrier Board PCB Schematics.....	85
11.2 - Sensor Board PCB Schematics.....	90
11.3 - Engineering Drawings.....	95
11.4 - Thermal Benchmark Test SOP.....	109

I. Executive Summary

The objective of this project is to develop a low-cost, lightweight visual navigation payload capable of collecting flight data during the operation of a First Person View (FPV) drone for guidance in global positioning system (GPS) denied environments. This project is sponsored by Teledyne FLIR, which will be providing the software component of the system.

Most modern-day drones rely on GPS technology for flight stability and position accuracy. However, there are many scenarios in which the GPS signal strength may be weak or blocked entirely, such as in indoor spaces or heavily forested areas, resulting in erratic flying behavior. NASA states that “one of the FAA’s requirements will be the ability to navigate in the event of a GPS outage . . . because GPS signals can often be degraded at low altitudes in urban areas—exactly where the highest precision is needed” [1].

The solution to this problem is the VIPER payload – Visual and Infrared Position Estimation for Reconnaissance – a navigation capability for drones in GPS-denied environments. This system integrates sensor readings with visual and infrared images from onboard cameras to determine precise position and orientation. This project is a low-cost, lightweight version of this system, integrating a FLIR Boson thermal camera into a small, sleek enclosure. The end goal is to conduct a 15-30 minute flight, comparing the operational capacity of the GPS-denied system with an onboard GPS to determine the degree of accuracy of the product.

VIPER is intended to serve as a functional payload that can be universally mounted to the underside of any drone. The full product is pictured below in figure 1. For the scope of this project, VIPER will be designed around a single drone platform and optimized for search and rescue operations. It operates independently with its own power and sensor architecture.

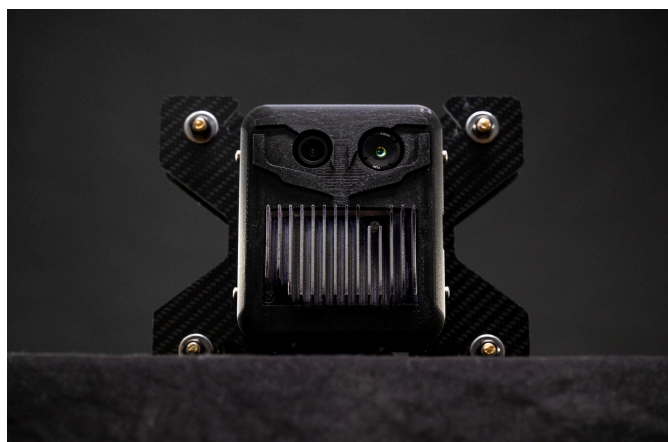


Figure 1. VIPER Payload

II. Background and Research Summary

Modern unmanned aerial systems (UAS) rely primarily on Global Positioning System (GPS) signals for navigation, stabilization, and autonomous flight. When GPS signals are degraded, obstructed, or intentionally disrupted, a UAS can no longer determine its position reliably. This problem occurs in many real environments, including dense urban areas, partially collapsed or indoor rescue sites, forests, and regions with electronic interference. As UAS continue to be used for mapping, inspection, defense, and especially emergency response, there is an increasing need for navigation methods that remain dependable when GPS information is unavailable.

Urban rescue operations are one of the most demanding contexts for GPS-denied navigation. In these missions, a UAS may be required to fly near tall structures, operate in smoke or darkness, or access confined spaces where satellite visibility is limited. Rescue teams rely on UAS for situational awareness and for locating people or hazards, so navigation systems must continue to function even when GPS cannot be used. This has motivated the development of sensing-based navigation techniques that estimate the vehicle's motion using onboard sensors rather than external positioning signals.

A major area of research addressing this need is visual-inertial navigation. This approach combines imagery from onboard cameras with measurements from an inertial measurement unit (IMU), which records acceleration, angular velocity, and sometimes magnetic field data. Visible-light cameras provide detailed images that are useful for tracking features and referencing structures in the environment. However, their performance decreases in low-light situations, fog, smoke, glare, or regions with limited texture. Infrared (IR) cameras detect thermal signatures instead of reflected light, allowing operation in darkness and visually degraded conditions. This capability is particularly valuable in urban rescue missions, where identifying human heat signatures or detecting hotspots can support search and assessment efforts. IR imagery can be less informative when temperature variation is low, so visible and IR imaging are often used together to provide a broader range of environmental information.

Other sensing technologies, such as Light Detection and Ranging (LiDAR), are also used for navigation when GPS is unreliable. LiDAR sensors generate accurate three-dimensional measurements of the surroundings, but they tend to be heavier, more expensive, and more power-intensive than camera-based systems. LiDAR performance can also decline in rain, fog, smoke, or airborne debris. These limitations make LiDAR less suitable for small UAS platforms that impose strict constraints on mass, size, and power consumption.

Across all sensing-based navigation methods, a key technical challenge involves aligning sensor data in time. Image frames and IMU measurements must be accurately synchronized to maintain consistent motion estimates and reduce drift that naturally accumulates in inertial measurements. High-quality synchronized datasets are also essential for developing and validating algorithms

that compare observed imagery to reference maps, which helps correct navigation errors in GPS-denied environments.

Although extensive research exists in visual-inertial and sensor-based navigation, there remains a need for compact, lightweight payloads that can collect synchronized visible-light, infrared, and inertial data specifically tailored for small UAS platforms. Many existing systems are proprietary, sized for larger aircraft, or not optimized for low-light and degraded-visibility conditions that appear frequently in urban rescue operations.

This project aims to address that need by supporting reliable aerial navigation and situational awareness in environments where GPS access is limited. The next section describes the proposed system and outlines the sensing components, operating requirements, and design considerations guiding the development of the payload.

III. Design

3.1 - Product Definition

VIPER is a visual navigation payload designed as an alternative navigation system for drones in situations where GPS is denied or difficult to obtain. It is equipped with a suite of sensors, including two IMUs, an altimeter, a compass, and a visual and IR camera, that are responsible for collecting positional information such as acceleration, heading, and altitude. Each sensor reading is timestamped by a common microcontroller, which is also responsible for timestamping camera images and compiling sensor data in a usable format for the Raspberry Pi. Meanwhile, camera images are collected and stored on the on board raspberry pi, which matches up camera timestamp data to the camera frame using the information from the microprocessor.

After the position timestamps are created, the raw sensor and image data are stored in a Solid-State Drive (SSD) as well as input into a map-matching algorithm developed by FLIR. This algorithm is responsible for comparing features across images to develop a virtual map, utilizing the position information and comparisons to known satellite imagery to correct for inertial drift. The algorithm then outputs a positional coordinate for the drone, which will be cross-referenced with a GPS (not pictured) incorporated into the V.I.P.E.R payload for an accuracy comparison. Once powered on pre-flight, there are no other operational maneuvers needed to run V.I.P.E.R. However, the payload is attached to an external rechargeable battery that will require charging between flights.



Figure 2. VIPER Mounted Underside the DJI Matrice M300 Drone

3.2 - Updated Product Specifications

Characteristic Needs	Engineering Characteristics	Target Specifications	Current Specifications	Need
Minimize overall weight	Mass	300 grams	260 grams (with damper: 547.4 grams)	Must Have
Minimize physical footprint	Dimensions	75 x 75 x 50 mm	112.01 x 93.98 x 59.94 mm (with damper: 180.09 x 145.03 x 82.80 mm)	Must Have
Ensure manufacturing affordability	Price (BOM)	< \$1000	\$851.32	Must Have
Operate reliably across the temperature range	Operating temperature	-15°C to 40°C ambient	-20°C to 25°C	Must Have
Optimize power efficiency	Power	$\leq 10\text{W}$	7W Average, 9W Peak	Must Have
Reduce vibration during operation	Vibrations	Filter 30-300 Hz	20Hz Damper Resonance	Must Have

Acquire and record operational data	Sensor/Data Capability	9-axis IMU Altimeter Visual Camera Flir Boson IR Camera	2 6-axis IMU Compass Altimeter Visual Camera Flir Boson IR	Must Have
Withstand environmental exposure	IP Rating	IP56	N/A	Stretch
Accurate position estimation compared to ground truth	Accuracy	±10m	±4m	Nice to have
Can integrate with multiple drone platforms	Modularity	Mount to 80% of drones in market	Mounts to DJI M300	Stretch

Table 1. Product Specifications

The specifications for this product were developed taking into consideration the primary drone platform as well as the use case for the product.

Based on the requirements of this project as well as the drone options available through FLIR, the DJI Matrice 300 was chosen. The maximum payload capacity for this drone is 1 kg [2], and based on preliminary weight calculations of the sensors on board, a target mass of 300 grams was needed to ensure minimal impact on the drone's flight time and maneuverability. With the finalized design, the mass of the payload is about 229 grams, which is well within the target mass and meets the mass specification of the product.

Table 1 also lists the payload's physical footprint, which was similarly constrained to 75 x 75 x 50 mm, so the system can be easily mounted on a wide variety of UAV platforms without obstructing the aerodynamics of the drone or requiring additional structural support. The final product specification is about 112.01 x 93.98 x 59.94 mm, which is slightly larger than the target

size of the full system, but the PCB system fits into this design specification. As the current dimension maintains the mass specification (which was the most important product specification to meet with the design), the current footprint was finalized as the final footprint.

Another core requirement included affordability with a bill of materials (BOM) target of under \$1000 to support scalable deployment across a wide variety of missions such as search-and-rescue missions. The current prototype comes in at a BOM cost of \$851.32, which fits comfortably in the price budget.

The payload must also operate reliably in varying environmental conditions, specifically between -15°C and 40°C , to accommodate both cold-altitude operations and heat exposure during extended flight missions. The payload is confirmed to operate in -20°C to 25°C . In order to confirm that the system would be able to operate in a higher temperature range, the payload needs to be manufactured out of ASA plastic and tested in a controlled thermal environment, which was outside of the scope of this project.

Power efficiency is critical due to strict limits on UAV energy constraints. The current estimates total to approximately 9.3W (the minimum specification); however, the FLIR Boson Camera and the Raspberry Pi 5 Compute Module may draw more power under heavy computation or peripheral load, meaning power margins must be carefully managed to remain at or below the 10W design target. The set-up of the PCB design maintains system operation under 10W, with max power draw occurring at 10W, satisfying the power budget requirements for the system. Typically, the payload operates at 7W.

The system must also significantly reduce vibration transmission to maintain image clarity and accurate inertial measurements when mounted onto the drone. A pre-existing study on the DJI Matrice 600 drone showed vibrational resonance in frequencies from 30-300 Hz [3], which was used to define the target frequency damping range of this project. However, further testing will need to be done on the DJI M300 drone platform to fully qualify the resonant nodes. While physical testing is still needed to verify the design, the drone damper system was designed for a resonance of about 20Hz, based on calculations of the drone as a plate-beam system, the details of which are outlined in the vibrational analysis section. The flight testing showed that the camera images remain sufficiently stable during flight with no significant motion blur, showing that the damper is able to sufficiently damp large vibrations.

The sensors required include a 9-axis IMU, altimeter, visual camera, and FLIR Boson to capture all the necessary data such as position, temperature, and landscaping. Ultimately, a second IMU was added to the system to provide redundancy for the ViSlam software.

As the drone will be operating in a variety of severe weather conditions, an IP56 rating will be required to ensure that the sensitive electronics on board can be protected. However, this is a stretch goal for the project, and an IP45 rating is still acceptable, as that is what the DJI M300 is

rated to [4]. To properly test the IP rating of this system, the housing needs to be manufactured using ASA, which was out of the scope of this project.

The universal mounting specification comes from the use-case scenario. With a universal mounting payload, any search and rescue operation that uses drones can easily mount the V.I.P.E.R payload for maximum operational capacity. V.I.P.E.R's size and mass allow it to be easily mounted to any drone larger than 250 grams. However, its mounting system is designed specifically for the DJI Matrice 300 drone due to time constraints on the project.

The final specification for this product will be the accuracy of the capture system. Components were chosen and designed specifically for high accuracy, with an aim of being within 10m of accuracy. Flight testing confirmed that the system is within a $\pm 4\text{m}$ position drift accuracy during the course of a 256m flight, revealing that the system is able to successfully function and accurately determine its location.

3.3 - Subsystems

3.3.1 - Sensors Subsystem

The data capture system is responsible for compiling, timestamping, and saving data collected during a flight. This includes all onboard sensor data, camera images, as well as position data from an external GPS module.

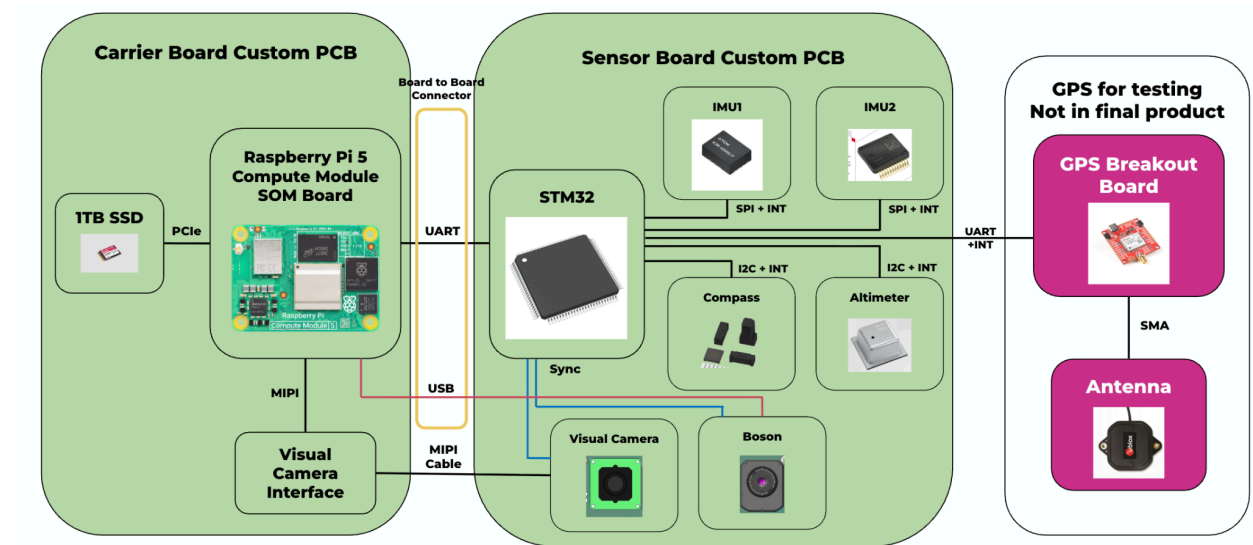


Figure 3. Capture System Block Diagram

Shown in Figure 3, the capture system block diagram consists of three main areas: compute and storage on a custom Carrier Board PCB, cameras and sensors on a custom Sensor Board PCB, and a separate GPS module for testing purposes only

A. Carrier Board PCB

The Visual-Inertial Odometry Simultaneous Localization and Mapping (VIO SLAM or VISLAM) software is run on a Raspberry Pi 5 Compute Module (RPCM). This is mounted onto a custom Carrier Board PCB shown in Fig. 4 and Fig. 5 [5]. Due to the vast amount of data collection required for this project, we have a 1 TB SSD memory over a Peripheral Component Interconnect Express (PCIe) interface. This is critical for the high speed, high volume storage of images and sensor readings.

Below is a 3D view of the Carrier Board:

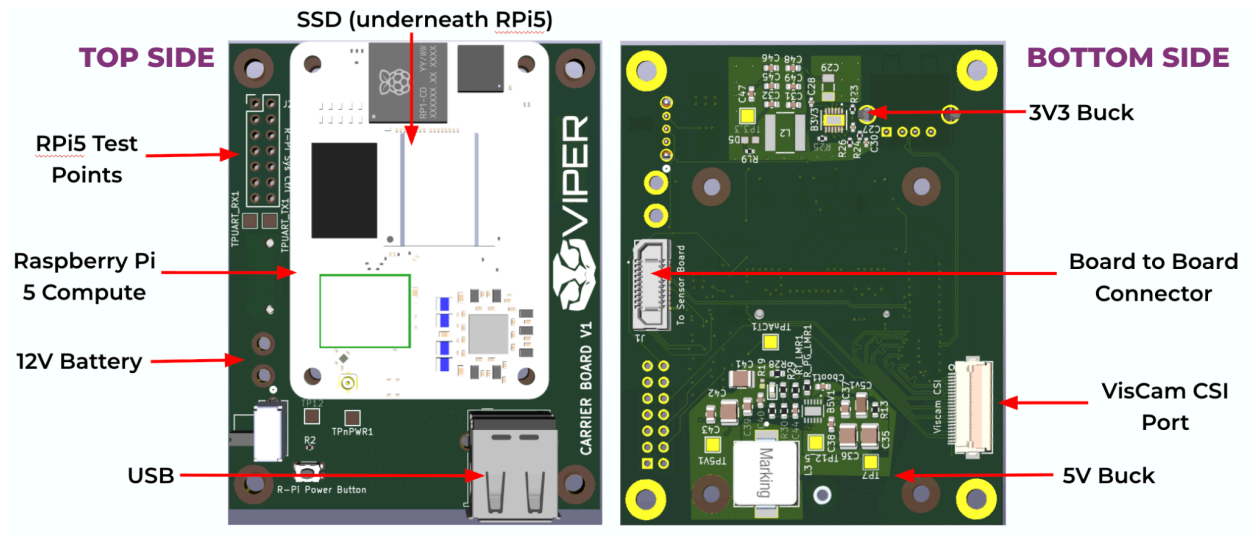


Figure 4. 3D view of Carrier Board PCB

The Carrier Board is the first custom PCB in the system and is designed as a two-sided assembly. The top side, which faces away from the bottom of the drone, contains the RPCM with the SSD mounted underneath it, along with the USB port and the 12V battery input connection. The bottom side, which faces the Sensor Board, includes the 3.3V and 5V buck converters, the VisCam CSI port, and the board-to-board mezzanine connector.

The Carrier Board interfaces with the Sensor Board through the board-to-board mezzanine connector, enabling the STM, Boson, and VisCam on the Sensor Board to communicate with the RPCM for data capture and control. The visual camera's MIPI signals are routed from the Sensor Board to the Carrier Board through a flex cable. The SSD communicates over PCIe, while the CSI port operates using MIPI.

These interfaces operate at very high speeds, with data rates ranging from hundreds of megabits per second to several gigabits per second [1]. Because of this, the signals are highly sensitive to impedance mismatches and routing discontinuities. As a result, the Carrier Board contains very few test points, since adding them to high-speed lines can degrade signal integrity. The existing test points are primarily allocated to the 3.3V and 5V buck converters and the Universal Asynchronous Receiver-Transmitter (UART) signal from the Sensor Board.

B. Sensor Board PCB

The Sensor Board shown in Figure 5 is the second custom PCB that contains the cameras and sensors, controlled by a high performance model of STM microcontroller (STM32H7VIT6). This allows us to use hardware interrupts, meaning the sensors and cameras tell the STM when they are ready to send data or images and the STM generates a timestamp. In addition, this board mounts the FLIR Boson infrared camera and Innomaker OV9281 global shutter visible camera. The Sensor Board also has the sensor suite of two IMUs, an altimeter, and a compass. The second IMU improves the accuracy of the visual odometry software. In terms of communication protocols, the visible camera uses a MIPI interface to send image data directly to the RPCM, while the FLIR Boson sends images directly to the RPCM over USB 2.0 [6], [7]. The altimeter and compass use I2C while the IMUs use SPI [8], [9]. Refer to the trade study and comparison in the previous section for the selection of sensors, cameras, STM, and compute module.

The following is the current 3D view of the Sensor Board

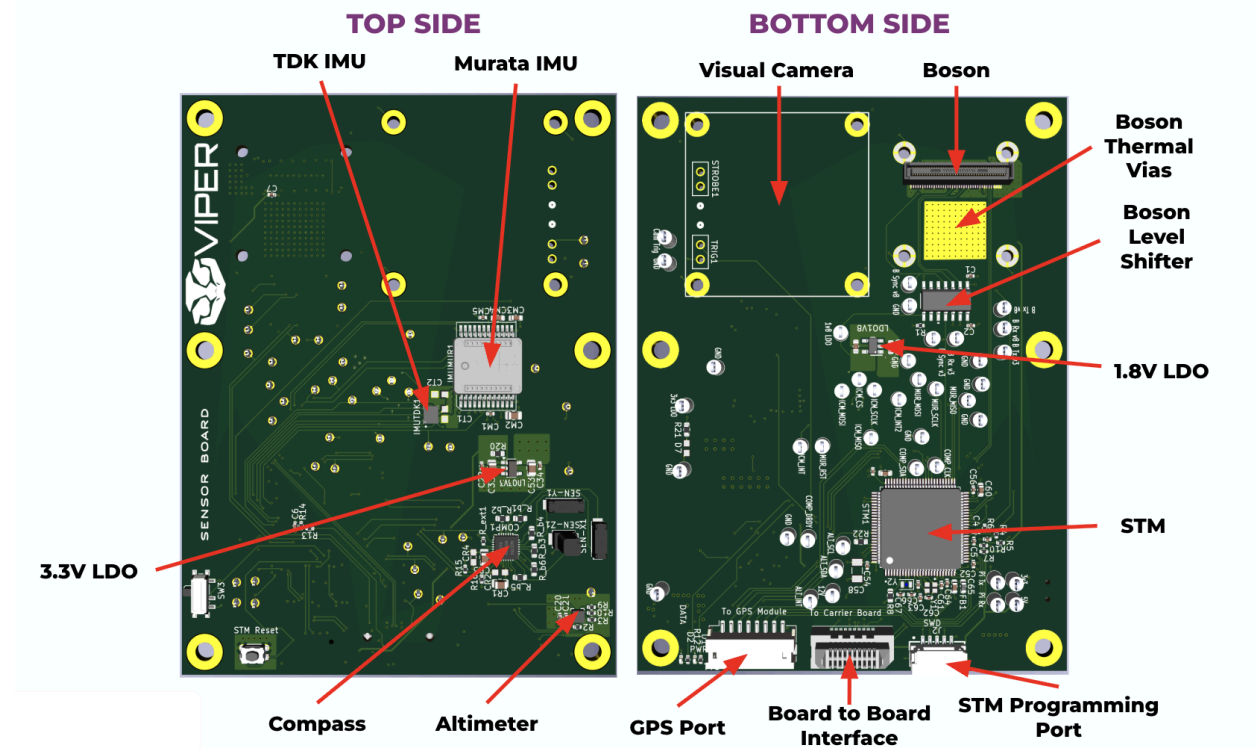


Figure 5. Sensor Board PCB

The Sensor Board as shown in Figure 5 interfaces with the Carrier Board through a board to board mezzanine connector. This allows the STM to communicate the timestamped data to the RPCM over UART and allows the Boson image frames to be sent directly to the RPCM. The visual camera MIPI signals are carried from the Sensor Board to the Carrier Board over a flex cable.

Finally, a separate GPS module with an external antenna is required to record the coordinates of the drone's location every time that the cameras take a photo and sensor readings are taken. This is because we need to use the data capture system to calibrate the VIOSLAM visual odometry software. This will serve as a ground truth comparison point to calculate the error between the coordinates determined by the software versus GPS. Figure 6 shows the physical connection between our external GPS and sensor board.

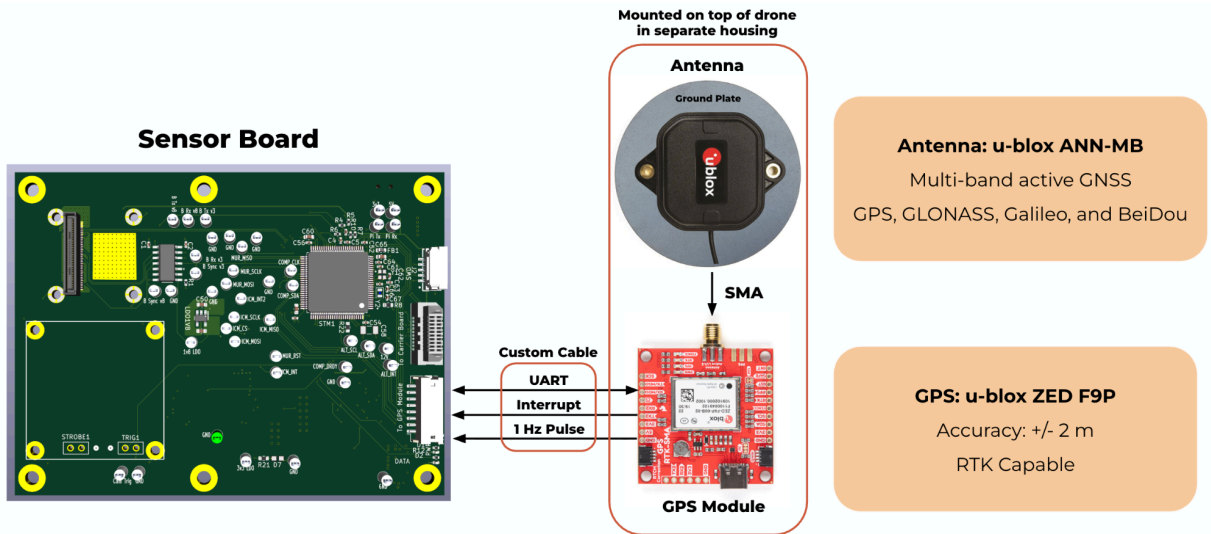


Figure 6. GPS module and antenna

In summary, all the captured data is saved to the SSD connected to the Raspberry Pi Compute Module (RPCM) 5 through a PCIe interface. This data is *post processed after a flight* to update and train the VIOSLAM visual odometry software. The data collected from the GPS will serve as the ground truth position, and the VIOSLAM AI model will be tuned to predict positions given the other collected sensor inputs. The difference between the GPS coordinates and the output of the VIOSLAM software will measure accuracy in the final product. Ideally, real time kinematic (RTK) mode on the GPS should be used to achieve centimeter level accuracy, but this was not implemented in this project. Thus, our ground truth GPS has a +/- 2m error bound built in.

C. Sensor Trade Studies

The sensors subteam is responsible for purchasing and integrating the electronic components for the payload. The following components are being used in the project:

- 1. IMUs (2)**
- 2. Compass**
- 3. Altimeter**
- 4. FLIR Boson Infrared Camera**
- 5. Global Shutter Visual Camera**
- 6. GPS Module and Antenna**
- 7. Raspberry Pi 5 Compute Module**
- 8. 1 TB Solid-State Drive (SSD)**

A visual block diagram of the basic integration of all the electronic components is shown previously in Figure 5 and 6.

The system architecture is centered around a custom carrier board PCB that hosts the Raspberry Pi 5 Compute Module as the central processing unit. The Raspberry Pi is responsible for taking in data from all the attached sensors and facilitating seamless communication between them. In order to have sufficient storage capacity for the high-throughput data logging, a 1 TB SSD is integrated through a high-speed PCIe interface.

The payload incorporates dual imaging to capture both thermal and visual data, critical for a wide range of applications such as night and day-time vision. The FLIR Boson IR camera is connected to the system via USB, capturing detailed thermal data while the visual camera is connected to the system via MIPI CSI, providing high-resolution optical imagery. To ensure accurate synchronization between the two (2) cameras, the thermal and visual systems are linked via hardware sync lines. This allows for gram-accurate alignment of the two (2) datasets, which will be essential for the visual odometry algorithm responsible for overlaying the thermal and visual information in real-time.

The payload also integrates a series of navigation and telemetry sensors to provide precise state estimation. The Inertial Measurement Unit (IMU) provides real-time data of the payload's acceleration, angular velocity, and orientation using accelerometer, gyroscopes, and magnetometer. Additionally, the altimeter incorporates barometric pressure and temperature sensors, providing altitude data. Both of these sensors interface with the STM32 through SPI.

Lastly, the system includes a GPS module, which connects via UART and interfaces with an external antenna through a coaxial cable (SMA) connector for reliable signal acquisition. This GPS module serves to provide ground-truth positional data for validation and training the VISLAM AI.

Selection of the specific electronic components for the payload are listed below.

1. IMU

An inertial measurement unit (IMU) measures motion by converting detected inertia into data describing an object's movement and orientation. It typically combines a 3-axis accelerometer and a 3-axis gyroscope (6-axis IMU), and in some cases adds a 3-axis magnetometer (9-axis IMU) to assist with heading correction and drift compensation. [2]

Search and Rescue (SAR) missions impose demanding requirements on inertial sensing hardware. A SAR drone must remain stable in strong winds and under continuous vibration while navigating GPS-degraded environments such as canyons, forests, or indoor spaces. Selecting an appropriate IMU therefore requires careful evaluation of dynamic range, noise density, bias drift, long-term stability, power consumption, and operating temperature. Another key consideration is whether to use raw sensor outputs or pre-processed (sensor-fused) data. While pre-processed data simplifies integration, it limits flexibility in tuning custom sensor-fusion algorithms. For this reason, a high-performance 6-axis IMU paired with a dedicated, high-quality compass was selected rather than relying on a fully integrated 9-axis solution.

Below, in Table 3, is a Pugh Chart comparing three (3) different IMU. The Bosch BMI088 will serve as the baseline IMU.

Criteria	Bosch BMI088 (Baseline)	Murata SCH16T-K10-1	TDK InvenSense ICM-45686
Gyroscope Accuracy (absolute)	0	+++	++
Acceleration Accuracy (absolute)	0	+++	++
Operating Temperature Range	0	+	+

Power Consumption	0	-	-
Weight	0	-	+
Package Size (Footprint)	0	-	+
Cost (Unit Price)	0	-	0
Redundancy Compatibility	0	++	+
Total Score	0	5	7

Table 2. IMU Pugh Chart [24]

The Bosch BMI088 serves as the baseline in the Pugh comparison. This industry-standard 6-axis IMU offers low gyro noise (~ 0.014 °/s/ $\sqrt{\text{Hz}}$) and high-quality accelerometer performance ($\sim 160\text{--}190$ $\mu\text{g}/\sqrt{\text{Hz}}$). It operates from -40 °C to $+85$ °C, consumes approximately 5 mA during full-rate gyro operation, and features a compact $3 \times 4.5 \times 0.95$ mm footprint at a cost of about \$5 per unit. Although classified as consumer-grade MEMS hardware, its performance is sufficient for stable flight control and basic navigation.

In contrast, the Murata SCH16T-K10-1 represents a tactical-grade solution with substantially improved stability. It offers gyroscope bias instability on the order of ~ 0.5 °/h (typical) and gyroscope noise density around ~ 0.0003 °/s/ $\sqrt{\text{Hz}}$, which is an order of magnitude improvement over consumer MEMS devices. The accelerometer also exhibits extremely low bias and high stability, supporting precise dead-reckoning over longer durations. The sensor supports measurement ranges up to ± 2000 °/s (gyro) and ± 16 g (accelerometer). These gains come at the expense of increased size, higher power consumption, and significantly higher cost, placing it in the professional/tactical hardware category.

The TDK InvenSense ICM-45686 provides precision-grade performance that bridges consumer and tactical solutions. It achieves a gyroscope noise density of approximately 0.0038 °/s/ $\sqrt{\text{Hz}}$ and an accelerometer noise density near 70 $\mu\text{g}/\sqrt{\text{Hz}}$, significantly improving signal clarity compared to the BMI088. It supports wide full-scale ranges from ± 15.625 to ± 4000 °/s (gyro) and ± 2 to ± 32 g (accelerometer), making it well suited for high-dynamic drone applications. This device offers improved stability and noise performance while maintaining moderate size and power requirements relative to tactical-grade systems.

As shown in the Pugh comparison, both the Murata SCH16T-K10-1 and the TDK ICM-45686 outperform the BMI088 in noise performance and bias stability. However, these improvements must be weighed against increased cost, power consumption, and system complexity when selecting the most appropriate IMU for SAR drone deployment. Since these two models work hand-in-hand, they were both implemented on the Sensor Board for redundancy and accuracy of data.

2. Compass

A digital compass, or magnetometer, is essential for determining a drone's absolute heading. In SAR missions, while IMUs track rapid movements, they suffer from "yaw drift" over time. Therefore, the compass is responsible for correcting this, ensuring the drone maintains its orientation. However, SAR drones generate significant electromagnetic interference (EMI) from high-current motors, which can cause navigation failure in standard sensors. To combat this, a sensor with high dynamic range and minimal magnetic hysteresis is required to distinguish the Earth's field from the drone's own internal noise.

Below, in Table 4, is a Pugh Chart comparing three (3) different compasses. The Bosch BMM150 will serve as the baseline compass.

Criteria	Bosch BMM150 (Baseline)	PNI RM3100	Honeywell HMC5883L
Resolution/Sensitivity	0	+++	+
Noise Floor	0	+++	+
Magnetic Hysteresis	0	+++	0
Sample Rate (ODR)	0	+	+
Power Consumption	0	-	0
Package Size (Footprint)	0	--	0
Cost (Unit Price)	0	-	0
Total Score	0	6	3

Table 3. Compass Pugh Chart [3]

As shown in Table 4, the PNI RM3100 was selected over the Bosch BMM150 baseline and the Honeywell HMC5883L due to its superior Magneto-Inductive technology. Unlike the Hall-effect baseline, the RM3100 offers a massive jump in resolution (13 nT) and is virtually free from magnetic "memory" (hysteresis), allowing it to remain accurate even after exposure to power surges from the motors. While it is larger and more expensive, its ability to provide stable heading data in metal-heavy environments makes it the only viable choice for high-reliability missions. Integrated alongside the Murata and TDK IMUs, it completes a tactical-grade navigation suite capable of precision flight without GPS.

3. Altimeter

The type of altimeter we plan to use is a barometric pressure sensor. We can convert the atmospheric pressure and air temperature into an altitude using the barometric formula [27]. The following is a Pugh Chart comparing two different altimeter models to the baseline Bosch BMP390, a standard for drone applications.

Criteria	Bosch BMP390 (Baseline)	TE Connectivity MS5611	ST Micro LPS22HB
Relative Accuracy (Noise/Precision)	0	-	-
Absolute Accuracy	0	-	-
Power Consumption	0	+	S
Package Size (Footprint)	0	-	S
Cost (Unit Price)	0	- -	+
Interface Options	0	S	S
Total Score	0	-3	-1

Table 4. Altimeter Pugh Chart [8], [28], [29]

The Bosch BMP390 stands out as the best choice of altimeter, as shown in the Pugh Chart in Table 5. It has best-in-class specifications, such as an incredibly low noise floor of 0.02 Pa and relative accuracy of ± 3 Pa (about ± 25 cm). It also has an absolute accuracy of ± 50 Pa (± 4 m).

Physically, the sensor is very low power and compact, drawing only $\sim 3.2 \mu\text{A}$ at 1Hz and fitting into a tiny $2.0 \times 2.0 \times 0.75 \text{ mm}$ package, all while maintaining a reasonable unit price around \$2.50. [30]

In contrast, the TE Connectivity MS5611 falls behind the Bosch sensor in nearly every other category. Its resolution is roughly 1.2 Pa compared to the BMP390's 0.02 Pa, and its absolute accuracy is significantly lower at $\pm 150 \text{ Pa}$ ($\pm 12 \text{ m}$). Furthermore, the MS5611 is physically larger at $5.0 \times 3.0 \text{ mm}$ and more expensive, costing between \$6.00 and \$9.00 per unit. [28]

Finally, the ST Micro LPS22HB main advantage is price below \$2.00, making it the cheapest option of the three. It has the same compact $2.0 \times 2.0 \text{ mm}$ footprint as the BMP390, but it is less impressive for high-precision applications. The LPS22HB has an RMS noise level of around 0.75 Pa and an absolute accuracy of $\pm 100 \text{ Pa}$ ($\pm 8 \text{ m}$); while decent for general use, the Bosch is an order of magnitude quieter, making the LPS22HB worse for critical tasks like getting a stable reading on the payload altitude. [29]

4. FLIR Boson

The thermal camera that will be used is the FLIR Boson camera. This is a camera that was given to us by FLIR to be built around. It has 60mK thermal sensitivity, 90 degree FOV, and a FPS of 60. [6]

5. Visual Camera

The first of our visual feeds will come from a selected visual camera module. We wanted our selection to be lightweight, have a similar frame rate as the thermal camera, a wide FOV, and capture images with global shutter. Typically, a camera uses a rolling shutter method to capture images, where one line of the image is captured in one instant, because it is more simple to implement. A rolling shutter capture will lead to image distortions undesirable for our position estimation algorithm as shown in Figure 7. For this project, we need to select a global, or total, shutter capture sensor in order to avoid distortions for images captured in flight.

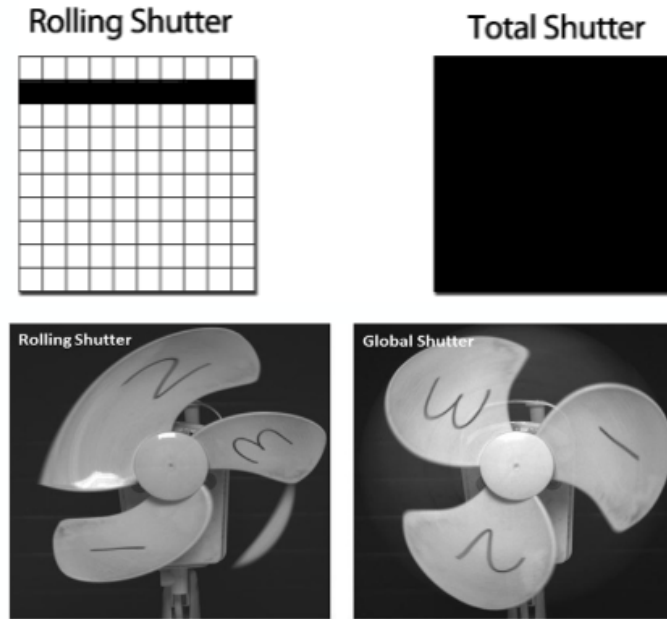


Figure 7. Rolling shutter and Total Shutter Comparison

The following is a Pugh chart comparing some global shutter camera modules, using the Raspberry Pi's global shutter camera module as a baseline, as it was our initial choice. However, we ran into some issues with lens sizes, since its CS mount uses physically larger lenses. Each entry indicates the metric relative to our baseline.

Criteria	Raspberry Pi Global Shutter (Baseline)	IQL-IMX296 Camera	OV9281 Mono Sensor Module
Price	0 : 50\$	- : 135\$	+ : 30\$
Size	0 : 38x38x19.8mm	+ : typically smaller	+ : 32x32x
FOV	0 : lens dependent	+ : presold with wide (130°) lens options	+ : presold with wide (130°) lens options
FPS	0 : 60 FPS	+	+
Lens Mount	0 : CS mount	+ : Option for CS or M12 mount	+ : M12 Mount

Table 5. Visual Camera Pugh Chart [31], [32], [33]

Based on the Pugh Chart, the OV9281 Mono Sensor Module appears to be the best choice, as it is cheaper, smaller, has a wide FOV, and comes with a lens. We will be prototyping with this camera next quarter.

6. GPS Module and Antenna

A GPS Module is required for the payload to provide a ground truth comparison point for the software. *The GPS module and its antenna are not part of the final product*, but only for calibrating and training the Spectacular AI VISLAM software. There are plenty of off-the-shelf options available, a Pugh Chart comparison is shown below.

Criteria	u-blox ZED F9P (Baseline)	Quectel LC76F	STMicro Teseo-LIV3F
Power Consumption (Tracking)	0	S	+
Tracking Sensitivity	0	–	–
Concurrent Constellations	0	S	S
Form Factor / Size	0	S	S
Supply Voltage Range	0	–	–
Net Score	Baseline	-3	-2

Table 6. GPS Module Pugh Chart [34], [35], [36]

The u-blox ZED-F9P draws ~330 mW in continuous tracking mode. It supports the reception of four Global Navigation Satellite System (GNSS) constellations (GPS, GLONASS, Galileo, and BeiDou) with a tracking sensitivity of -167 dBm, which will ensure strong performance in deep urban environments. [34]. The ZED-F9P also has a breakout board for easy interfacing.

The LC76F is also a compact chip (10.1 x 9.7 mm) that matches the u-blox footprint and supports multi-constellation reception, but it is slightly worse in sensitivity. Its sensitivity is rated at -165 dBm, which is 2 dBm less sensitive than the ZED-F9P, potentially resulting in slower location fixes in weak signal areas [35].

The Teseo-LIV3F is another popular GNSS module with a similar 9.7 x 10.1 mm form factor. Its tracking sensitivity is specified at -163 dBm, which is the lowest of the three, making it worse for challenging indoor or urban scenarios [36].

Based on this analysis, our chosen GPS module is the u-blox ZED-F9P, which has very high sensitivity and is able to communicate with multiple satellite systems. It also has a user-friendly breakout board currently in use by the team. In addition, we are using the standard u-blox ANN-MB multi band antenna, which allows for the ZED-F9P to communicate with all GNSS constellations.

7. Raspberry Pi 5 Compute Module

The compute module that will be used in this project is the Raspberry Pi 5 Compute Module. The compute module will be responsible for interfacing with the on-board sensors, storing data, and running the Visual navigation software. The Raspberry Pi 5 Compute Module was chosen due to its low price, low weight, and compatibility with the software. A Pugh chart comparing the Raspberry Pi 5 Compute Module and the Nvidia Jetson Nano is shown in Table 8.

Criteria	Raspberry Pi CM5 (Baseline)	Nvidia Jetson Nano
RAM	0	-
Price	0	-
Weight	0	-
Wireless Connections	0	-
CPU	0	0
GPU	0	+

Table 7. Pugh chart comparing the Pi CM5 and Jetson Nano [39], [40], [41]

As shown in Table 8, the Nvidia Jetson Nano and the Raspberry Pi 5 Compute Module are very similar to each other having similar CPUs and capabilities, the main differences are the GPU, weight, and price. When compared to each other, Jetson Nano is more optimized for AI with its improved NVIDIA Maxwell architecture GPU, but was significantly heavier with a weight of 141g when compared to the Pi with 20g, more expensive with its price being nearly double that of the raspberry pi, and did not have wireless connection capabilities. Based on these factors, the Pi seemed the more sensible choice since its lower weight and price are very important in meeting the weight and price specifications of the project. The final consideration for choosing the Raspberry Pi 5 Compute module over the Nvidia Jetson Nano was its known compatibility with the software that will be used during this project. With the Jetson Nano, more code would be needed to properly run the program. Based on this, the Raspberry Pi is much easier to program while being capable of running the algorithm, which eliminates the need for the

increased GPU power of the Jetson Nano. Based on price, weight, and software compatibility, the Raspberry Pi CM5 has been chosen as the compute module for the project [39], [40], [41].

8. 1 TB Solid-State Drive (SSD)

To save the data for our project, the Raspberry Pi needs to be connected to a storage device. This is primarily done through an external micro SD card but can also be done through a SSD card or USB sticks. Due to the large amount of data required for our project, it was decided to go with the largest SSD available which was a 1TB SSD. An SSD is more expensive, but is significantly more durable, reliable, and quicker when compared to any alternatives. Despite the cost, the SSD was chosen for the storage device due to its more compact design, increased storage capacity, and its fast communication with the Raspberry Pi [42], [43].

D. PCB Manufacturing and Stackup

The chosen PCB manufacturer is JLCPCB. To ensure successful PCB fabrication, we have ensured that both the Carrier Board and Sensor Board have passed an Electrical Rules Check (ERC) and a Design Rules Check (DRC) in KiCAD. The ERC checks for errors in the schematic and the DRC checks for errors in the PCB layout as well as parity with the PCB schematic.

In addition to the ERC and DRC, we are using a standard JLCPCB stackup (the board layering) shown in Fig. 9 (JLC04161H-3313). This is a 4 layer stackup, and we are using the top and bottom layers for signal traces, the second layer for a single ground plane, and the third layer for power planes. This stackup also sets the dimensions for impedance controlled differential pair traces required for USB (90 Ω), MIPI D-PHY (100 Ω), and PCIe (85 Ω). An example for USB is shown below in Fig. 9. This further ensures the boards meet the design requirements of JLCPCB.

* The first stack-up calculated below is highly recommended as it has lowest cost and quickest turn-around.

Layers
PCB Thickness
Inner Copper Weight
Outer Copper Weight
Unit

Impedance Configure
+ New Impedance
Duplicate Impedance
Calculate

Impedance (Ω)	Type	Signal Layer	Top Ref	Bottom Ref	Trace Spacing (mm)	Impedance trace to copper (mm)
90	Differential Pair (Non coplanar)	L1	/	L2	0.2032	/

<
JLC04161H-3313(Finished thickness1.56mm $\pm$ 10%)
JLC04161H-3313A(Special/Finished thickness1.58mm $\pm$ 10%)
JLC04161H-7628(Standard/Finished thickness1.5
>

Impedance (Ω)	Type	Signal Layer	Top Ref	Bottom Ref	Trace Width	Trace Spacing	Impedance trace to copper
90	Differential Pair (Non coplanar)	L1	/	L2	0.1582	0.2032	/

3) JLC04161H-3313 Stackup

Layer	Material Type	Thickness	
Top Layer	Copper	0.035mm	
Prepreg	3313*1	0.09940mm	
Inner Layer L2	Copper	0.0152mm	1.3mm H/HOZ with copper
Core	Core	1.265mm	
Inner Layer L3	Copper	0.0152mm	
Prepreg	3313*1	0.09940mm	
Bottom Layer	Copper	0.035mm	

Figure 9. JLCPCB stackup and impedance control example for USB (90Ω diff pair)

With the carrier board and sensor board put together, the finalized stackup is the following.

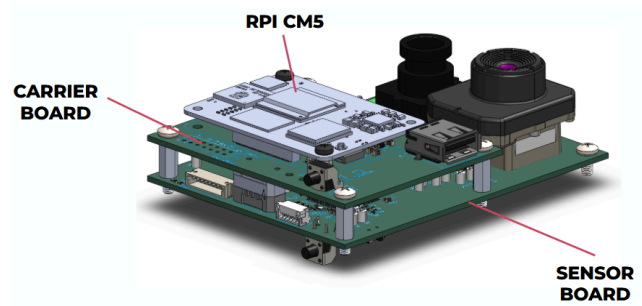


Figure 10. PCB CAD stackup

The physical stackup is the following.

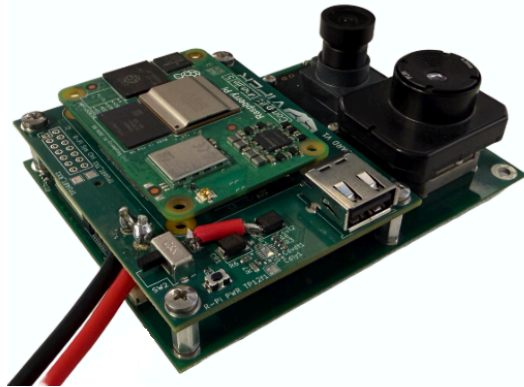


Figure 11. PCB physical stackup

3.3.2 - Software Subsystem

A. STM Dataflow: Sensor Data and Timestamp Generation

This section gives an overview on how our data acquisition system will be collecting and timestamping all received data, which includes all inertial sensors on our sensor board, as well as the visual camera.

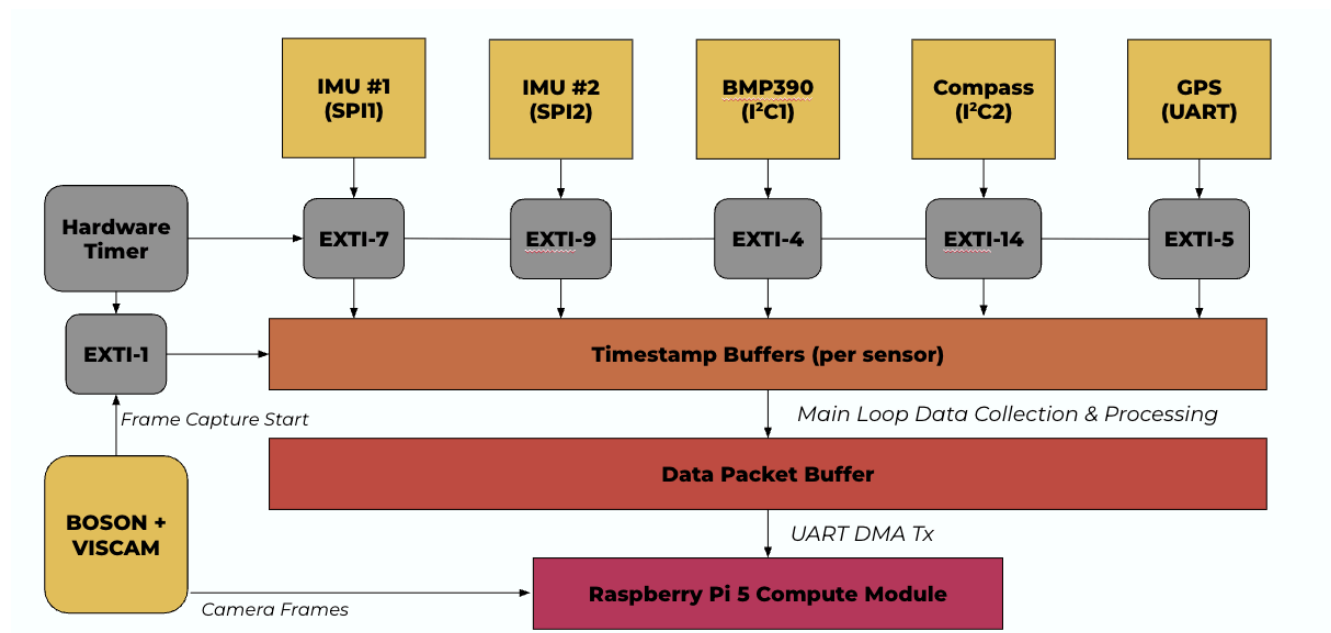


Figure 12. Interrupt Architecture for Timestamp Generation

B. Timestamp Generation

The STM32H7, the microcontroller selected for our system due to its high performance and timing accuracy, uses a single hardware timebase for all sensor timing. TIM2, a hardware timer, is configured to run as a 32-bit free running microsecond counter (high speed external (HSE) clock set so the counter increments at 1 μ s per tick). This clock is started during system init and never reset in nominal operation, so every timestamp is tied to the same time domain. This gives a common reference for IMU, barometer, magnetometer, GPS data, and visual/infrared camera frame starts. This is critical for the VISLAM application within the sensor fusion steps.

C. Interrupt-Driven Architecture

All of the chosen sensors provide a Data Ready (DRDY) or sync output that corresponds to when data has been captured by a sensor and is ready for export. These DRDY/sync signals are wired to EXTI pins on the STM32H7. On each relevant GPIO edge, the EXTI callback runs and immediately reads the current TIM2 count; this value is the sensor timestamp for the corresponding sample. The interrupt handler is designed in a way to do minimal work (only push timestamp into a small circular buffer) so latency and jitter are kept low. While the actual sensor reads (SPI & I²C are carried out in the main loop and paired with the pre-captured timestamp. The act of storing timestamps in individual buffers per sensor decouples the timing requirements ensuring no data is lost up to the buffer size.

D. Per-Sensor Dataflow

- **IMUs (SCH16T and ICM-45686):** Each has a dedicated DRDY pin (MUR_DRDY, ICM_INT1). The ISR pushes the TIM2 value into a 16-entry timestamp buffer; the main loop drains this buffer, reads the sensor over SPI, attaches the timestamp to the sample, and pushes the sample into a ring buffer. Main then pops samples, builds binary packets (sync, type, length, header timestamp, payload with **sensor_ts_us**, CRC), and enqueues them into the stream buffer. This decouples the 800 Hz DRDY rate from variable main-loop execution
- **BMP390 (barometer):** DRDY is connected to ALT_INT. The EXTI callback pushes the TIM2 value into a 128-entry timestamp queue. The main loop pops timestamps, reads compensated pressure and temperature over I2C, and streams a baro packet that carries the popped value as **sensor_ts_us** (Output Data Rate (ODR): 25 Hz).
- **RM3100 (magnetometer):** COMP_DRDY triggers the EXTI callback, which stores the TIM2 value and a flag. The main loop reads the sample over I2C, assigns the stored timestamp to the sample, and enqueues it; stream packets include **sensor_ts_us** for that sample.
- **VISCAM (OV9281):** The camera's frame-sync (strobe) is connected to VISCAM_TRIG. The EXTI callback pushes the TIM2 value into a timestamp queue; the main loop pops it

and sends a viscam packet containing that strobe timestamp and a frame index for alignment with Raspberry Pi-side frames.

- **GPS:** The GPS receiver is polled at 10 Hz. When a valid fix is available, the stream packet includes both GPS time (week, time-of-week) and a **system_timestamp_us** from TIM2 when the GPS data was accepted, for alignment with the rest of the sensor stream.

E. Stream Path to the Raspberry Pi

The main loop produces data packets for downline streaming to the Raspberry Pi-5, all packet types are written into a single 8 KB circular stream buffer. A DMA driven UART (UART5 921600 baud, TX-only) is the consumer. A function called `stream_process()` runs each main-loop iteration and if DMA is non-busy sends up to 512 bytes of data from the buffer to `HAL_UART_Transmit_DMA`. When the transfer completes, the UART TX-complete callback advances the buffer read index and clears the DMA-busy flag so that the next data chunk can be sent. This architecture keeps the main loop non-blocking while still sustaining high throughput to the Pi. Packets contain sync bytes (0xAA 0x55), packet type, payload length, a header timestamp (TIM2 at packet creation time, for latency/debugging purposes), CRC16 over header and payload and finally the actual sensor reading & timestamp.

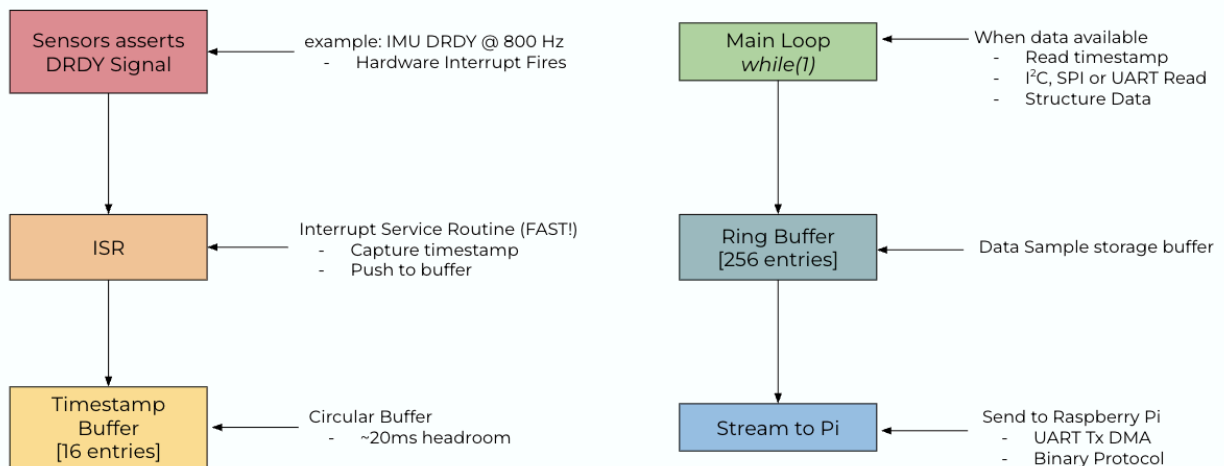


Figure 13. Parallel Data Streams: Interrupt Generating Timestamps and UART stream to Pi

F. Interrupt Driven Architecture Summary

Sensor data flows from hardware DRDY events → TIM2-captured timestamps and lightweight ISR buffers → main-loop sensor read and sample assembly → typed packets into stream buffer → DMA over UART to the Raspberry Pi. Timestamp generation is centralized on TIM2 and captured on DRDY ensuring consistent, microsecond resolution timing for multi-sensor fusion and camera alignment.

G. Raspberry Pi Dataflow Overview: The IR and Visual Cameras

In the final version of the position estimation payload, the Raspberry Pi Compute module will be responsible for running the VISLAM algorithm, feeding it the proper sensor and camera inputs to estimate GPS position in real time. For this capstone project, we are limiting the Compute Module's responsibilities to saving all the timestamped data onto an external SSD for offline processing. The offline processing is covered in the [Data Post Processing](#) section. Here, we will detail the data being received by the Pi, as well as the necessary threads to ensure that no data is lost when saving the dataset to the SSD.

The following figure shows the data flow passing through the Pi to the external SSD

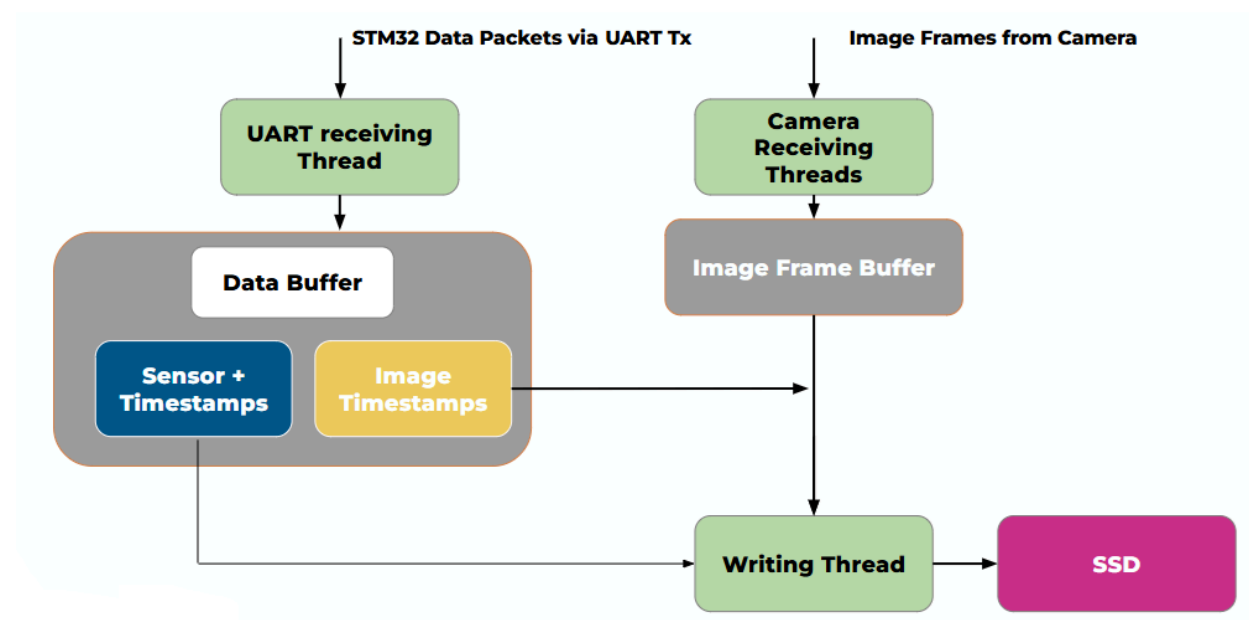


Figure 14. Data flow from Microcontroller UART stream and Visual Cameras to Dataset stored on SSD

The data flows from the top, through the threads, and finally ends up at the SSD.

The left branch represents the UART connection between the STM and CM5. Again, the STM collects data from sensors, as well as generates timestamps corresponding to each sensor's data. This set of data is pushed into a Data Buffer to ensure efficient data throughput, and is represented by the dark blue box labeled "sensor + timestamps". This data can directly be fetched by a writer thread that saves the data as .log files on the external SSD.

The camera image frames, both IR and visual, follow a slightly different path, since the image data can only be sent to the CM5, because the STM can't interface with the cameras. However, one requirement for the timestamped data capture system is for timestamps to be generated on

one common clock, which will be the STM, whose real time MCU capabilities is able to generate precise timestamps on hardware interrupts.

3.3.3 - Housing Subsystem

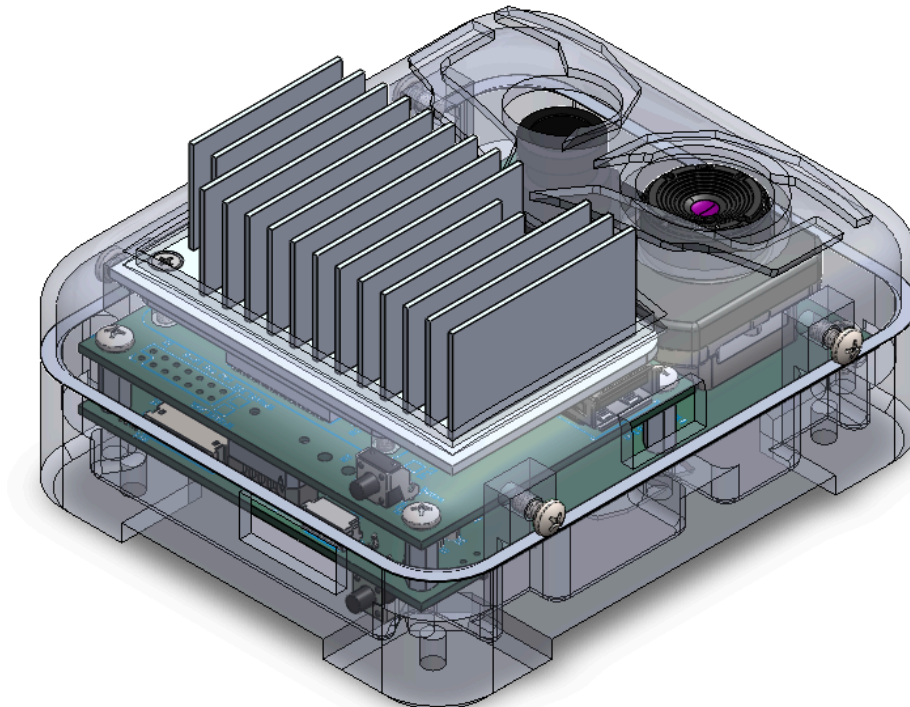


Figure 15. CAD Render of Full Assembly

A. Enclosure

The housing enclosure serves as the primary mechanical structure for the navigation payload, integrating all electrical, optical, and thermal components into a compact, drone-mountable system that can withstand real-world environments. Its design must balance four competing requirements: structural rigidity, low mass, environmental protection, and efficient heat dissipation. Because the payload will be carried by drones operating in hazardous and potentially degraded environments, the enclosure must remain lightweight enough to avoid overloading the drone while still providing sufficient stiffness to maintain optical alignment, protect internal components, and preserve long-term durability.

The enclosure is designed around the footprint of the sensor board, which houses the visible camera, FLIR Boson, and supporting sensors. The internal geometry, including standoffs,

mounting bosses, and component clearances, is configured to ensure secure integration of all subsystems while supporting compact packaging and repeatable assembly. Component placement is strongly influenced by thermal behavior, with heat-generating elements positioned to interface directly with the heat sink in order to maintain safe operating temperatures under the expected 10 W of electrical power draw.

The enclosure must also preserve optical performance for both the visible global-shutter camera and the FLIR Boson. These sensors require unobstructed, rigidly aligned viewports with minimal distortion. The window interfaces must also support the environmental sealing strategy required for the system IP rating. Material selection for these viewports balances optical clarity, environmental durability, and manufacturability.

Environmental protection is a core requirement given the intended operating conditions. The package must operate in ambient temperatures from -15 °C to 40 °C while resisting dust and water ingress. To achieve this, the enclosure uses a two-piece clamshell architecture with gasketed joints that enable tool-accessible disassembly for maintenance. In addition, the housing incorporates two PE13 adhesive Gore vents for pressure equalization. These vents allow the enclosure to regulate internal pressure during temperature and altitude changes while maintaining protection against liquid water and particulate ingress. This pressure regulation reduces stress on seals and optical windows which improves enclosure reliability during field operation.

To support rapid data collection and servicing without compromising environmental sealing, the enclosure also includes an externally accessible USB interfacing port sealed with a silicone cover or compression seal. This allows wired data access and system interfacing while preserving the enclosure's sealing performance when the port is not in use. This feature is particularly useful during bench validation, field debugging, and dataset transfer.

Thermally, the enclosure interfaces directly with the heat sink, which serves as the primary heat-transfer pathway for the payload. Rather than acting as the main thermal path itself, the housing is designed to support and package the heat sink assembly, which removes heat from critical components and rejects it to the surrounding environment. This approach allows the enclosure to maintain its structural and environmental functions while the dedicated heat sink manages the majority of thermal dissipation.

The current enclosure considers a material selection of ASA plastic with eventual production through injection molding. ASA was selected for its favorable balance of low mass, outdoor durability, UV resistance, dimensional stability, and suitability for weather-exposed applications. This makes it appropriate for a search-and-rescue payload that may be exposed to sunlight, dust, moisture, and temperature variation during repeated field deployment. All mechanical interfaces,

including drone mounting, optical window attachment, port sealing, and internal component placement, are being designed for repeatable assembly and scalable production.

At the current stage of development, the enclosure geometry remains relatively box-like because the design effort has prioritized packaging, manufacturability, sealing, and thermal performance. However, exterior form development is ongoing to better reflect the payload's intended search-and-rescue use case. Current concept sketches are exploring a more refined outer shell with a stronger mission-specific visual language while preserving the internal layout and functional requirements. These sketches can be included in this section to illustrate the direction of the next enclosure iteration.

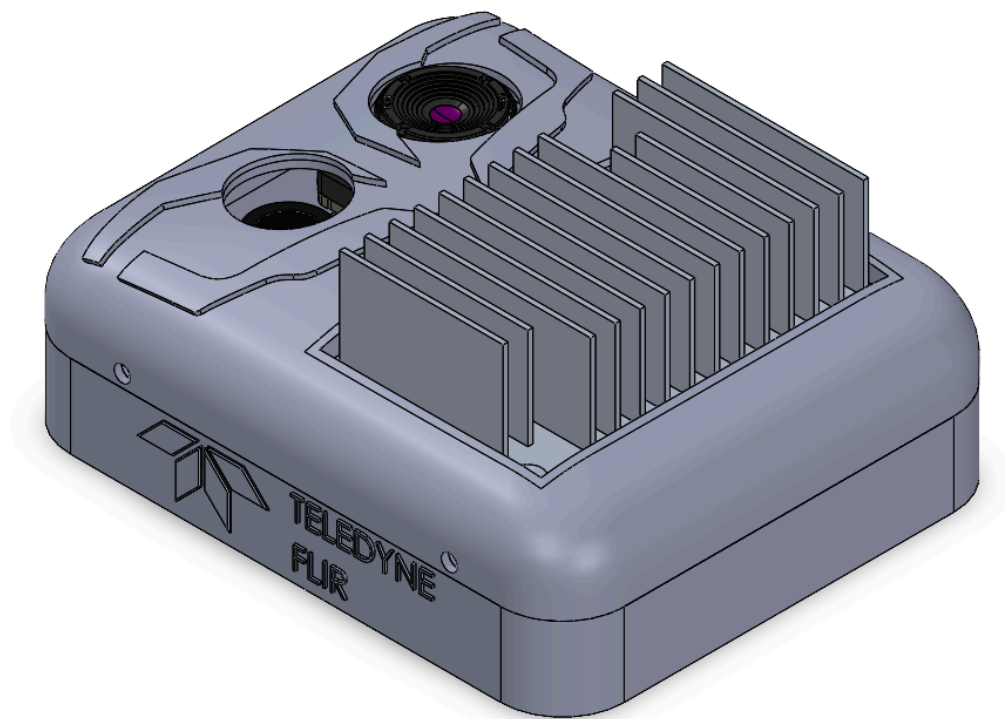


Figure 16. CAD rendering of conceptualized design

Beyond the current packaging-focused housing, the enclosure has been refined through a search-and-rescue-oriented design iteration intended to improve its external character and mission-specific appearance. As shown in **Figure 15**, this updated concept introduces a more purposeful and application-driven outer form while preserving the functional constraints imposed by the internal layout, optical viewports, and heat sink integration. Relative to the earlier box-like configuration, the revised housing presents a more cohesive and field-appropriate

design language that better aligns with the payload's intended operational environment. This iteration represents a meaningful step toward a final enclosure that integrates mechanical performance, environmental protection, and visual identity into a unified housing design.

B. Heat Sink

The heat sink uses passive convection to transfer heat from the electrical components to the environment. Its design must accomplish this task while remaining as lightweight as possible. The final design will be CNC milled out of Aluminum 6061, and it has a 1.75mm thick base plate with 14, 1mm fins that are spaced such that there is 3.93mm separating each fin. Crucially, the design has a flat area around the edges without any fins to make space for a gasket for environmental sealing. The design also incorporates two M2.5 clearance holes to securely mount the heat sink to the Raspberry Pi CM5.

Standoffs were added to eliminate interference in the assembly. However, the addition of the standoffs necessitated additional extrusions that allow the thermal pads to make contact with the CM5. The bevelled corners of the design were added to make the assembly process easier, as sharp corners interfered with access to two crucial screws on the carrier board.

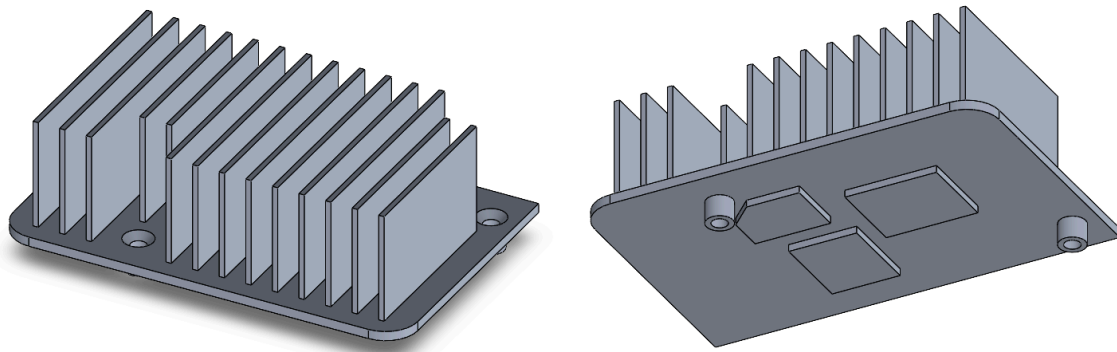


Figure 18. Custom Heat Sink CAD Render

This final design maximizes the rate of heat transfer at low air speeds (<2mph), which are expected when hovering or making small adjustments. Additionally, this design brings the weight of the heat sink down to 47.7g from the preceding design's ~70g.

C. Environmental Sealing

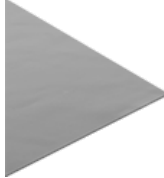
Visual Camera	Pressure Regulation	External Cables	Data Access Ports	Gaskets	Exposed Screws
					
1mm Thick Acrylic Sheet sealed with Silicone Caulk	PE13 Adhesive Gore Vents	Hot Glue	Silicone Rubber Dust Covers	Laser Cut Silicone	O-rings

Table 8. Environmental Sealing Methods

Environmentally sealing the housing will be accomplished with laser-cut silicone gaskets for larger areas (around the Boson, heat sink, and main body), o-rings for any exposed screws, custom-molded silicone dust-covers for exposed ports, hot glue to seal external cable ports, silicone caulk to seal the visual camera's acrylic window, and PE13 adhesive gore vents to provide pressure regulation while preventing any dust or water ingress. This will give the system an IP56 or better rating.

D. Vibration Isolator

The vibration isolator consists of silicone dampers sandwiched between two plates that connect to the drone and the payload. The plates provided stiffness for the mass-damper system, while the soft silicone dampers absorb high-frequency vibrations to ensure the capture system is not significantly impacted. The damper has a resonance of 20 Hz, which is half the lowest continuous excitation frequency for the system. This ensures that when the drone is operating at its general range of 40-300 Hz, vibrations will be damped by the system and ensure critical sensor data is not significantly impacted. Figure 23 below shows the sandwich design of the mounting system, as well as a look at the damper attached to the full payload assembly.

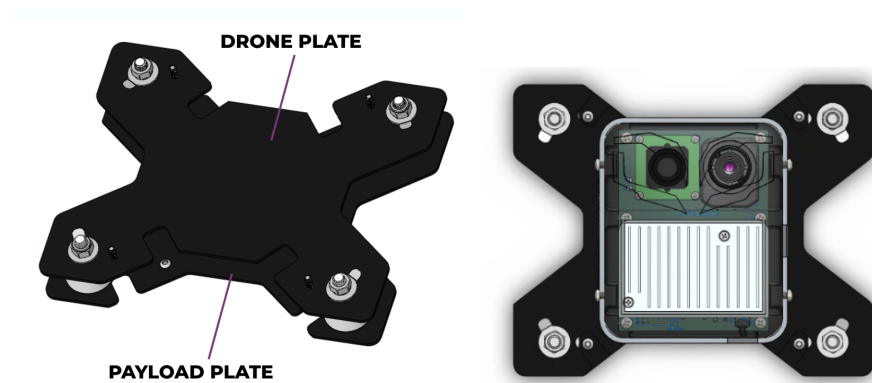


Figure 26. a) CAD Render of Vibration Damper Mount b) Payload attached to Mount

E. Manufacturing

The enclosure is being designed for eventual manufacture through injection molding in ASA plastic. ASA was selected for its favorable combination of low mass, UV resistance, environmental durability, and dimensional stability, making it appropriate for outdoor deployment in search-and-rescue applications. In support of this manufacturing path, the enclosure geometry is being evaluated with attention to standard molding constraints such as draft angle inclusion, parting strategy, and mold release behavior.

These considerations are illustrated in Appendix 10.3, specifically Case M and Case F, which present engineering drawings of the enclosure developed to assess manufacturability. The drawings highlight the use of draft angles to support clean mold release while also informing the separation of the enclosure into manufacturable sections. This work helps ensure that the current design can transition from prototype geometry toward a production-feasible molded housing without sacrificing sealing interfaces, structural features, or internal packaging requirements.

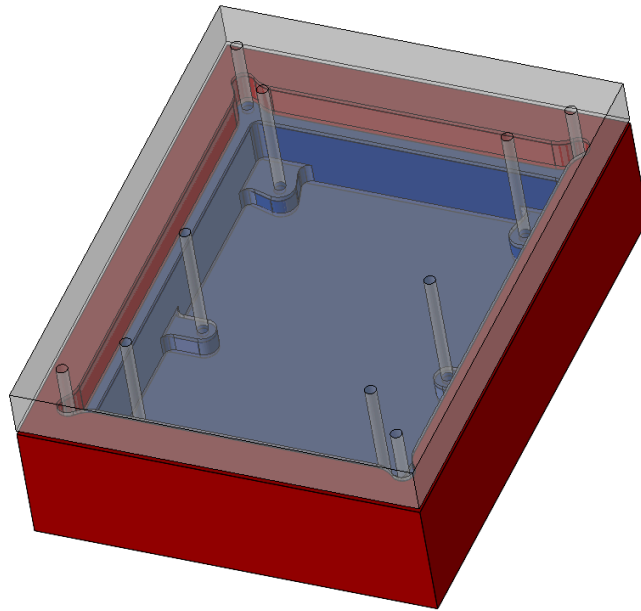


Figure 27. CAD Render of Full Injection Molding Mold Assembly for Prototype Housing Case. The Prototype Case is Blue, and the Other Pieces are the Mold.

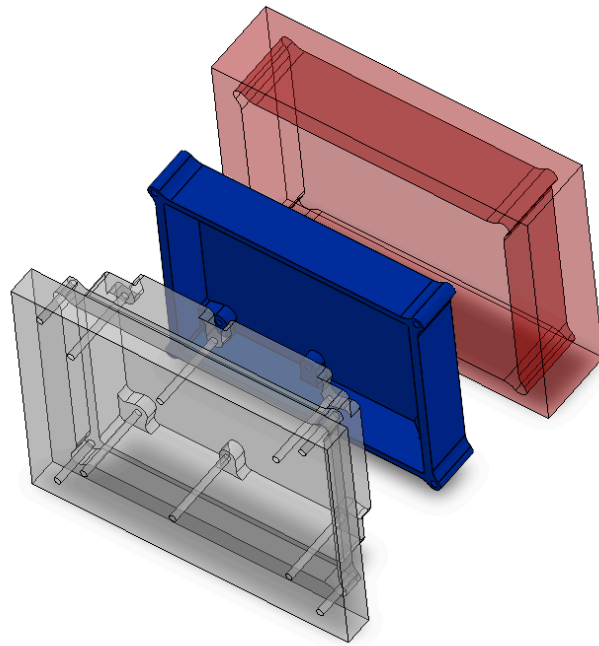


Figure 28. Exploded View of Injection Molding Assembly

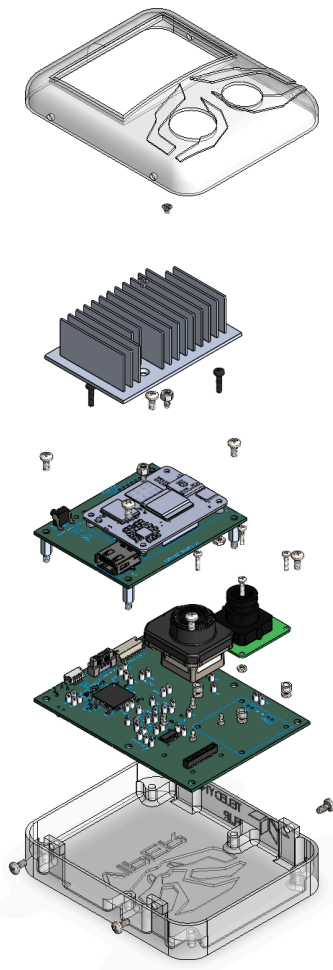
A preliminary concept rendering of the enclosure with its mold arrangement is referenced in Figure 25 to visualize how the current housing form may be translated into a molded part and corresponding tooling strategy. This concept image serves as an early manufacturing study, identifying where additional geometric refinement may be needed to better support molding and assembly.

The heat sink is currently being considered as a custom-manufactured component due to the need for mounting features and thermal interface surfaces tailored specifically to the Raspberry Pi CM5 compute module. Because the geometry must align closely with the module footprint and surrounding packaging constraints, an off-the-shelf solution is less suitable than a purpose-built design. The present manufacturing approach under consideration consists of CNC milling the primary body to establish the external dimensions, mounting locations, and critical thermal contact surfaces, followed by wire EDM to produce the fin structure. This combination of processes is intended to provide both the dimensional precision required for assembly and the fine-feature control needed for efficient fin fabrication. A corresponding engineering drawing and process concept are referenced in Appendix 10.3.

The damping plate subassembly is being designed around simple, repeatable fabrication methods that support both prototyping and future refinement. At present, the damping plates are intended to be water jetted from Carbon Fiber Reinforced Plastic, providing a practical balance of

geometric accuracy, manufacturability, and cost-effective iteration. The vibration-isolation grommets are being considered as molded silicone elements with embedded standard bolts or screws, allowing the isolators to serve as both damping features and mounting interfaces. This approach is intended to simplify implementation, reduce assembly complexity, and improve the repeatability of the final installation. A preliminary drawing and manufacturing concept for the damping plate and grommet assembly are also referenced in Appendix 10.3.

F. Assembly



The full assembly of the VIPER payload consists of attaching two subassemblies, the housing subassembly and the damper assembly. The process is outlined below.

Housing Assembly:

- 1) Attach SSD to corresponding connection on the carrier board and screw into place
- 2) Attach Raspberry Pi to carrier board above the SSD and screw into place at two diagonal holes (the remaining corner holes will attach to the heat sink)
- 3) Attach the BOSON and VisCam to the sensor board and screw into place
- 4) Screw down sensor board to the bottom housing using standoffs
- 5) Connect VisCam ribbon cable to the underside of the carrier board and screw carrier board onto sensor board standoffs
- 6) Place thermal pads onto Raspberry Pi and other high heat components
- 7) Screw heat sink onto the Raspberry Pi
- 8) Place top housing over all components and screw into place

Figure 21. Exploded View of Housing Assembly

Damper Assembly:

- 1) Attach vibration dampers onto the payload plate using hex nuts
- 2) Place drone mount plate on top and fasten using hex nuts

Full Assembly:

- 1) Flip over housing enclosure and screw damper assembly onto the payload
- 2) Position full assembly underside the drone and screw into place

IV. Analysis

4.1 - PCB

4.1.1 - PCB Simulations: Buck Converter

To verify that the selected buck converter design would properly provide the expected output supply voltage, as well as be capable of reliably supplying the expected worst-case load conditions, we simulated the selected IC in PSpice. Initially, we expected to use LTSpice, with Spice models provided by Texas Instruments. However, the specific regulators selected in the design only had encrypted simulation models, resulting in using “Pspice for TI”, a Pspice simulator including a large library of TI components.

The following details the process used to validate the 5V buck converter, which is placed on the carrier board. These simulations focused on confirming closed-loop regulation under maximum anticipated current draw, which for the 5V supply line was 5A (refer to power diagram). This meant confirming the values of the feedback network to output the proper 5V, as well as simulating a load of 5A. In the simulation, a resistive load was used to emulate the multiple components that would draw a total of 5A. The simulation’s circuit is shown below:

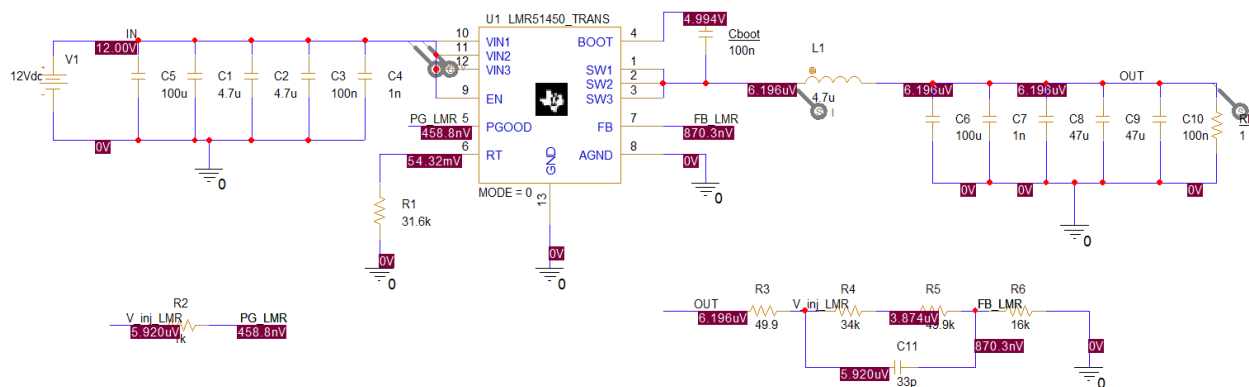


Figure 29. 5V Buck Simulation Circuit

The screenshot above illustrates the complete buck circuit and feedback network that we placed in the PCB as implemented in PSpice for TI. The following plotted waveforms capture both the inductor switching current and regulated output voltage. The inductor current, shown in the top pane, is a high frequency triangular ripple about the DC operating point of 5A, while the output voltage reaches a 5V in steady state.

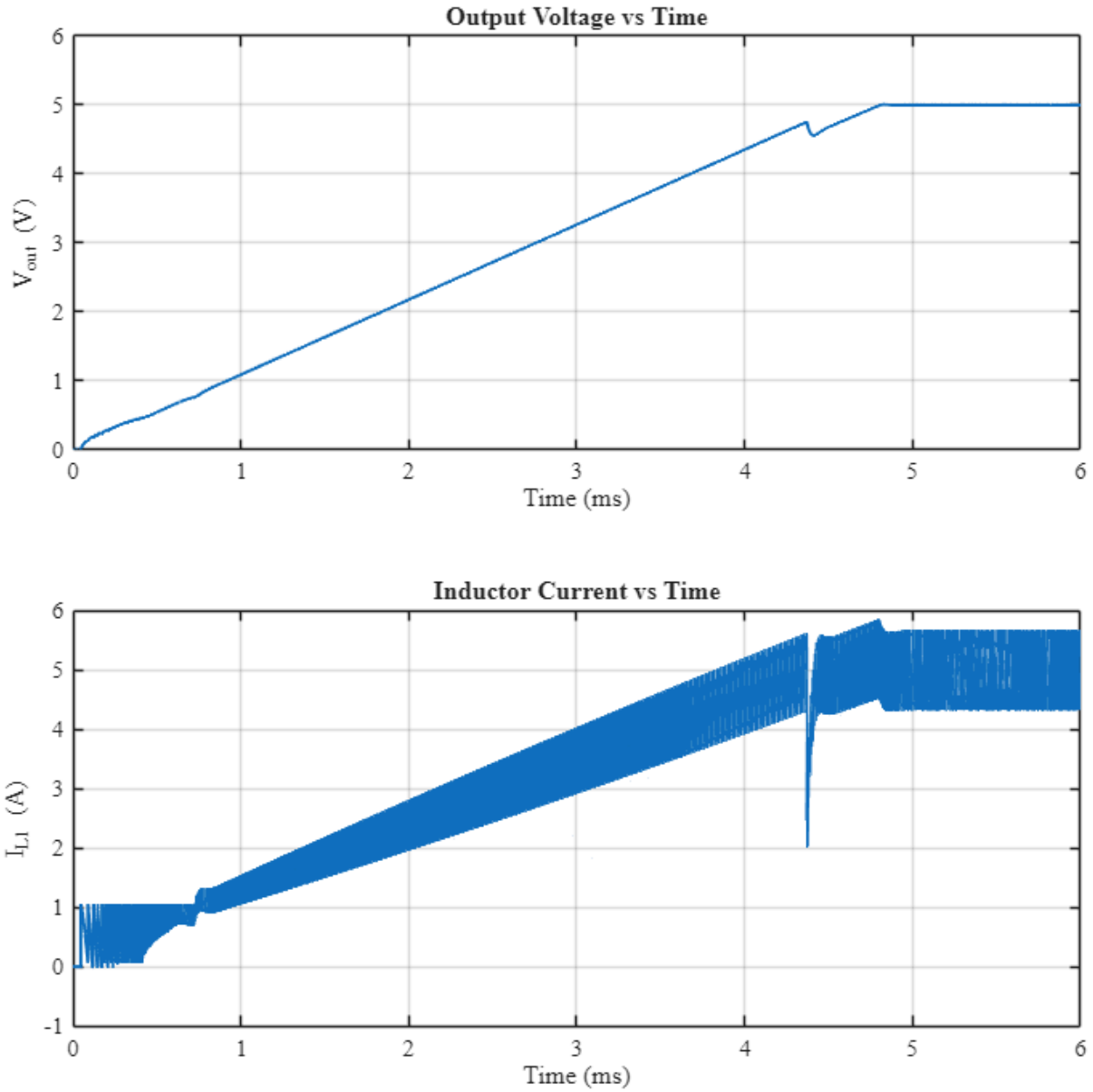


Figure 30. Output Waveform of 5V Buck Simulation; Top Trace is Buck Output Voltage, Bottom Trace is switching inductor current

This has confirmed the resistive values of the feedback network, as well as confirming that the output can be loaded up to our expected 5A. The simulation results also help us characterize the output slew rate, for possible analysis of power rails on power up.

4.2 - Software

4.2.1 - Data throughput

One figure of merit of our system is the data rate at which we are able to stream over UART from the Microcontroller to the Pi. A large amount of data, including the most crucial timestamps, are sent over this UART connection. The following table details the parameters associated with the setup of the transmission link.

UART Link (STM32 → Raspberry Pi-5)

Parameter	Value
Interface	UART5 (DMA TX)
Baud Rate	921,600
Word Format	WordLength: 8, StopBits: 1, Parity: None
Theoretical Max	~92 KB/s
DMA Chunk Size	512 B (max per call)
Stream Buffer Size	8192 B

Table 9. Parameters of UART link between STM and Compute Module

We have then done an analysis of the maximum data rate as captured from the sensors, and ensure that the UART transmission does not limit our data acquisition.

Per-Packet Size & Rate (Steady State)

Packet Type	Sensor	Packet Size (Bytes)	Rate (Hz)	Bandwidth (KB/s)
IMU1	SCH16T	29	800	23.2
IMU2	ICM-45686	29	800	23.2
Baro	BMP390	22	25	0.55
Mag	RM3100	27	Variable	-
Viscam	OV9281 Strobe	19	Frame Rate	-

IR-Cam	Boson	19	Frame Rate	-
GPS	ZED-F9P	71-80	10	0.71
Total				47.7

Table 10. Estimated required bandwidth for UART link

Total Utilization: $\sim 47.7/92 = 52\%$ of UART capacity

As we find, we are only utilizing around 52% of the UART link's capacity. The remaining throughput margin ensures that the Compute module will receive all sent data without any packet losses.

4.2.2 - Cyclic Redundancy Check (CRC) for Data Transmission

To further ensure the reliability of our transmission link, we implemented a Cyclic Redundancy Check, which adds 16 more bits of redundancy per packet to assist with error detection. This implementation will allow us to detect and keep track of errors.

Below is a table detailing the implementation of our CRC structure.

Item	Value
Algorithm	CRC-16-CCITT (MSB-first)
Polynomial	0x1021
Initial Value	0xFFFF
Width	16 bits

Table 11. Chosen CRC implementation

- Implementation: CRC is computed over bytes 2-7 of the header (type, len, timestamp) then over the entire payload itself. Sync and CRC bits are excluded.
- Receiver: Raspberry Pi uses the same algorithm and initial value. Computed CRC is compared to received CRC. On mismatch the packet is discarded and `crc_errors` is incremented

4.2.3 - Buffers and Overflow Handling

The system uses fixed size and circular buffers in both the STM32 firmware and the Raspberry Pi. Overflow is handled by checking space before writes, rejecting or dropping data when full

and counting events for diagnostics. Before any write could fill a buffer, the code checks for space. On full: the write is rejected, and a debug counter is incremented ('overflow_count', 'bmp_ts_dropped', 'imu_timestamp_overrun', ect). In nominal operation the system has proven to run with no overflows.

4.2.5 - Data Post Processing

As discussed in [Raspberry Pi Dataflow](#) section, our sensor, image frames, and associated time stamp data is saved to an external SSD, where we will format into proper data sets after a flight, and send them into our associated algorithms. The first software we will need to package the data for is *Kalibr*, which calibrates the orientation of IMUs and camera locations relative to each other. *Kalibr* will give us extrinsic parameters, relating to the camera projection transform. These “extrinsics” are required for the final Spectacular AI Position Estimation algorithm to accurately estimate position. The following sections outline the requirements for our sensor data to work with both necessary softwares.

A. Kalibr Calibration

Kalibr is used to estimate the extrinsic spatial and temporal relationships between the cameras and the IMU in the system. These parameters describe the rigid-body transformation between sensor coordinate frames and are required for accurate visual-inertial odometry. The calibration process uses synchronized image and IMU measurements and performs a batch optimization to estimate the transformation between the sensors.

Kalibr requires sensor data to be provided in a ROS bag format containing image streams and IMU measurements. Robot Operating System (ROS) bags are a common file format in controls and robotics applications as they are from the ROS ecosystem. This file format allows for recording and storing timestamped data from multiple sensors, such as our IMUs and cameras, and a “playback” system, to simulate live data for post processing tasks.

The Kalibr documentation provides a bagcreator script that constructs a ROS bag from image files and an IMU CSV file. In this format, images are organized into camera folders (e.g., cam0, cam1), with filenames corresponding to timestamps in nanoseconds. The IMU data is stored in a CSV file containing timestamped measurements of angular velocity and linear acceleration.

During calibration, Kalibr uses the recorded sensor data together with configuration files describing the camera model, IMU noise characteristics, and calibration target. The calibration process estimates the transformation between the camera frame and the IMU frame, producing output files including a YAML file containing the estimated extrinsic parameters, a text summary of results, and a detailed calibration report.

Calibration setup

Two cameras were calibrated independently in production:

Sensor	Resolution	Camera model	Calibration target
Viscam (visible)	1280×720	Pinhole + radial–tangential distortion	AprilGrid (5×6 tags, 45 mm tag size, 14 mm spacing)
Boson (thermal)	640×512	Pinhole + radial–tangential distortion	Asymmetric circle grid (4×11 , 30 mm spacing)

Table 12. Calibration Setup

Prior to camera–IMU calibration, IMU noise parameters were characterized using Allan variance analysis on $\sim 9,800$ s of static IMU data at ~ 735 Hz. The resulting noise densities (after applying safety scaling factors of $5\times$ on white noise and $10\times$ on random walk) were:

Parameter	Value
Gyroscope noise density	3.69×10^{-5} rad/s/ $\sqrt{\text{Hz}}$
Gyroscope random walk	1.12×10^{-6} rad/s ² / $\sqrt{\text{Hz}}$
Accelerometer noise density	3.33×10^{-3} m/s ² / $\sqrt{\text{Hz}}$
Accelerometer random walk	2.40×10^{-4} m/s ³ / $\sqrt{\text{Hz}}$

Table 13.: IMU Noise Parameters

Camera intrinsics

Viscam

Parameter	Value	Uncertainty ($\pm$)
Focal length f_x, f_y	1095.3, 1093.1 px	2.5, 2.4 px
Principal point c_x, c_y	661.2, 429.4 px	1.8, 1.8 px
Distortion (k_1, k_2, p_1, p_2)	$-0.442, 0.205, -6.77 \times 10^{-4}, 6.55 \times 10^{-4}$	—
Reprojection error (std)	0.40 px (u), 0.33 px (v)	—

Table 14. OV9281 Camera Intrinsic Parameters

Boson

Parameter	Value
Focal length f_x, f_y	410.4, 410.2 px
Principal point c_x, c_y	314.5, 261.2 px
Distortion (k_1, k_2, p_1, p_2)	$-0.384, 0.146, -2.81 \times 10^{-4}, -2.46 \times 10^{-4}$
Reprojection error (std)	0.16 px (u and v)

Table 15. FLIR Boson Camera Intrinsic Parameters

Both cameras show sub-pixel intrinsic calibration accuracy. The Viscam exhibits moderate radial distortion ($k_1 = -0.44$), consistent with a wide-angle lens.

Camera–IMU extrinsics and timing

The primary VIO sensor is the Viscam paired with IMU0. Kalibr estimates the rigid transform T_{ic} (camera to IMU) and the temporal offset between image timestamps and IMU samples.

Viscam $\leftrightarrow$ IMU0

Quantity	Value
Translation (camera origin in IMU frame)	[25.5, 1.4, -33.4] mm
Baseline magnitude	42.1 mm
Camera–IMU time shift	+4.46 ms ($t_{imu} = t_{cam} + \text{shift}$)
Gravity estimate	$[-0.12, -9.81, -0.03]$ m/s ²

Table 16. Visual Camera-IMU Extrinsic Parameters

Viscam–IMU calibration residuals

Residual	Mean	Median	Std
Reprojection error	0.45 px	0.39 px	0.28 px
Gyroscope	0.0085 rad/s	0.0074 rad/s	0.0055 rad/s
Accelerometer	0.057 m/s ²	0.047 m/s ²	0.060 m/s ²

Table 17. Visual Camera-IMU Quality Metrics

These residuals indicate a well-constrained calibration: reprojection error is well under 1 pixel and inertial residuals are small relative to expected motion during the calibration sequence.

The resulting camera–IMU extrinsics define the relative pose between sensors and are required for downstream visual-inertial algorithms. These parameters allow the system to correctly project visual measurements into the inertial frame and fuse them with IMU measurements for motion estimation.

B. Spectacular AI Position Estimation

The final stage of data processing uses the Spectacular AI visual-inertial odometry (VIO) system to estimate the six-degree-of-freedom pose of the platform. This algorithm combines camera imagery with inertial measurements from the IMU to perform visual-inertial SLAM, producing a continuous estimate of device position and orientation over time.

Spectacular AI expects sensor data to be organized in a structured dataset format that includes synchronized camera and IMU measurements. The recording format is designed to efficiently store high-frequency multi-sensor data using encoded video streams together with metadata files describing sensor measurements and timestamps.

Each dataset typically includes camera frames, associated timestamps, and IMU measurements consisting of accelerometer and gyroscope readings. These synchronized streams allow the algorithm to track visual features between frames while using inertial data to estimate motion between observations. By fusing these measurements, the VIO system estimates the device’s trajectory and orientation relative to the environment.

For accurate position estimation, the algorithm also requires the extrinsic calibration parameters obtained from Kalibr. These parameters define the rigid transformation between the IMU coordinate frame and the camera coordinate frame, enabling the algorithm to correctly combine visual feature observations with inertial measurements. After calibration and formatting, the recorded datasets can be processed by the Spectacular AI pipeline to generate pose estimates and reconstructed trajectories from the flight data.

C. Dataset Formats

Following the requirements of the two softwares, we opted to save our data such that we could flexibly post-process and organize the data for *Kalibr*’s ROS bag and Spectacular AI’s inputs.

On each test flight, a new directory will be created, with three total folders.

None

```
flight_0xx/
├── boson/
├── sensors/
│   ├── baro.csv
│   ├── boson.csv
│   ├── gps.csv
│   ├── imu1.csv
│   ├── imu2.csv
│   ├── mag.csv
│   ├── shutdown.csv
│   ├── sync_diag.csv
│   └── viscam.csv
└── viscam/
```

The *sensors* folder contains many .csv files. Each csv corresponding to a polled sensor contains both the sensor data along with the interrupt generated timestamp associated with the sensor data. baro.csv, imu1.csv, gps.csv are a few examples. boson.csv and viscam.csv only contain timestamps associated with an image index, which is matched up with their corresponding image frames in post processing.

The *Boson* and *viscam* folders contain all the camera image frames. File writing size and speed is optimized using the .pnm image file format. Each image is indexed by their sequence number since program start, for use to match to appropriate image timestamps as mentioned earlier.

We choose this file format so that all image frames can be linked to their associated timestamp by their sequence number after the data system has finished running. This reduces the stress on the system, reducing the chance that the CPU is busy and misses any other data collection responsibilities.

4.3 - Thermal

4.3.1 - Material trade studies

Material	C10100 Copper	C21000 Copper	C11000 Copper	C12200 Copper	6061 Aluminum	6063 Aluminum
Density (g/cm ³)	--	--	--	--	0	0
thermal conductivity	++	+	++	++	0	+

(W/m*k)						
machinability	0	0	0	0	0	+
corrosion resistance	0	+	-	-	0	+
total +	2	2	2	2	0	2
total -	2	2	3	3	0	0
score	0	0	-1	-1	0	2

Table 18. Heat Sink Material Trade Study[2], [3], [4], [5], [6], [7]

A trade study was done to establish the best material for the manufacture of the heat sink. Aluminum 6061 was used as the baseline and remains the first choice due to availability, however Aluminum 6063 is the optimal candidate because of its slightly better corrosion resistances, machinability, and conductivity.

4.3.2 - Benchmark test

A thermal benchmark test was done to establish a baseline for internal temperatures inside the housing without any proper heat sinking. Three load resistors were placed inside the housing to model the expected heat output of the system, and six thermocouples were placed throughout the box to measure temperatures of each resistor, the air inside the box, as well as both the internal and external walls of the box itself. Refer to [Appendix 10.4](#) for full SOP.

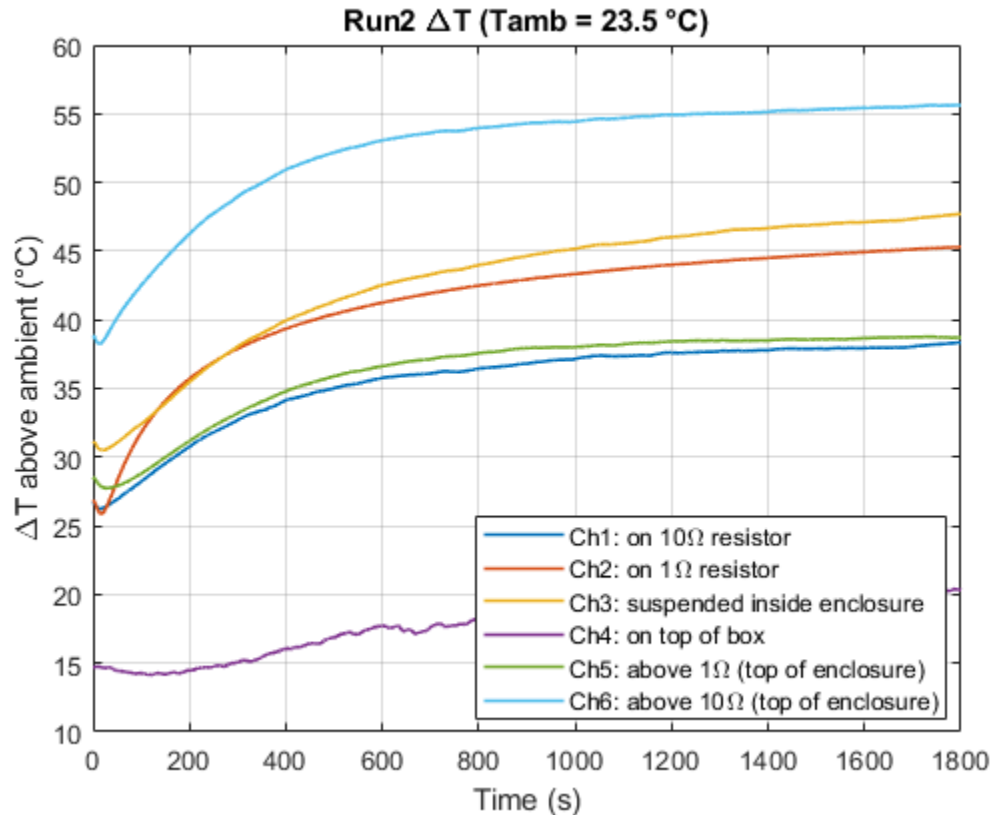


Figure 31. Results of Thermal Benchmark Test

These results proved that a dedicated heat sink was necessary for the success of this project, as the maximum internal temperature exceeded 45 °C above ambient. The altimeter must remain below 65 °C, and the remaining sensors max out at 85 °C, including the CM5 processor. So, these temperatures would greatly degrade the operation of all internal components at the maximum end of the required operational temperature range (40 °C) without intervention.

4.3.3 - Heat Sink Optimization

A. Purpose

Earlier designs used an off-the-shelf heat sink that would have made manufacturing easier, but it had two critical flaws: it was far too heavy (~70g), and it did not efficiently transfer heat to the environment because the fin spacing prevented laminar flow across the fins. So, an improved, custom design was necessary. However, the footprint was kept the same to assist the overall housing design.

For any custom design, it is necessary that the maximum temperature never surpasses 80 C since the Raspberry Pi CM5 begins throttling its processor speed beyond that point.

B. Methods

Using FEA in Solidworks Flow Simulation, three different variables were tested for their effects on maximum temperature and weight: fin thickness, fin spacing, and base plate thickness.

The first series of tests looked to optimize the fin thickness and spacing for thermal efficiency. The initial temperature of the solid was set to 40C; the ambient temperature of the air was also set to 40C to represent the worst possible operational conditions; airflow over the fins was restricted to 0.5m/s; convection was restricted to the fin surfaces only, so the base plate would not influence the study; and 10W of heat was applied uniformly across the bottom surface of the heat sink. The geometries were tested for the maximum temperature at steady state. The best performing fin configuration with the lowest maximum temperature moved on to the second test.

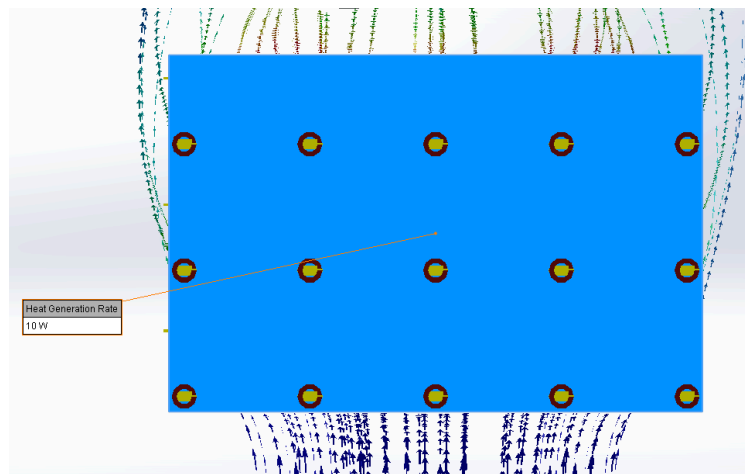


Figure 32. Heat Sink Optimization Test 1 Setup. Red Marks for Heat Generation, and Yellow for No Convection.

The second series of tests analyzed the cost of reducing baseplate thickness for weight savings versus the temperature increase. To properly test for spreading capabilities, the input heat was

reduced to 7.5W and the area of the heat input was reduced to a 20mm x 20mm region approximating where the Raspberry Pi CM5 would be located. The initial and ambient conditions remained the same between tests, as did the restriction on convective surfaces. The different geometries were tested for weight savings and maximum steady state temperatures, as compared to the original 3mm design.

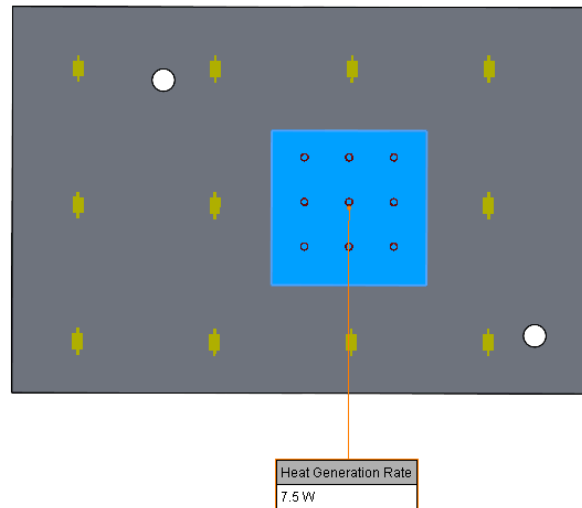


Figure 33. Heat Sink Optimization Test 2 Setup. Red Marks for Heat Generation, and Yellow for No Convection.

C. Results

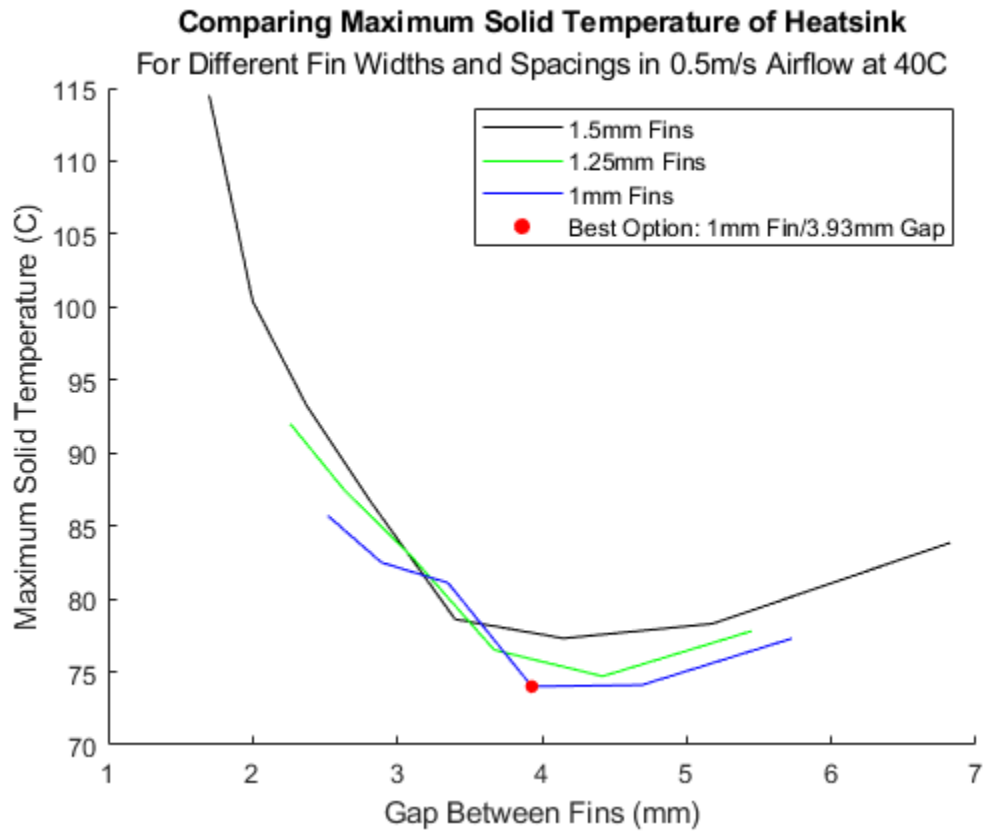


Figure 34. Results of Heat Sink Optimization Test 1

The results of the first test indicated that the most efficient geometry for convective heat transfer had a fin thickness of 1mm with 3.93mm of space between each fin. This geometry had a maximum solid temperature of 74.0 C, which is a significant improvement over the base design that had a maximum solid temperature of 114.5 C.

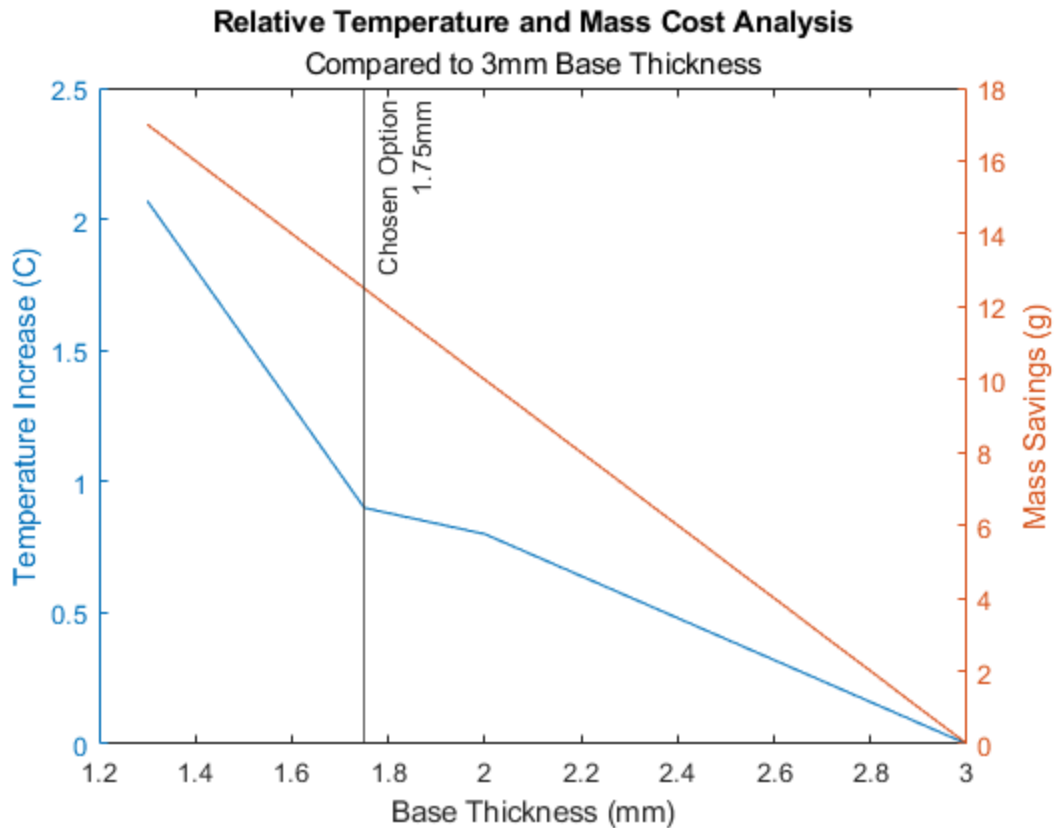


Figure 35. Results of Heat Sink Optimization Test 2 as compared to the original 3mm thick baseplate design that weighed 59g and had a maximum temperature of 71.7C

The results of this test indicated that a base plate thickness of 1.75mm had the best tradeoff between weight savings and maximum temperature. Although it resulted in a 0.9 C increase in maximum temperature, the 12.5g of weight reduction justified that cost.

These results were directly implemented into the final design of the heat sink.

4.3.4 - Full System Simulations

In order to accurately simulate the heat transfer of our system, two separate simulations were run inside COMSOL Multiphysics. The first simulation was a heat transfer in solids and fluids simulation with an ultra simplified housing design which was a single solid block with our optimized heat sink attached. The solid housing block was set as a heat source, generating 11.5 watts uniformly. The laminar fluid flow was set at 2 mph and at a 15 degree angle relative to the housing. This is to model the worst case scenario, equivalent to when the drone is flying very slow or awaiting takeoff in the field. From this simulation, we were able to calculate the heat transfer coefficients of each individual side of the housing as well as for the sink. Also, this simulation taught us about the airflow and pressure field created by our payload.

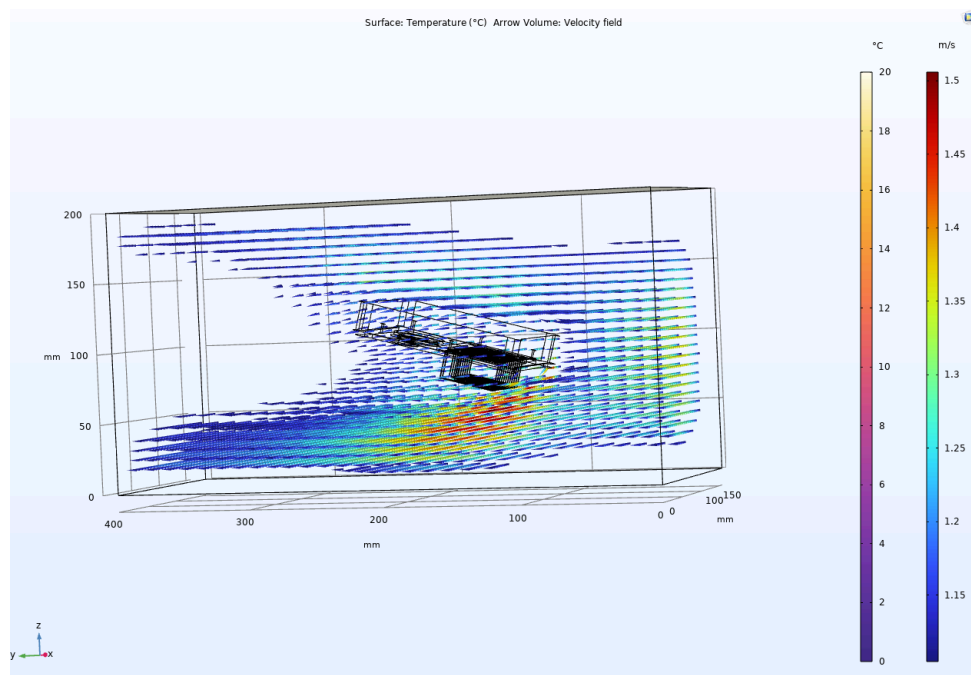


Figure 36. Wind tunnel simulation. 2 mph wind, 15 degree angle

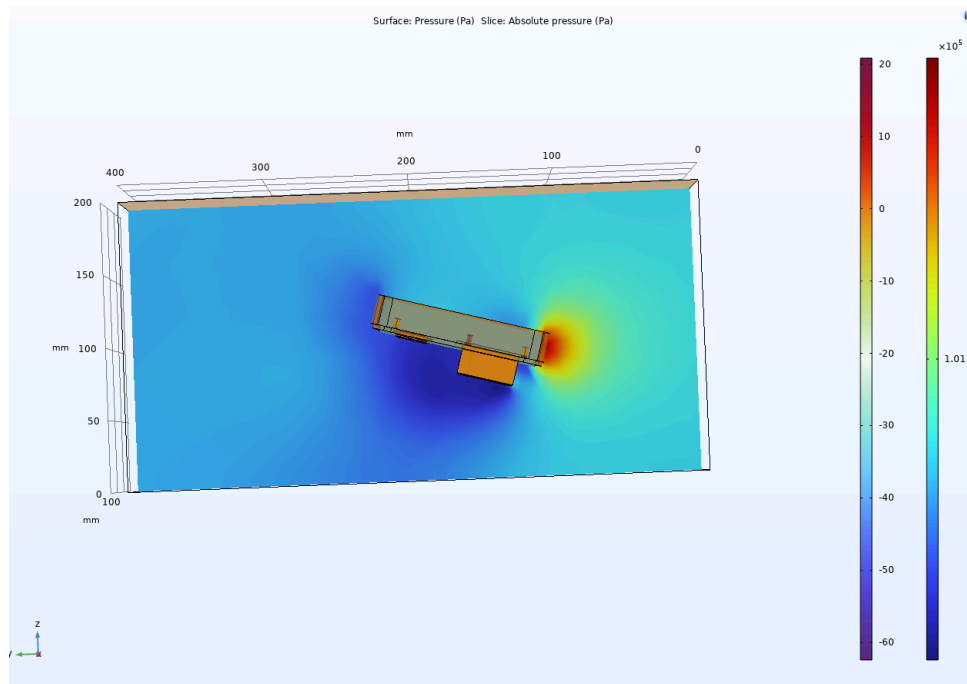


Figure 37. Pressure map around payload during flight

The connective coefficients are:

Surface	h (W/(m ² * K))
Front	99.4
Back	24.4
Sides	37.7
Top	17.2
Bottom	32.4
Sink	9.1

Table 19. Convective coefficients of individual surfaces

After finding the convective coefficients of each surface of the housing, we could move to a more complex solid model while ditching the fluid simulation. The full assembly stackup was modified for simplicity which included removing most components from the PCBs. Only 2

components remained on the PCBs which were the CM5 and Bozon. These components will generate 10 watts and 1.5 watts respectively for simulation purposes. Another simplification which had to be made is that the air inside the enclosure was modeled as a solid block. It has the same heat properties as air but the simulation does not model any circulation of the air inside.

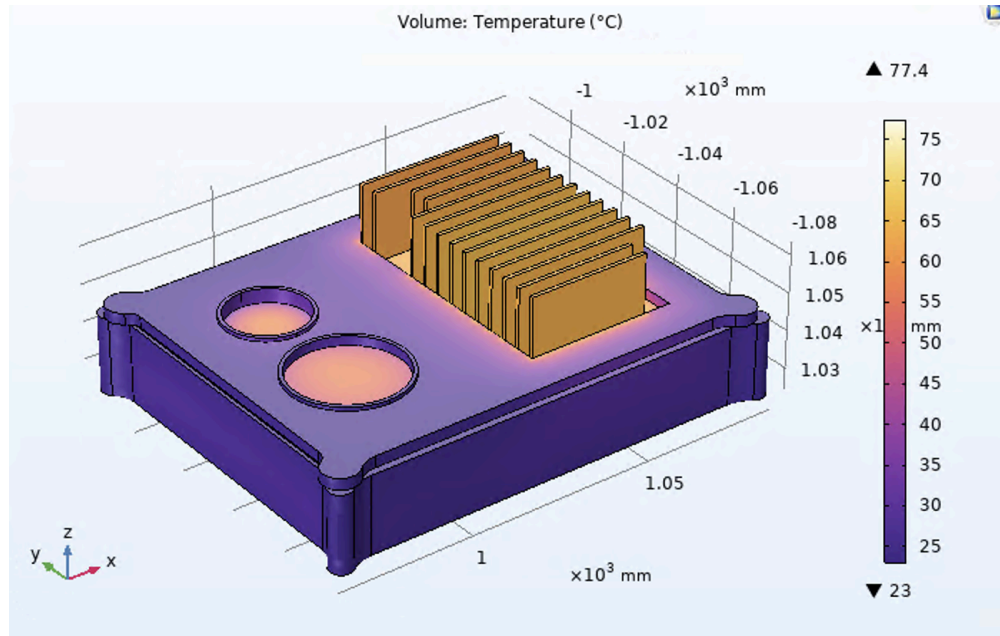


Figure 38. Temperature map of payload during flight

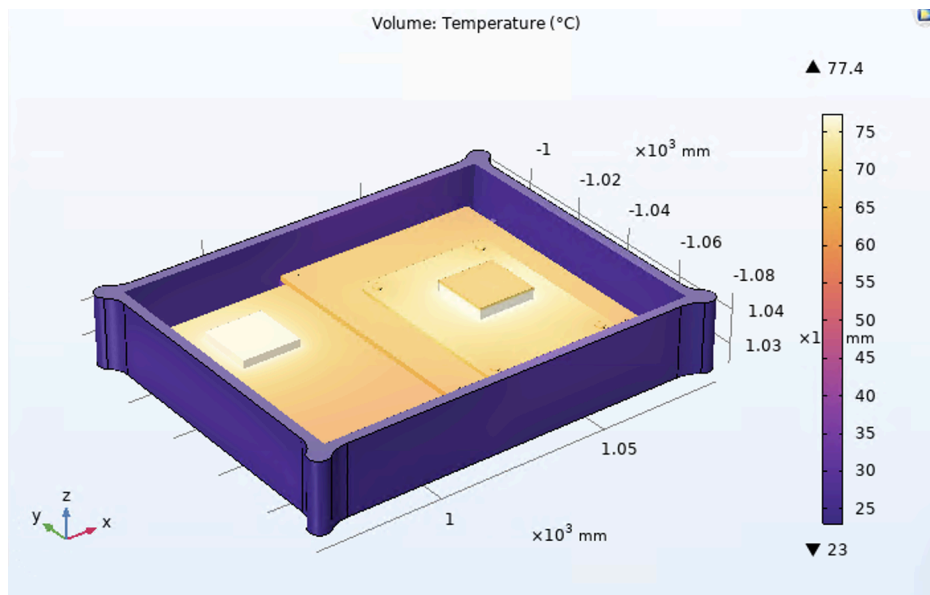


Figure 39. Temperature map of internals during flight

The simulated steady state temperatures of the housing are:

Probe location	Temp (C)
Boson chip max	76.5
CM5 chip max	77.4
Sink max	67.5
Housing max	71.2
Internal air average	49.7

Table 20. Temperature of payload during flight

A transient study was also run to find the time constant of the system. It was found that the system reaches steady state in about 30 minutes which is important to keep in mind when testing of the system begins.

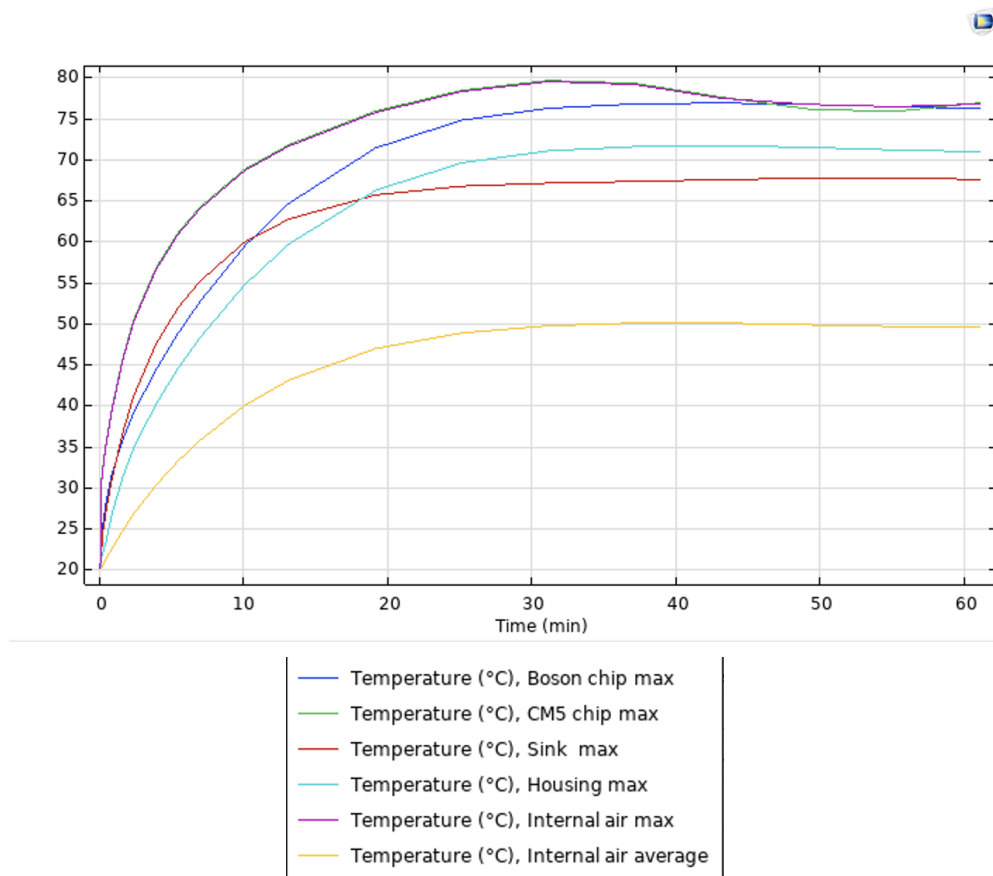


Figure 40, 41. Startup temperature plot

4.4 - Vibrational

The vibrational analysis begins with characterizing the lowest continuous operational frequency of the DJI Matrice 300, which can be done by modeling the drone as a plate (mass) and 4 cantilever (spring) arms. This calculation allowed for a stiffness to be determined for the drone.

Drone modeled as a PLATE (mass) + 4 CANTILEVER (spring) ARMS $\rightarrow k = 557.3 \text{ kN/m}$

Using the maximum takeoff weight specification for the drone payload and plugging in the calculated values into the frequency equation gave a drone frequency of about 40Hz.

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

$$k = \text{stiffness}$$

$$m = \text{mass}$$

$$f_n = \text{natural frequency}$$

$$f_n = 40 \text{ Hz}$$

To maintain best vibrational damping, a factor of safety of 2 was chosen, which results in a 20Hz resonant frequency for the damper.

$$f_n = \frac{1}{2} (40 \text{ Hz}) = 20 \text{ Hz}$$

Again, rearranging the equation above and substituting the mass of the payload, the required stiffness for the damper is:

$$k = m \cdot (2\pi f_n)^2$$

$$k_{\text{isolator}} = 5021.65 \frac{N}{m}$$

Dividing per four dampers, this value comes out to be:

$$k_{\text{grommet}} = \frac{k}{4} = 1255.41 \frac{N}{m}$$

As this is a relatively soft stiffness, the next calculation is to determine how much deflection the dampers will experience using the deflection equation below:

$$\delta = \frac{W}{k}$$

$W = \text{weight}$

$k = \text{stiffness}$

$\delta = \text{deflection}$

$$\delta = 0.25mm$$

These parameters allow for the selection of the dampers. Based on the calculations above, for a stiffness of about $1255.41 \frac{N}{m}$, a damper with a shore hardness of about 30A is required. The deflection parameter determines the general width and height of those dampers

4.5 - Weight Analysis

The current mass budget for the VIPER payload is summarized in Table 15, with the component distributions shown in Figure 34 and Figure 35. The updated enclosure and internal electronics assembly has a total mass of 259.7 g without the damper assembly. This value includes the Boson module, visual camera, sensor board, carrier board, Raspberry Pi Compute Module 5, housing components, heat sink, cabling, and fasteners. When the full damper assembly is included, consisting of the damper plates, isolator cylinders, and mounting hardware, the fully mounted assembly mass increases to 547.4 g.

Item	Mass (grams)
Boson	14.0
Visual Camera	4.0
Sensor Board	23.0
Carrier Board	13.8
RPi CM5	15.0
Housing Top	32.4
Housing Bottom	71.5
Heat Sink	47.7
Cabling	18.5
Fasteners	19.8
TOTAL	259.7

Table 21. Current mass budget for VIPER enclosure and internal electronics assembly, excluding damper assembly.

Mass Budget (g)

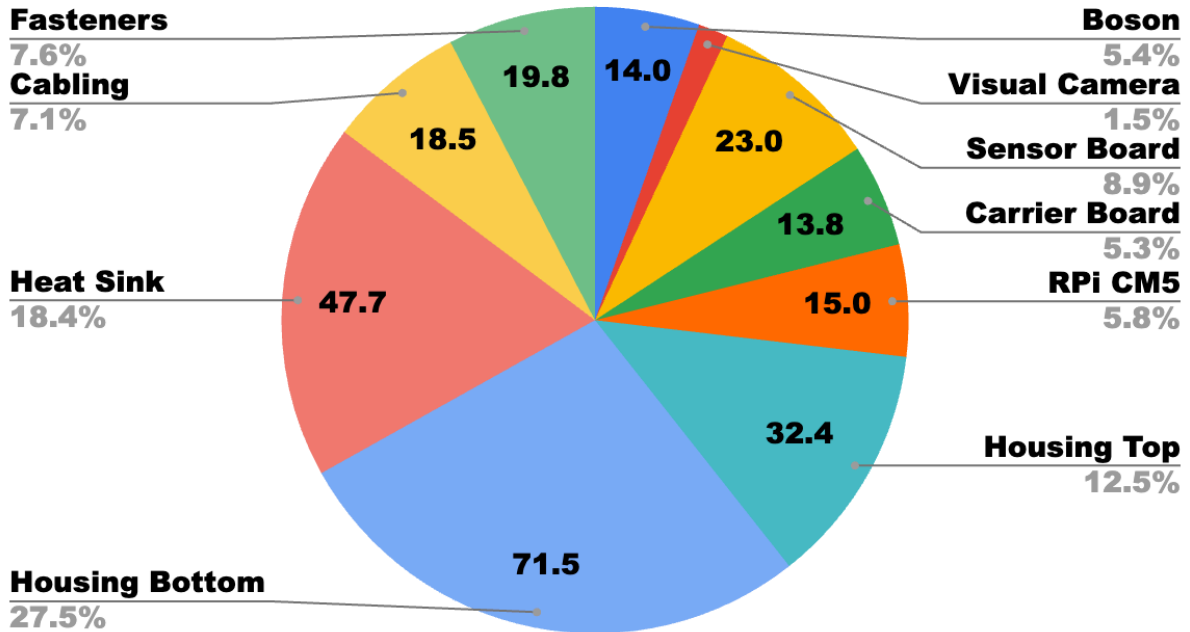


Figure 42. Mass distribution of VIPER enclosure and internal electronics assembly, excluding damper assembly.]

Item	Mass (grams)
Boson	14.0
Visual Camera	4.0
Sensor Board	23.0
Carrier Board	13.8
RPi CM5	15.0
Housing Top	32.4
Housing Bottom	71.5
Heat Sink	47.7
Cabling	18.5
Fasteners	19.8
Damper Plates	287.7
TOTAL	547.4

Table 22. Current mass budget for fully mounted VIPER assembly, including damper assembly.

Mass Budget (g)

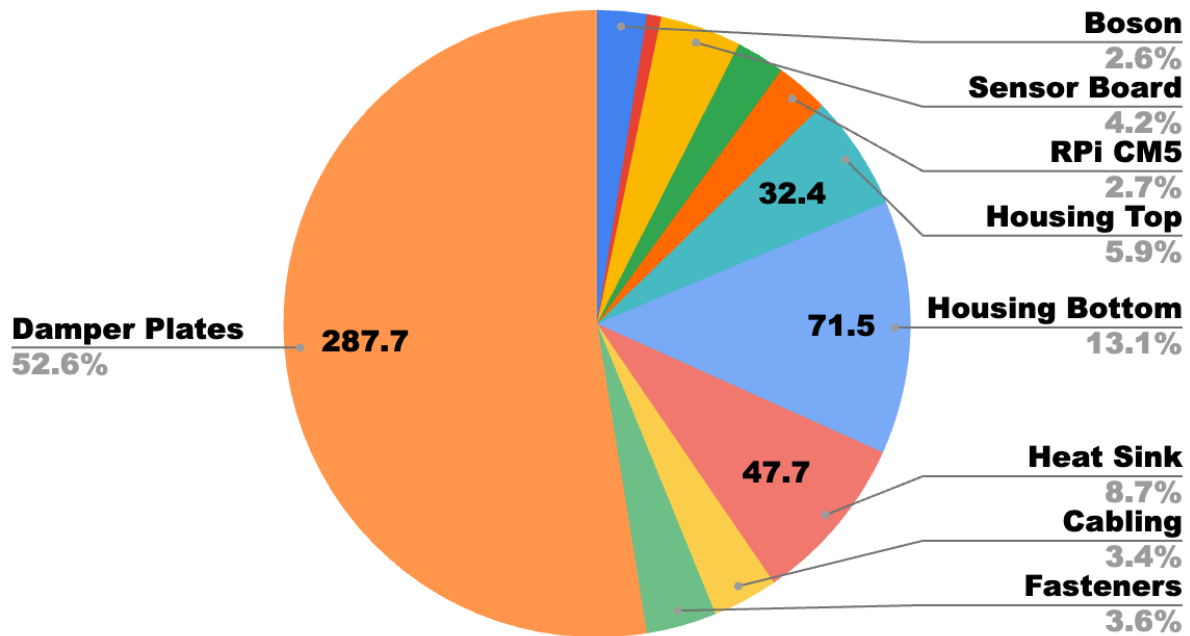


Figure 43. Mass distribution of fully mounted VIPER assembly, including damper assembly.

The original target mass for the VIPER payload was 250 g. The current enclosure and internal electronics assembly is slightly above this target at 259.7 g, but this result is still considered highly successful because the current design is significantly more mature than the earlier design iterations. Previous versions were mainly focused on fitting the electronics into an enclosure and proving the basic layout. The current version has been refined into a more complete final product form, with cleaner exterior geometry, improved packaging, added cabling, updated fastener estimates, and integrated interfaces for sealing, mounting, and assembly. Even with these added design features, the payload remains within 9.7 g of the original target mass, which is only 3.9% above the target.

For the enclosure and internal payload hardware alone, the largest mass contributor is the housing bottom at 71.5 g, which makes up 27.5% of the 259.7 g assembly mass. The next largest contributor is the heat sink at 47.7 g, or 18.4% of the total. The housing top contributes 32.4 g, or 12.5%, while the sensor board contributes 23.0 g, or 8.9%. The fasteners and cabling also make up noticeable portions of the mass budget at 19.8 g and 18.5 g, respectively. The remaining components include the Raspberry Pi CM5 at 15.0 g, Boson module at 14.0 g, carrier board at 13.8 g, and visual camera at 4.0 g. These values show that the mechanical hardware, especially the housing and heat sink, remains the largest contributor to the internal payload mass.

The heat sink optimization played a major role in keeping the final payload mass close to the original target. Earlier heat sink concepts reached 70.9 g, while the current optimized heat sink is 47.7 g. This is a reduction of 23.2 g, or approximately 32.7%. Without this reduction, the current enclosure and internal electronics assembly would have been approximately 282.9 g instead of 259.7 g, assuming all other component masses stayed the same. This shows that the heat sink redesign was one of the most important weight-saving changes in the final mechanical design.

The optimized heat sink also helped balance thermal performance with mass. Rather than simply reducing material everywhere, the redesign focused on lowering unnecessary heat sink mass while maintaining useful fin geometry for airflow and heat rejection. This was important because the heat sink still needed to support the thermal requirements of the electronics while staying compatible with the compact enclosure. As a result, the current heat sink represents a better design tradeoff than the earlier version because it lowers mass while still supporting the payload's thermal management needs.

The current design also represents a major progression from the earlier fall and winter quarter iterations. The fall quarter payload estimate was 253.4 g, while the winter quarter estimate was reduced to 229.0 g after the heat sink redesign. The current internal payload mass increased to 259.7 g, but this increase reflects a more complete and refined design rather than a loss of weight control. The current housing has a more polished final-product appearance, more intentional packaging, added cabling, updated fastener estimates, and improved assembly and sealing interfaces. Because of this, the current mass budget is more representative of the final VIPER payload than the earlier design estimates.

The fully mounted system mass is significantly higher once the damper assembly is included. With the damper assembly, the total mass becomes 547.4 g. The damper assembly contributes 287.7 g, which is 52.6% of the fully mounted assembly mass. This means that more than half of the final mounted system mass comes from the vibration isolation and mounting structure rather than the enclosure, electronics, or thermal hardware. Although this adds significant mass, the damper assembly is necessary because it reduces vibration transmission to the cameras and IMUs, which directly affects image quality and navigation accuracy.

The current VIPER mounting plates were designed specifically for the selected drone platform. Because of this, the mounted assembly includes geometry needed to interface with this drone while also supporting the vibration isolation system. For future versions, the mounting plate geometry could be adapted or redesigned for other drone interfaces. Depending on the mounting requirements of each platform, this could provide additional opportunities to reduce mass while improving modularity across different drone systems.

The mass distribution changes significantly depending on whether the damper assembly is included. Without the damper assembly, the housing bottom, heat sink, housing top, sensor

board, cabling, and fasteners are the dominant contributors. With the damper assembly included, the relative percentage of every other component decreases because the damper assembly dominates the total system mass. For example, the housing bottom decreases from 27.5% of the internal payload mass to 13.1% of the fully mounted mass, while the heat sink decreases from 18.4% to 8.7%. This shows that the payload enclosure and internal hardware were kept relatively lightweight, while the mounting and vibration isolation system became the largest contributor to the final integrated assembly mass.

The housing remains one of the clearest areas for future mass optimization. The housing top and bottom together contribute 103.9 g, which is 40.0% of the enclosure and internal payload mass. Because the housing has now been developed into a more complete product-like form, future refinements can focus on removing unnecessary material while preserving stiffness, sealing performance, durability, and manufacturability. Potential improvements include localized wall thickness reduction, refined internal supports, lighter mounting bosses, and more efficient corner and fillet geometry.

Cabling and fasteners are also important secondary contributors. Together, cabling and fasteners add 38.3 g, or 14.7% of the internal payload mass. While these components are smaller than the housing and heat sink, they are still large enough to matter in a lightweight drone payload. Future improvements could include shorter cable runs, simplified routing, reduced excess connector length, and lighter fasteners where appropriate. These changes would need to be balanced against strength, reliability, grounding, corrosion resistance, and ease of assembly.

Overall, the updated weight analysis shows that the VIPER payload remains impressively lightweight for the level of integration included in the current design. The enclosure and internal electronics assembly has a final mass of 259.7 g, which is only slightly above the 250 g target while incorporating a more refined final product geometry and added functional interfaces. The heat sink optimization was a key contributor to this result, reducing the heat sink mass by 23.2 g compared to the earlier 70.9 g concept. The fully mounted assembly mass is 547.4 g and is dominated by the damper assembly, which accounts for 52.6% of the total. This added mass is partly a result of designing the mounting plates around the selected drone platform and vibration isolation requirements. Moving forward, the largest opportunities for additional weight savings are the damper assembly and housing structure, especially if the mounting interface is adapted for other drone platforms. Smaller improvements may also be possible through cabling and fastener optimization.

V. Risk Mitigation

Table 11 outlines the risks of the project and their respective severity, probability, and priority.

Risk	Severity	Probability	Multiplied	Priority	Prevention	Failsafe
Software does not properly fuse camera and sensor data	5	5	25	1	Software buffers check for data fusion	PCB has USB input to upload updated versions of software to microcontrollers
Water seeps into the enclosure	5	4	20	2	System has ingress sealing; check for gaps before flying	Dry with silica packets
Electronics overheat	5	4	20	2	Heat sink cools system	Heat sink is removable and replaceable with a new heat sink design
Drone vibrations cause noisy sensor readings	5	4	20	2	Damper mount dampens high vibrations	Software filtering during data pre-processing
Electrostatic discharge buildup in the enclosure	5	2	10	3	Camera mounts are grounded	Add TVS diodes to 2nd PCB iteration if time permits

Sensor mounts break	4	2	8	4	Rigid fasteners, PEM inserts, and aluminum standoffs ensure system is secured	Replace with snaps in case of fastener issues/use stronger material
Insufficient power flows to internal sensors	3	2	6	5	Followed PCB power layout best practices.PCB test point check will indicate insufficient power; system has indication LEDs for low power warnings pre-flight	Use a breakout board instead of the non-functional sensor. PCB redesign if issue is diagnosable and time permits.
EMI from the drone or crosstalk between PCB traces	3	2	6	5	Followed signal integrity best practices. PCB test points will reveal issues during testing	Use a breakout board instead of the non-functional sensor. PCB redesign if issue is diagnosable and time permits.
Dust seeps into the enclosure	2	3	6	5	Ingress protection through silicone covers	Add a layer of foam behind cover
Enclosure is unable to handle landing shock	5	1	5	6	Damper system dampens landing shock; gently land drone	Replace vibration absorption dampers with harder mounts

Payload mount breaks	2	2	4	7	Tighten fasteners before flight to ensure system will not break	Replace with stronger fasteners
-----------------------------	---	---	---	---	---	---------------------------------

Table 23. Risk Analysis Chart

The project's risk profile can be broken into three (3) distinct categories: Structural, Technical, and External.

Structural risks originate from mechanical forces and physical stressors that may degrade performance or cause system failure. These risks include landing shock or mount failure, as well as vibration-induced distortion that can produce noisy sensor data readings. Continuous exposure to resonance and high-frequency oscillations can further propagate mechanical fatigue and introduce measurement inaccuracies. To mitigate these risks, a custom insulator system will be designed using dampening elements tuned to the drone's characteristic resonant frequencies. Additionally, high-impact materials and reinforced structural geometry will be incorporated into the enclosure design, with performance verified through Finite Element Analysis (FEA) to ensure survivability under shock loading conditions.

Technical risks refer to the failures due to the internal performance of the payload, namely overheating of the sensors, EMI from the drone or crosstalk between sensors, power distribution instability, and timing inconsistencies in the computational fusion of the sensor data. Thermal risks will be mitigated through active thermal interface control consisting of high-conductivity pads or thermal paste. Layout optimization of the PCB will mitigate EMI crosstalk, resulting in stronger isolation between signal pathways and improved sensing accuracy. Power distribution stability will be ensured through the use of blocking capacitors, buck converters or voltage regulators, and layout optimization to maintain clean and consistent power supply rails. Timing discrepancies between sensor outputs will be reduced using software-level synchronization using hardware triggers and interrupts, enhancing data fusion reliability.

External risks refer to environmental factors such as water ingress, dust contamination, electrostatic discharge (ESD), and external EMI from the drone and GPS module. Environmental ingress will be prevented through sealed interface utilizing gaskets rated for moisture and particulate exclusion. To counteract external EMI and suppress ESD, the enclosure will be coated with a conductive nickel or silver layer [44]. This way, the payload will have a continuous shielding barrier that functions as a Faraday cage, minimizing external signal coupling and protecting sensitive internal components.

VI. Testing

6.1 - Thermal Test

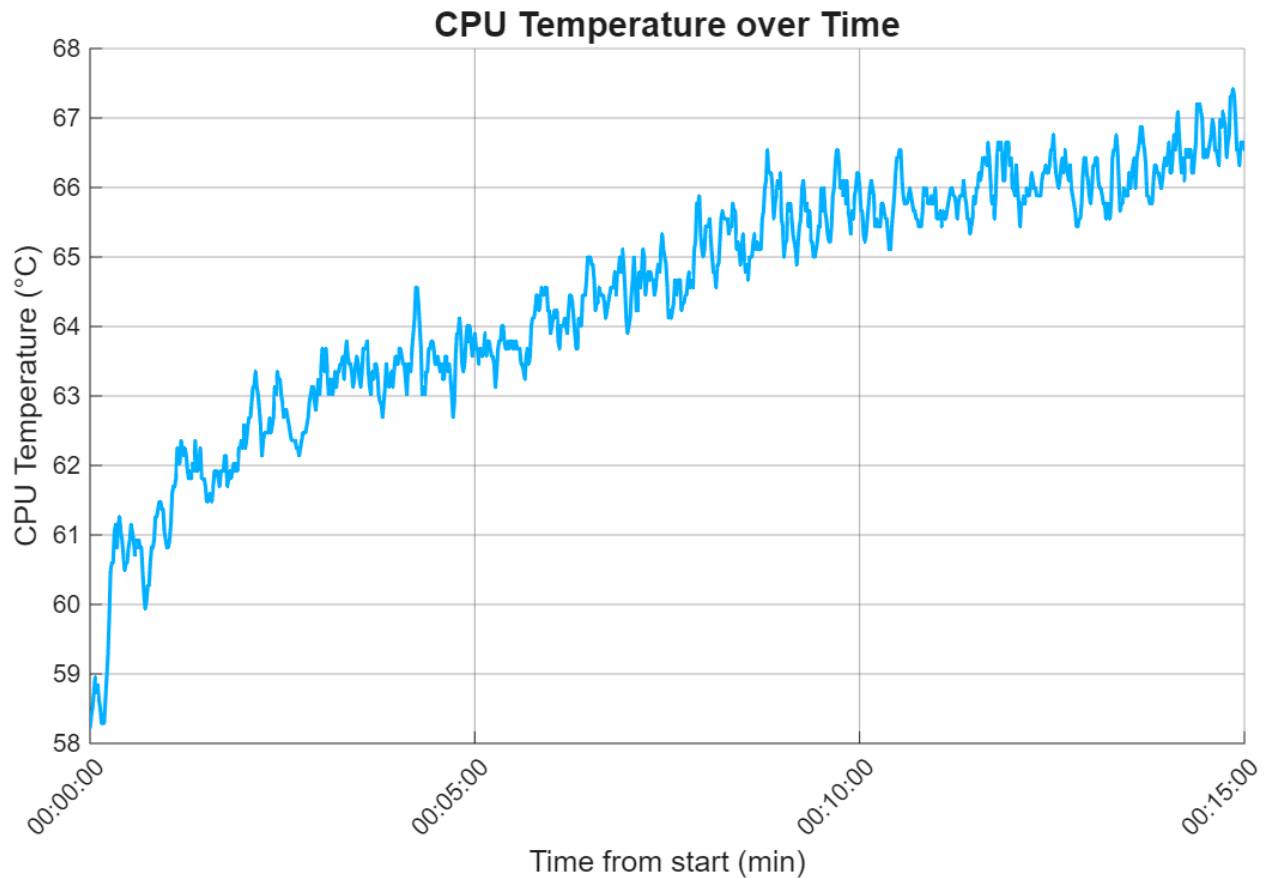


Figure 44. CPU Temperature of stationary 15 minute test (no forced convection)

Measuring CPU temperature was the main way of testing the effectiveness of the custom heat sink. This first test was done to test the worst case scenario for the payload: full load while remaining stationary, so no forced convection would occur. Although this test did not start from ambient conditions, it still shows that the temperature of the system only rose by 10 C over the course of 15 minutes. Additionally, the maximum temperature remained below 70 C, which is well within the operational limits of the Raspberry Pi CM5.

In actual use, the payload is not expected to remain stationary for that amount of time while outputting its maximum power. Therefore, the heat sink has accomplished its job.

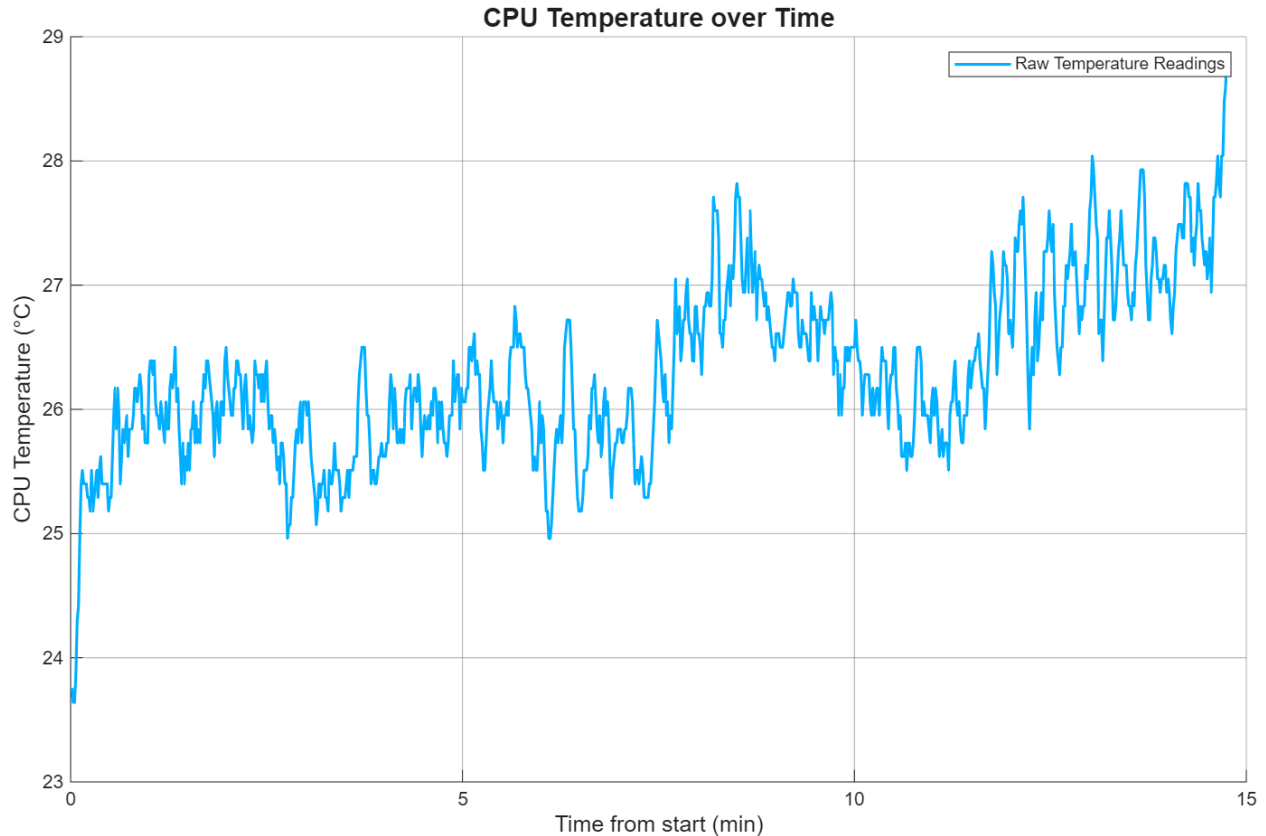


Figure 45. CPU Temperature of 15 minute full flight test (varying speed)

Similar to the stationary test, this test flight lasted about 15 minutes, but now the system has active airflow over the fins. This active airflow and the subsequent forced convection maintained significantly lower temperatures, as compared to the stationary test. In an ambient atmosphere of $\sim 24^{\circ}\text{C}$, the maximum temperature of the CPU only peaked at $\sim 29^{\circ}\text{C}$, which is a 5°C difference.

Extrapolating this out, the heat sink will properly cool the Raspberry PI CM5 in an ambient temperature environment exceeding 70°C , far surpassing the 40°C operating temperature goal of this project.

6.2 - Flight Test

The flight test was the main way that we tested the functionality of our system. The flight procedure went as such. We started by attaching the system to the drone and ensuring that none of the wires would get caught in the propellers. Once fully attached, plug in the battery or toggle the switch to turn on the system power. We then wait for the GPS lock which will be indicated with a blue LED turning on. Once the GPS is activated, take off and fly a predetermined route.

During this stage, we had to keep the drone in wide paths and at slow speeds due to the limitations of the system. While flying be sure that the blue LED does not turn off as that will mean that the system is no longer capturing data. Once enough data has been collected, land the drone in an open area far away from people. Once the drone lands wait for the GPS to shut off which will be indicated by the blue LED turning off and unplug the system. Once this is done, we have now saved a dataset which is then ready to be post processed by the spectacular AI SLAM algorithm. A visual of the data collection procedure is shown in figure X

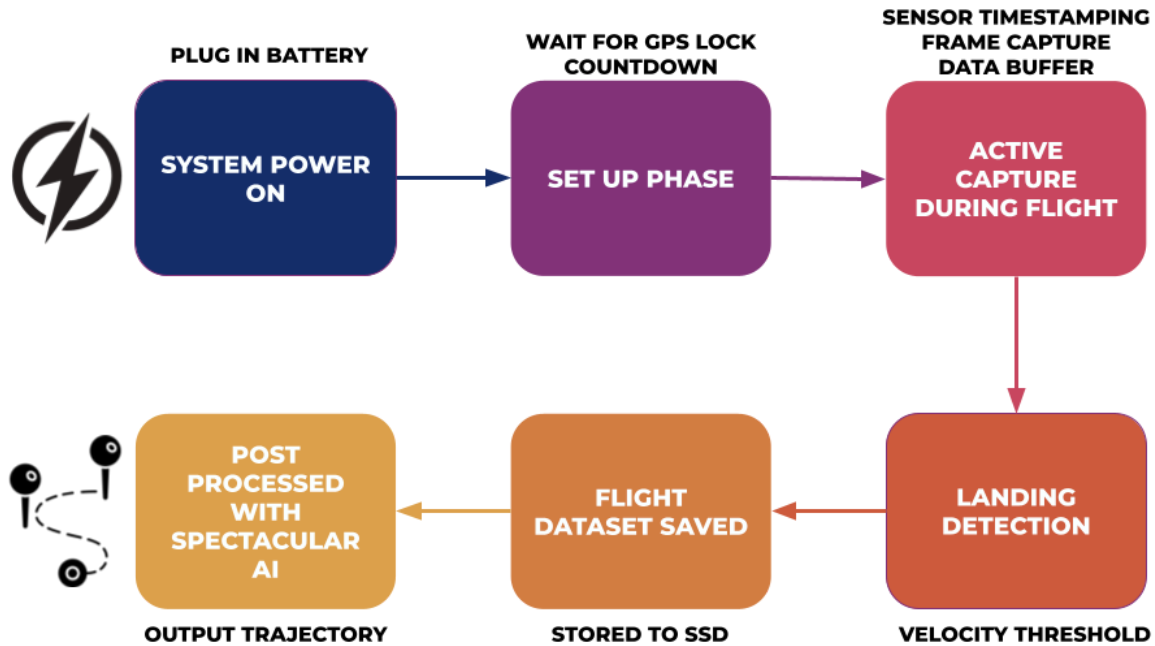


Figure 46. Flight Test Data Collection Operational Flowchart

6.3 - VIO Results and Metrics

Each flight test provides a dataset corresponding to the timestamped and organized sensor data and camera image frames into a single folder. Flight data is stored in the structure described from [Section: Dataset Formats](#). Before feeding the data into the Spectacular AI SDK, we carry out some data reorganization into the SDK's desired JSONL format as documented at github.com/SpectacularAI. Once formatted, we used the *Replay API*, allowing us to simulate real-time SDK behavior with our previously recorded flight datasets. This gives us a good estimate on how the position estimation algorithm will perform in real time, as well as the quality of our collected data and capture system hardware.

The following figure is a 3d visualization of the algorithm's estimated position during a test flight, plotted alongside the path recorded by our onboard GPS, acting as a ground truth.

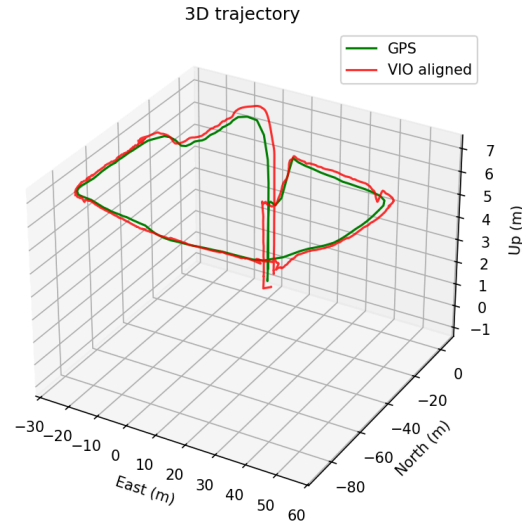


Figure 47. Estimated VIO trajectory comparison to Recorded GPS (RMSE 3.6m)

As we can see, the estimated position, labeled as *VIO aligned*, closely follows the route traced by GPS. It is important to note that the pure VIO output is an estimate of x, y, and z coordinates in meters, with the beginning of the flight dataset set at 0,0,0 meters. So there was a rotation transform applied to the VIO output to achieve the same orientation as the GPS recording, hence the label *VIO aligned*. Once this alignment rotation is applied, we compute quantitative error metrics associated with how close the position estimate was to our ground truth, which we are taking as the GPS readings. For example, the average deviation away from the location recorded by the GPS on this ~275m flight was 3.6m, roughly one percent of the total flight distance.

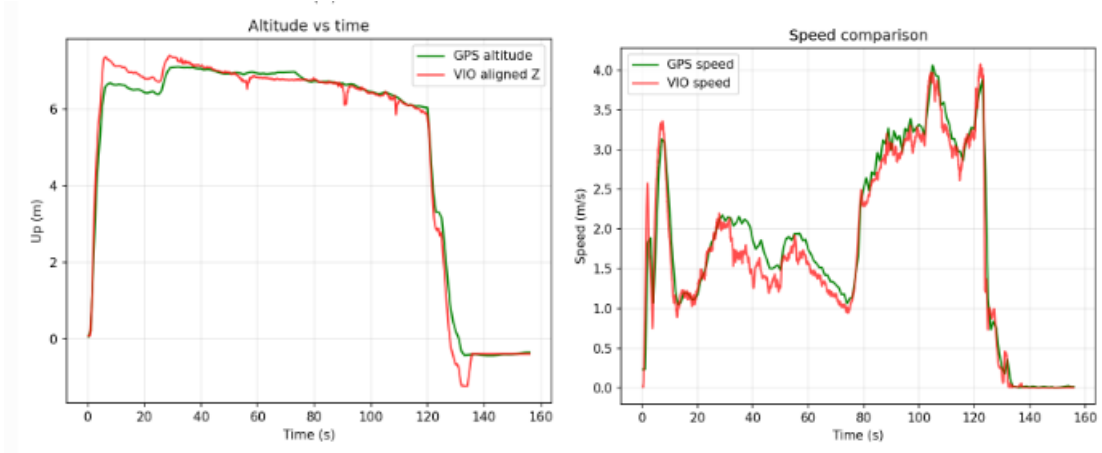


Figure 49. More flight data comparisons and error metrics

The plots above make more comparisons between the path traced by the GPS coordinates. The plot on the left compares the recorded and estimated altitude throughout the test flight. Visually, the estimated altitude tracks well, with an average error from the GPS recordings of 0.32 m. This error is much less than that the absolute path error most likely due to the barometric altimeter sensor whose data is a direct input in the VIO algorithm. The plot on the left compares the measured speed throughout the flight, showcasing that the position estimation algorithm also matches the speed profile throughout the flight well. Quantitatively, the speed recorded by GPS and speed from VIO output had a correlation of 0.982, very close to an ideal 1.

The last figure of merit for our VIO results is the percent of tracking uptime. While in a real time flight, the SDK may “lose tracking” during a flight, causing it to not produce any position estimates until it regains tracking. These events are caused by significant camera motion blur or erroneous sensor readings. For jittery and improperly calibrated systems and unreliable datasets, the algorithm may lose tracking for up to a few minutes. The Replay API used to analyze our data also indicates when the SDK would lose tracking if it was estimating in real time. For the same flight dataset we have been previously discussing, we had a tracking uptime of 99.98%, indicating a robust data collection throughout the flight, with the only tracking loss occurring on takeoff, where the images were blurry.

VII. Task Distribution

Based on the five subteams of Housing Design, Vibrations, Sensors, Software, and Power, our ten-person team breaks down into the following organizational chart, as shown in Figure 43.

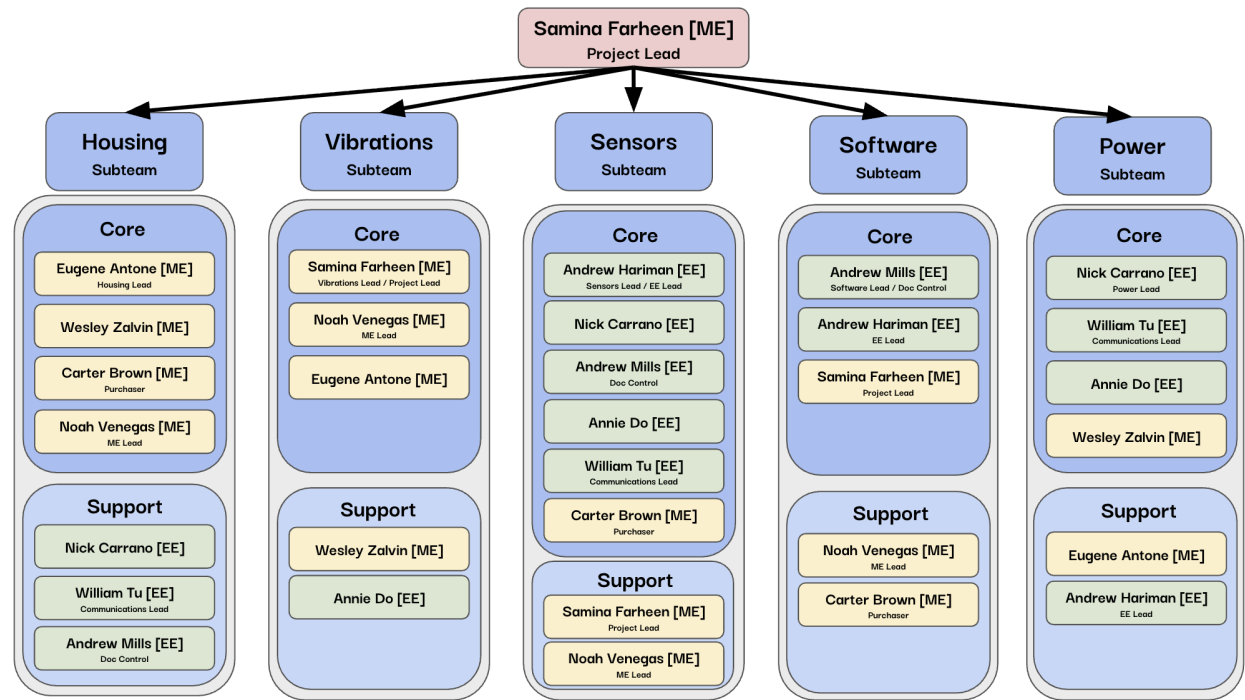


Figure 50. Organizational Chart

Housing Design is responsible for creating the enclosure for the system and the mounting structure. The Vibrations team is responsible for ensuring the vibrations experienced by the system are mitigated so that sensors can report accurate data. The Systems team is responsible for integrating the cameras and sensors with the Raspberry Pi Compute Module, which will involve PCB design. The Software team is responsible for ensuring our sensors can communicate with FLIR's map matching algorithm ensuring any necessary drivers work seamlessly. The Power team is responsible for ensuring all devices have adequate power distribution and interfacing with the battery, also a part of the PCB design.

The "Core" members do the work for the team while the "Support" members provide additional input. Within each subteam, there is a subteam lead indicated on the chart. The Electrical Lead (Andrew Hariman) and Mechanical Lead (Noah Venegas) are tasked with ensuring the subteams that they are on are successful. All teams are overseen by Samina Farheen, the Project Lead.

The final project timeline is pictured below:

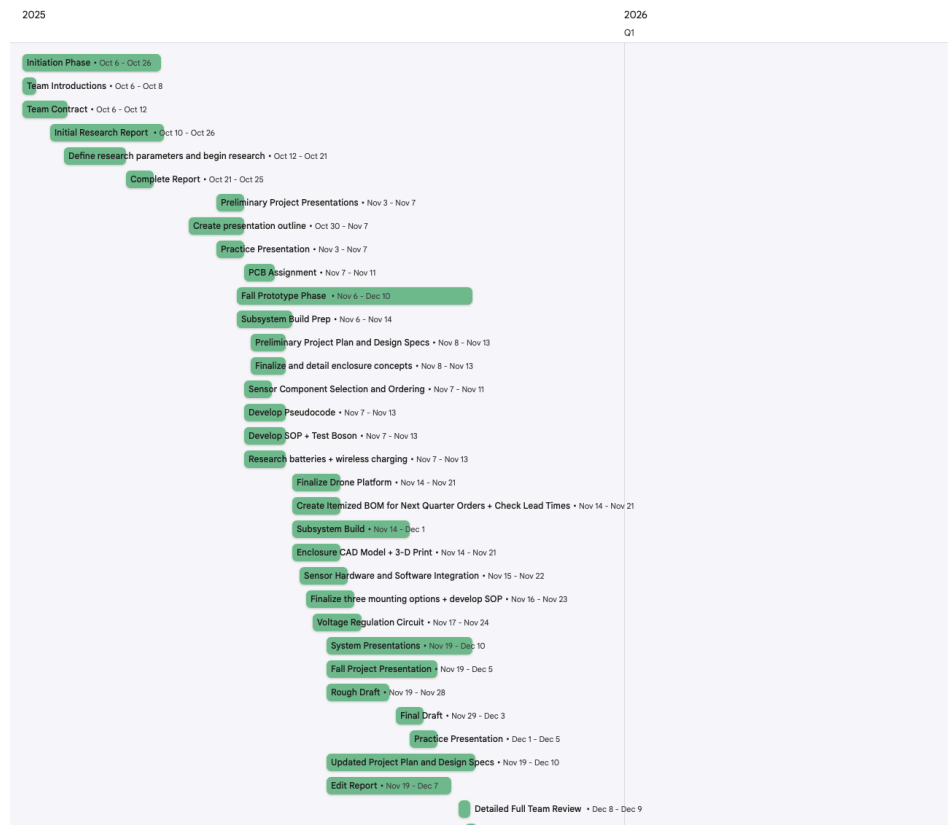


Figure 51. Gantt Chart Part 1

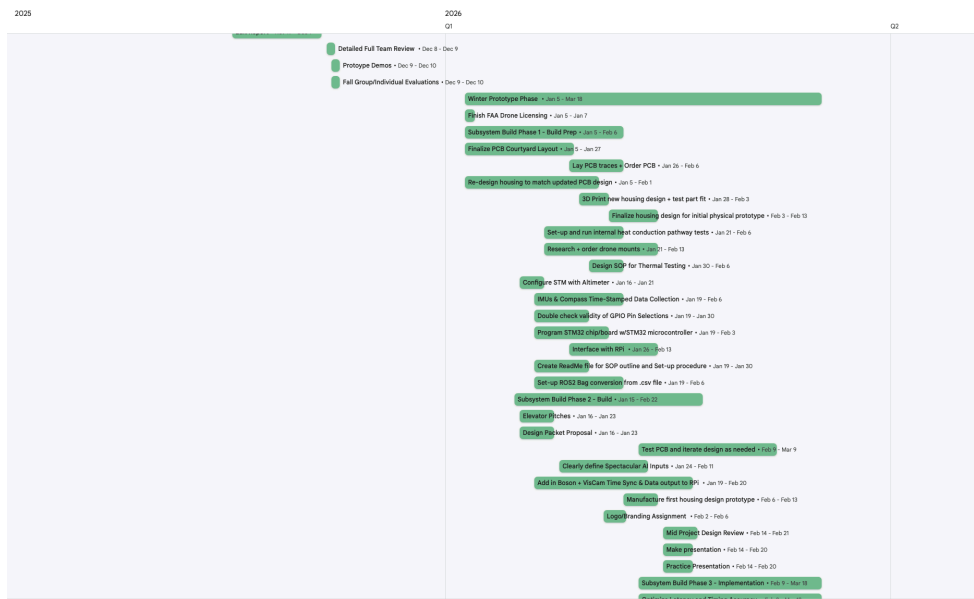


Figure 52. Gantt Chart Part 2

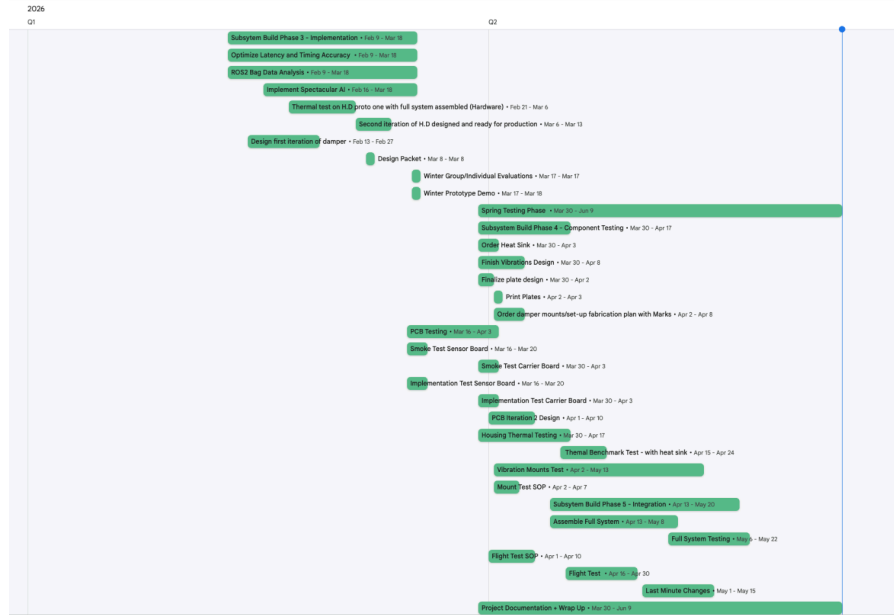


Figure 53. Gantt Chart Part 3

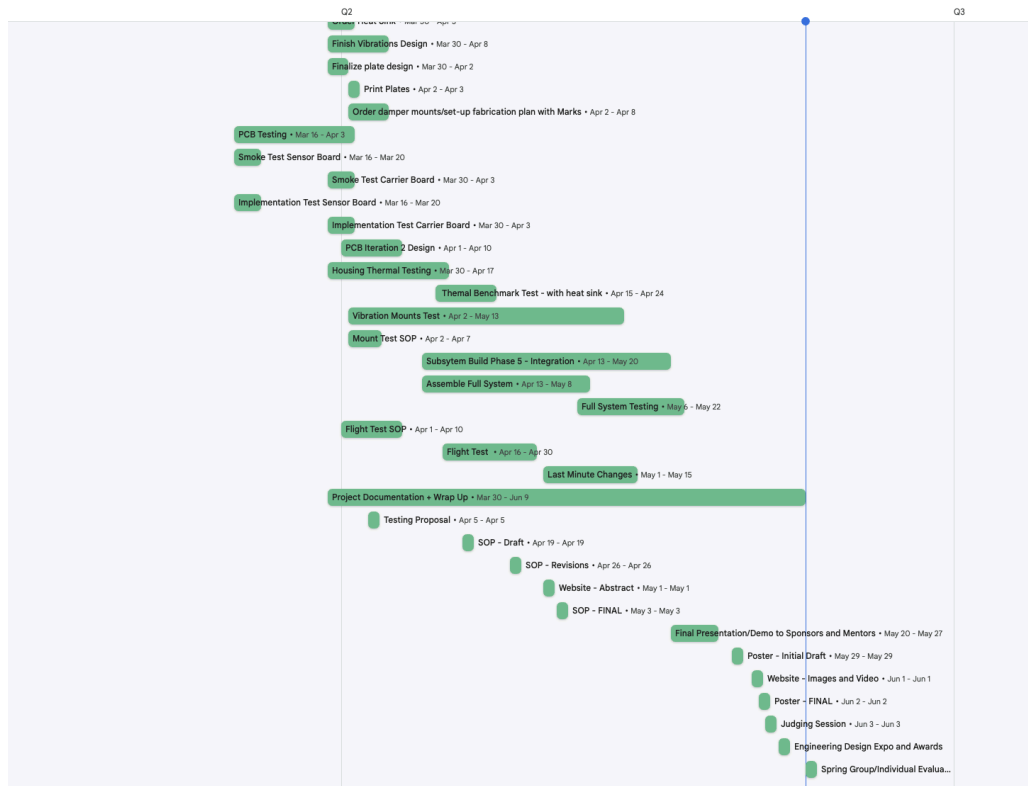


Figure 54. Gantt Chart Part 4

VIII. Cost

8.1 - R&D Expenditure

The total budget for the project is \$10,000. As of the spring quarter close-out, approximately \$8,250 of that budget has been expended, representing 82.5% of the total allocation.

Expenditures span the full project lifecycle, from initial sensor prototyping and PCB development through final assembly hardware, custom manufactured components, and iteration 2 PCB orders.

Category	Item	Cost
Compute & Processing	Raspberry Pi modules/boards	\$381.44
	IO Boards	\$59.43
	MCU Dev Board	\$29.72
Sensors & Navigation	GPS System	\$401.37
	IMUs	\$406.18
	Altimeter Breakout	\$19.72
	Compass System	\$412.30
Imaging System	Global Shutter Cameras	\$120.88
	Camera Cables	\$65.79
Power System	Batteries + Charger	\$89.69
	Battery Connectors	\$33.97
	Power Supplies	\$21.56
	Buck Converters	\$35.64
	Wiring Kit	\$20.99
	XT60 Extension Cable	\$19.38
Thermal Management	Custom Heat Sinks (Xometry)	\$301.21
	Active/Off-Shelf Heat Sinks	\$22.80
	Thermal Paste	\$31.59
Housing & Mechanical	Aluminum Stock	\$236.56
	Vibration Damper Hardware	\$103.24
	Fasteners & Hardware	\$193.59

	Custom Damper Plates (SendCutSend)	\$259.69
PCB & Electronics Assembly	Custom Sensor PCB — Iteration 1 (JLCPCB)	\$1,429.46
	Custom Carrier PCB — Iteration 1 (JLCPCB)	\$746.45
	Custom Carrier PCB — Iteration 2 (JLCPCB)	\$743.78
	Custom Sensor PCB — Iteration 2 (JLCPCB)	\$1,341.66
	Connectors & Push Buttons	\$73.16
	Standoffs & Crimping Hardware	\$86.95
	STM32H7 & Supporting Components	\$102.25
Storage & Accessories	1 TB SSD	\$135.75
	512 GB SSDs (×2)	\$216.24
	128 GB Flash Drive	\$12.88
	Scissors & Calipers	\$27.45
Total		\$8,249.79 / \$10,000

Table 24. R&D expenditure

8.2 - Production BOM

The production BOM lays out the cost of each production unit when manufactured in runs of 100 units. The target spec for each unit is \$1,000. The largest line item is the Boson thermal camera which will be purchased from FLIR at a reduced price of \$350.

Category	Item	Line Total
Compute & Control	Raspberry Pi CM5	\$195.00
	Custom PCB (Assembled)	\$40.00
	MCU (Integrated)	\$6.00
Navigation Sensors	IMU (Murata)	\$44.07
	Compass (RM3100)	\$25.00
	Altimeter	\$2.50
	GPS Module	\$17.00
Imaging System	Global Shutter Camera	\$50.00

	Boson Thermal Camera	\$350.00
	Camera Cable	\$2.00
Power System	3S LiPo Battery	\$15.00
	Wiring Harness	\$4.00
	Power Regulation	\$2.00
Thermal Management	Heat Sink	\$10.00
	Thermal Interface Material	\$2.00
Mechanical & Housing	Injection Molded Enclosure	\$2.00
	Fasteners / Seals	\$5.00
Storage & I/O	1 TB SSD	\$80.00
	External Connectors	\$5.00
Total		\$856.57

Table 25. BOM Expenditure for VIPER

IX. Conclusion

The goal of this project was to successfully develop a GPS denied drone payload capable of using visual and inertial navigation to output a trajectory within some degree of accuracy. Additional requirements were to create a small, lightweight, and completely modular system with a bill of materials cost under \$1,000. VIPER is able to match these metrics, surpassing the accuracy goal of $\pm 10\text{m}$ with an accuracy of $\pm 4\text{m}$.

A major contributor to the success of this platform is the custom two board PCB stackup design, which drastically downsized the system's physical footprint.. Through this project the team learned how to design complex custom PCBs, heat sinks, vibration dampers, and design for manufacturing of injection molded parts. Most significantly, the VIPER project taught the team the intricacies of multidisciplinary design, and how intertwined electronic and mechanical design components must be in order to produce a system that integrates and functions well. There is no component of this project that did not require input from both mechanical and electrical teams for its engineering.

To improve the current design of VIPER, the team would have liked more time and resources to properly test the mechanical components of the project such as the heat sink and vibration damper through more rigorous, industry standard testing. Ultimately the team ended up putting the system together and using the success of the flight tests (whether the data collected would provide VIO results) as the metric for measuring the success of the mechanical components, but they require more detailed testing to determine their effectiveness individually. Second, the accuracy metrics achieved by VIPER's VIO system were conducted on slow flights that moved in a circular trajectory. By improving these components, VIPER may be able to function in more aggressive flight maneuvers. Additionally, data is currently being processed on a separate computer, not the VIPER system. Next steps would include integrating the VIO software directly with VIPER for real time tracking.

Ultimately, VIPER is meant to function as a modular payload that autonomously guides drones in GPS denied situations. In addition to the improvements mentioned in the previous paragraph, the next major milestone of this project would be to autonomously navigate drones with VIPER's payload.

What began as a scattered collection of sensors and wires has been successfully transformed into a cohesive, modular VIO payload capable of high-accuracy trajectory estimation. Beyond meeting the technical metrics, the true success of this project lies in the multidisciplinary engineering growth of the team. While there are many milestones still ahead for the platform, the 2026 VIPER capstone project concludes as a fully realized proof-of-concept, ready for the next generation of autonomous flight development.

X. Works Cited

- [1] “Navigate Without Limits: Understanding GPS-Denied Drones.” Accessed: Oct. 25, 2025. [Online]. Available: <https://www.flyability.com/blog/gps-denied-drone>
- [2] “Payload SDK.” Accessed: Dec. 10, 2025. [Online]. Available: <https://developer.dji.com/doc/payload-sdk-tutorial/en/model-instruction/payload-develop-criterion.html>
- [3] C. Ge, K. Dunno, M. A. Singh, L. Yuan, and L.-X. Lu, “Development of a Drone’s Vibration, Shock, and Atmospheric Profiles,” *Appl. Sci.*, vol. 11, no. 11, p. 5176, Jan. 2021, doi: 10.3390/app11115176.
- [4] “Just How Weather Resistant is the Matrice 300 RTK?” Accessed: Dec. 10, 2025. [Online]. Available: <https://enterprise-insights.dji.com/blog/matrice-300-weather-resistance>
- [5] “raspberrypi-5-product-brief.pdf.” Accessed: Oct. 25, 2025. [Online]. Available: <https://datasheets.raspberrypi.com/rpi5/raspberrypi-5-product-brief.pdf>
- [6] “flir-boson-datasheet.pdf.” Accessed: Oct. 25, 2025. [Online]. Available: <https://f.hubspotusercontent10.net/hubfs/20335613/flir-boson-datasheet.pdf>
- [7] “Camera Serial Interface 2 (MIPI CSI-2) | MIPI.” Accessed: Nov. 11, 2025. [Online]. Available: <https://www.mipi.org/specifications/csi-2>
- [8] “Pressure Sensor BMP390,” Bosch Sensortec. Accessed: Oct. 25, 2025. [Online]. Available: <https://www.bosch-sensortec.com/products/environmental-sensors/pressure-sensors/bmp390/>
- [9] “Inertial Measurement Unit BMI088 | Bosch Sensortec.” Accessed: Oct. 25, 2025. [Online]. Available: <https://www.bosch-sensortec.com/products/motion-sensors/imus/bmi088/>
- [10] “Spectacular AI SDK documentation — spectacularAI documentation.” Accessed: Dec. 07, 2025. [Online]. Available: <https://spectacularai.github.io/docs/sdk/index.html>
- [11] “MAVLink Basics — Dev documentation.” Accessed: Oct. 26, 2025. [Online]. Available: <https://ardupilot.org/dev/docs/mavlink-basics.html>
- [12] “Material Datasheet.” Accessed: Dec. 10, 2025. [Online]. Available: <https://www.albis.com/en/products/download/doc/en/SI/basf/UltradurB4406G6.pdf>
- [13] “Aluminium Alloy 6063 Data Sheet.” Accessed: Oct. 24, 2025. [Online]. Available: <https://atlassteels.com.au/wp-content/uploads/2021/08/Aluminium-Alloy-6063-Data-Sheet-11-08-21.pdf>
- [14] “thyssenkrupp Industrial Solutions AG • Postfach 10 22 51 • 45022 Essen.” Accessed: Oct. 24, 2025. [Online]. Available:

https://ucpcdn.thyssenkrupp.com/_legacy/UCPthyssenkruppBAMXUK/assets.files/material-data-sheets/aluminium/aluminium-6061.pdf

- [15] “Teledyne FLIR BOSON 640 x 512 14mm 32° HFoV – LWIR Thermal Camera Core – OEMCameras.com.” Accessed: Dec. 10, 2025. [Online]. Available: https://www.oemcameras.com/product/flir-boson-640x480-14mm-htm/?gad_source=1&gad_campaignid=765631667&gclid=CjwKCAiAuIDJBhBoEiwAxxhgyFuRqoHSX11ozFYyAQYQo4cjrjlmwyAeh8qB1UtDj1EnFecLHin0T0hoCX7cQAvD_BwE
- [16] “Raspberry Pi NVMe SSD - 1TB,” PiShop.us. Accessed: Dec. 10, 2025. [Online]. Available: <https://www.pishop.us/product/raspberry-pi-nvme-ssd-1tb/>
- [17] “Raspberry Pi Compute Module 5”.
- [18] “Camera - Raspberry Pi Documentation.” Accessed: Dec. 10, 2025. [Online]. Available: <https://www.raspberrypi.com/documentation/accessories/camera.html>
- [19] “adafruit-bmp388-bmp390-bmp3xx.pdf.” Accessed: Dec. 10, 2025. [Online]. Available: <https://cdn-learn.adafruit.com/downloads/pdf/adafruit-bmp388-bmp390-bmp3xx.pdf>
- [20] “adafruit-9-dof-orientation-imu-fusion-breakout-bno085.pdf.” Accessed: Dec. 10, 2025. [Online]. Available: <https://cdn-learn.adafruit.com/downloads/pdf/adafruit-9-dof-orientation-imu-fusion-breakout-bno085.pdf>
- [21] A. Argoudelis, “Comparison of vibration isolators with silicone to anti-vibration products made of natural rubber compounds,” The Blog of Vibration Control Experts | Vibro and Alpha Acoustiki. Accessed: Dec. 10, 2025. [Online]. Available: <https://antivibration-systems.com/vibration-isolators-silicon-vibration-control-rubber/>
- [22] “Key Drone Components: The Invisible Power of Rubber and Silicone,” YINGKO RUBBER ENTERPRISE CO.,LTD. Accessed: Dec. 09, 2025. [Online]. Available: https://yingkorubber.com/en/knowledge_news/key-drone-components-the-invisible-power-of-rubber-and-silicone/
- [23] “Inertial Measurement Unit (IMU) – An Introduction,” Advanced Navigation. Accessed: Dec. 06, 2025. [Online]. Available: <https://www.advancednavigation.com/tech-articles/inertial-measurement-unit-imu-an-introduction/>
- [24] “BMI088 Datasheet.” Accessed: Dec. 09, 2025. [Online]. Available: <https://www.bosch-sensortec.com/media/boschsensortec/downloads/datasheets/bst-bmi088-ds001.pdf>
- [25] “ADIS16500 (Rev. A).” Accessed: Dec. 09, 2025. [Online]. Available: <https://www.analog.com/media/en/technical-documentation/data-sheets/adis16500.pdf>

- [26] “IMX-6_IMU_AHRS_GNSS-INS_Datasheet.” Accessed: Dec. 09, 2025. [Online]. Available:
https://docs.inertialsense.com/datasheets/IMX-6_IMU_AHRS_GNSS-INS_Datasheet.pdf
- [27] “Barometric Formula,” Wikipedia. Accessed: Dec. 07, 2025. [Online]. Available:
https://en.wikipedia.org/wiki/Barometric_formula
- [28] “Pressure Sensor MS5611,” TE Connectivity. Accessed: Oct. 13, 2025. [Online]. Available: <https://www.te.com/en/product-MS561101BA03-50.html>
- [29] “LPS22HB.” ST Microelectronics.
- [30] “Adafruit BMP390 - Precision Barometric Pressure and Altimeter,” Adafruit. Accessed: Oct. 13, 2025. [Online]. Available: <https://www.adafruit.com/product/4816>
- [31] “Amazon.com: innomaker Global Shutter Camera Module with OV9281 Mono Sensor, 309fps High Speed, Native Raspberry Pi OS Driver, Supports Rpi-Cam/Libcamera & External Trigger, for All RPi Models : Electronics.” Accessed: Dec. 10, 2025. [Online]. Available:
<https://www.amazon.com/Raspberry-External-Monochrome-Bullseye-libcamera/dp/B09WTP5GZH?th=1>
- [32] R. P. Ltd, “Buy a Raspberry Pi Global Shutter Camera,” Raspberry Pi. Accessed: Dec. 10, 2025. [Online]. Available:
<https://www.raspberrypi.com/products/raspberry-pi-global-shutter-camera/>
- [33] “IQL-IMX296/FF/Ibis,” DigiKey Electronics. Accessed: Dec. 10, 2025. [Online]. Available:
<https://www.digikey.com/en/products/detail/image-quality-labs-inc/IQL-IMX296-FF-Ibis/27791167>
- [34] “u-blox MAX-M10S Datasheet.” u-blox.
- [35] “Quectel LC76F Datasheet.”
- [36] “STMicro Teseo-LIV3F Datasheet.”
- [37] “u-blox Antenna Reference Page,” Taoglas. [Online]. Available:
<https://www.taoglas.com/taoglas-antenna-reference-guides/ublox-antenna-reference-page/?module=34559>
- [38] “SGPDF254B Datasheet.” Taoglas. Accessed: Dec. 09, 2025. [Online]. Available:
<https://www.taoglas.com/datasheets/SGPDF254B.pdf>
- [39] “Raspberry Pi 5 Compute Module Datasheet.” Raspberry Pi. [Online]. Available:
<https://pip-assets.raspberrypi.com/categories/944-raspberry-pi-compute-module-5/document/s/RP-008180-DS-6-cm5-datasheet.pdf?disposition=inline>

- [40] “NVIDIA Jetson Nano Development Kit.” Accessed: Dec. 09, 2025. [Online]. Available: https://openzeka.com/en/product/nvidia-jetson-nano-developer-kit-4gb/?srsId=AfmBOopW4U60p6Tq4Dqbh6xgxMAroQlrleFX7EQJCYgkw7_kXq7uejjO
- [41] “Nvidia Jetson Nano vs Raspberry Pi - Which one is better for your project?” Accessed: Oct. 26, 2025. [Online]. Available: <https://www.socketxp.com/iot/nvidia-jetson-nano-vs-raspberry-pi-which-one-is-better-for-your-project/>
- [42] “Raspberry Pi - Getting Started.” Raspberry Pi. [Online]. Available: <https://www.raspberrypi.com/documentation/computers/getting-started.html>
- [43] “Raspberry Pi SSD.” Raspberry Pi. Accessed: Dec. 09, 2025. [Online]. Available: <https://www.raspberrypi.com/products/ssd/>
- [44] C. Geer, “EMI Shielding Metals | Advanced Metal Etching®,” <https://metaetching.com/>. Accessed: Dec. 05, 2025. [Online]. Available: <https://metaetching.com/metal-selection-for-emi-shielding/>
- [45] “Methods of Circuit Protection,” Altium. Accessed: Dec. 07, 2025. [Online]. Available: <https://resources.altium.com/p/methods-protect-your-circuit>
- [46] “Simple Solutions for Reverse Polarity Protection,” Cadence PCB. Accessed: Dec. 07, 2025. [Online]. Available: <https://resources.pcb.cadence.com/blog/simple-solutions-for-reverse-polarity-protection>
- [47] “TPS54331.” Texas Instruments, Oct. 2023. Accessed: Dec. 07, 2025. [Online]. Available: https://www.ti.com/lit/ds/symlink/tps54331.pdf?HQS=dis-dk-null-digikeymode-dsf-pf-null-ww&ts=1765075215521&ref_url=https%253A%252F%252Fwww.ti.com%252Fgeneral%252Fdocs%252Fsuppproductinfo.tsp%253FdistId%253D10%2526gotoUrl%253Dhttps%253A%252F%252Fwww.ti.com%252Flit%252Fgpn%252Ftps54331
- [48] U. Waseem, “Trace Width vs. Current in PCB Design,” Wevolver. Accessed: Dec. 07, 2025. [Online]. Available: <https://www.wevolver.com/article/trace-width-vs-current-in-pcb-design>
- [49] “How Thermal Vias Enhance Heat Dissipation,” Sierra Circuits. Accessed: Dec. 07, 2025. [Online]. Available: <https://www.protoexpress.com/kb/how-thermal-vias-enhance-heat-dissipation-in-pcbs/>
- [50] C.-H. Kang, H.-S. Park, S.-W. Seo, and D.-G. Kwag, “Design and Experiment of a Passive Vibration Isolator for Small Unmanned Aerial Vehicles,” *Appl. Sci.*, vol. 14, no. 10, p. 4113, Jan. 2024, doi: 10.3390/app14104113.
- [51] M. minutes read, “How to Estimate Injection Molding Cost?,” Formlabs. Accessed: Dec. 10, 2025. [Online]. Available:

https://formlabs.com/blog/injection-molding-cost/?srsltid=AfmBOopyh_oedFkCAEIU-ayJTjU_9otA27wQNJeQ1Ik1h8k5MGAScBia

11.1 - Carrier Board PCB Schematics

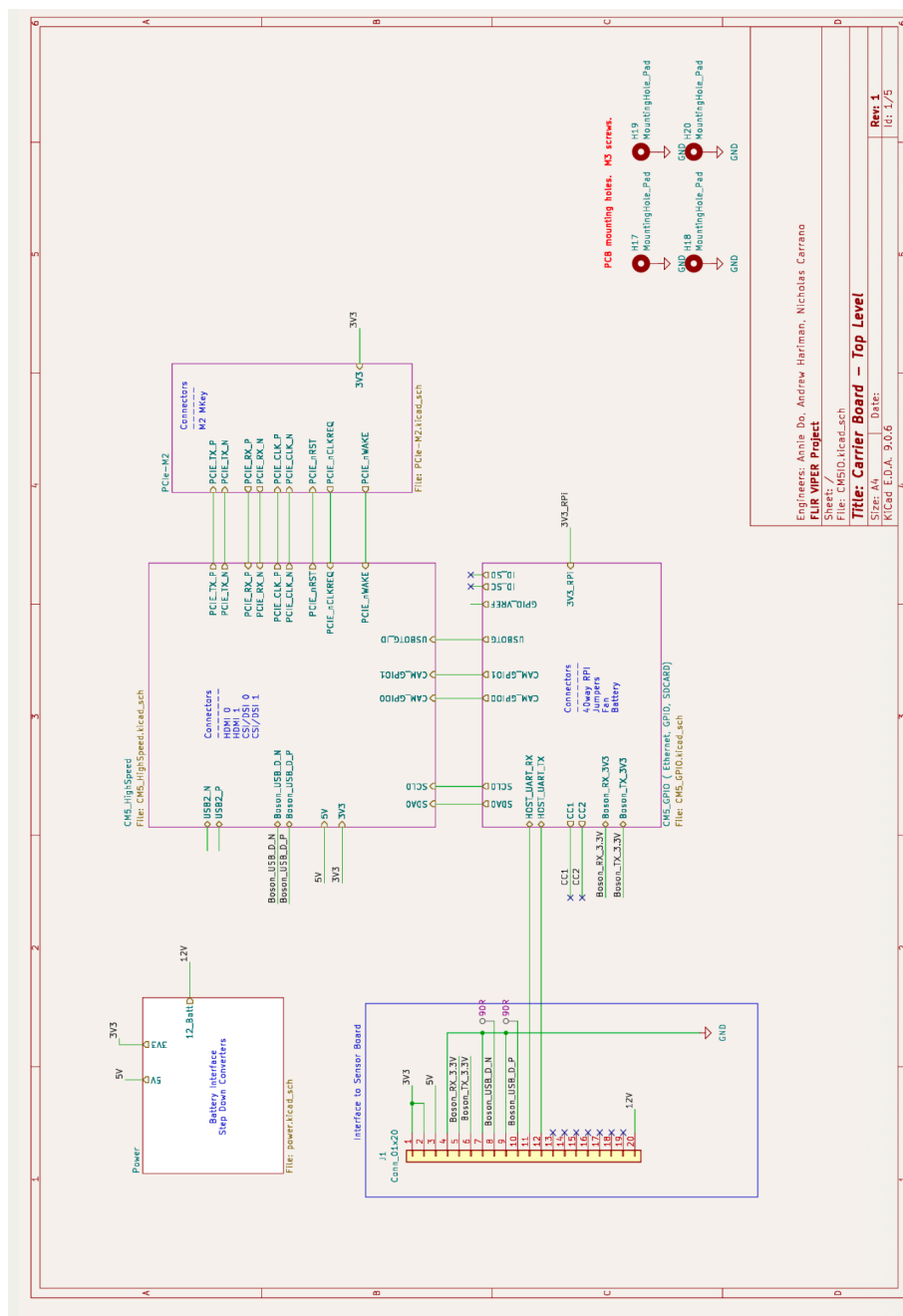
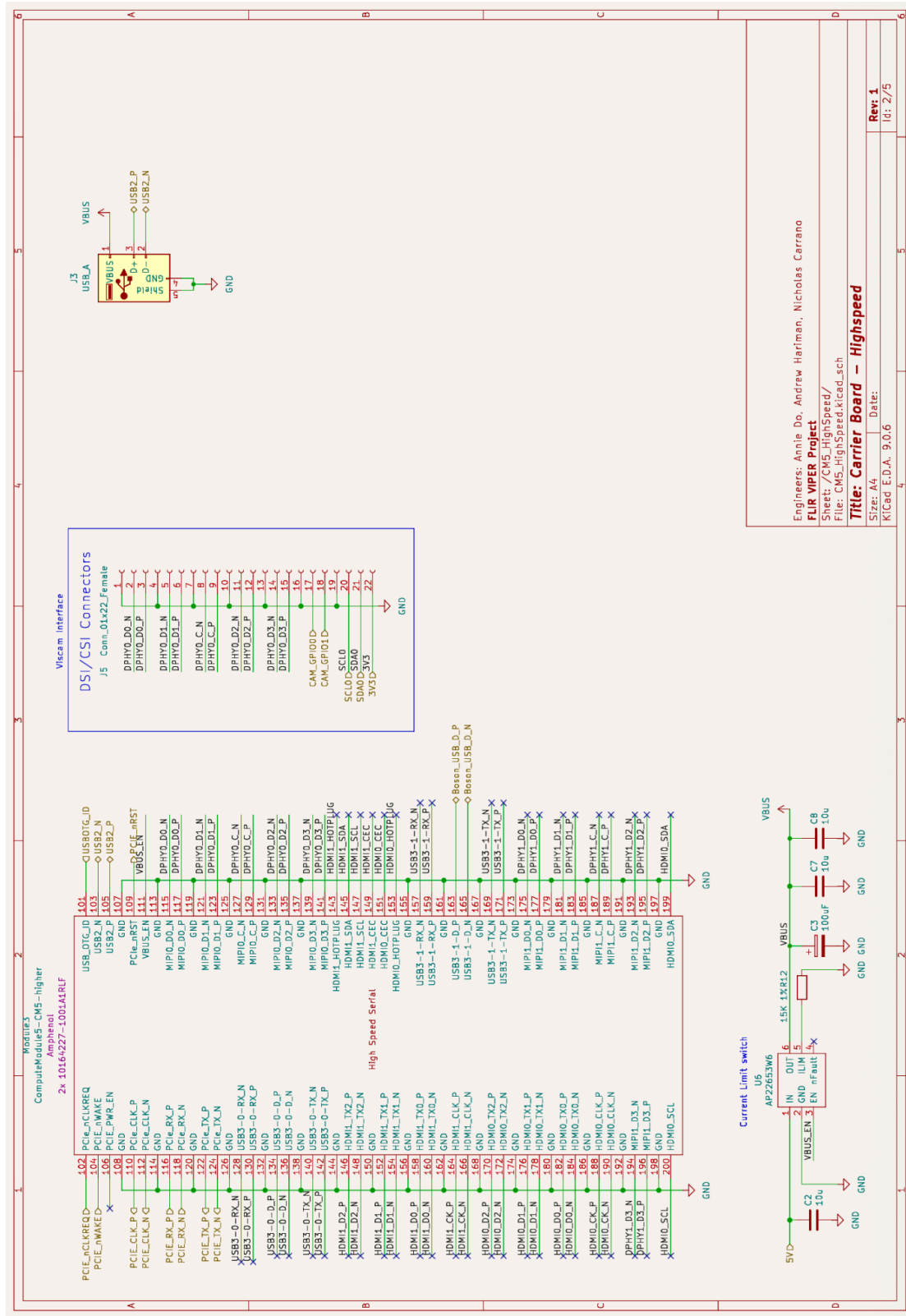


Figure 55.



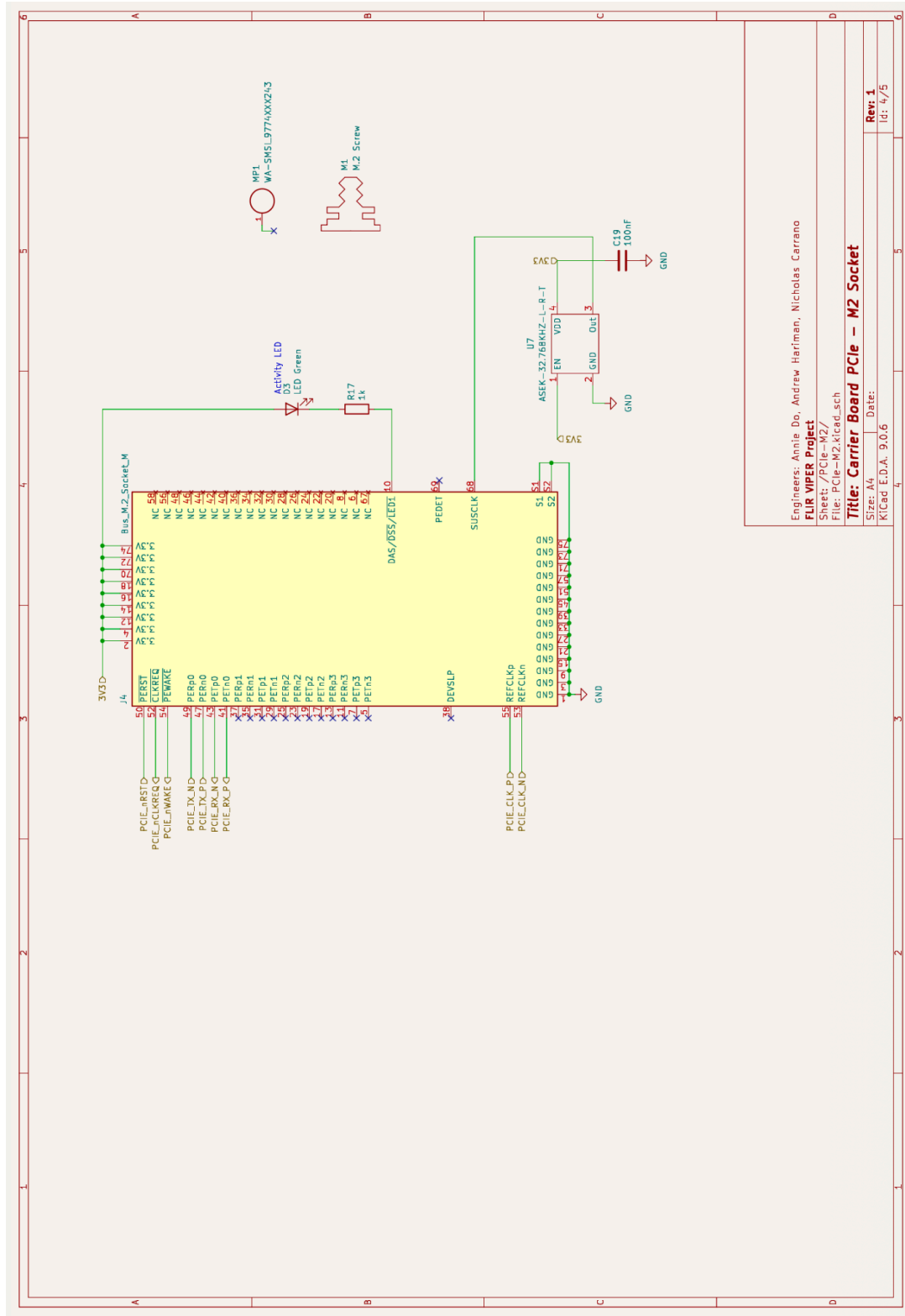


Figure 58.

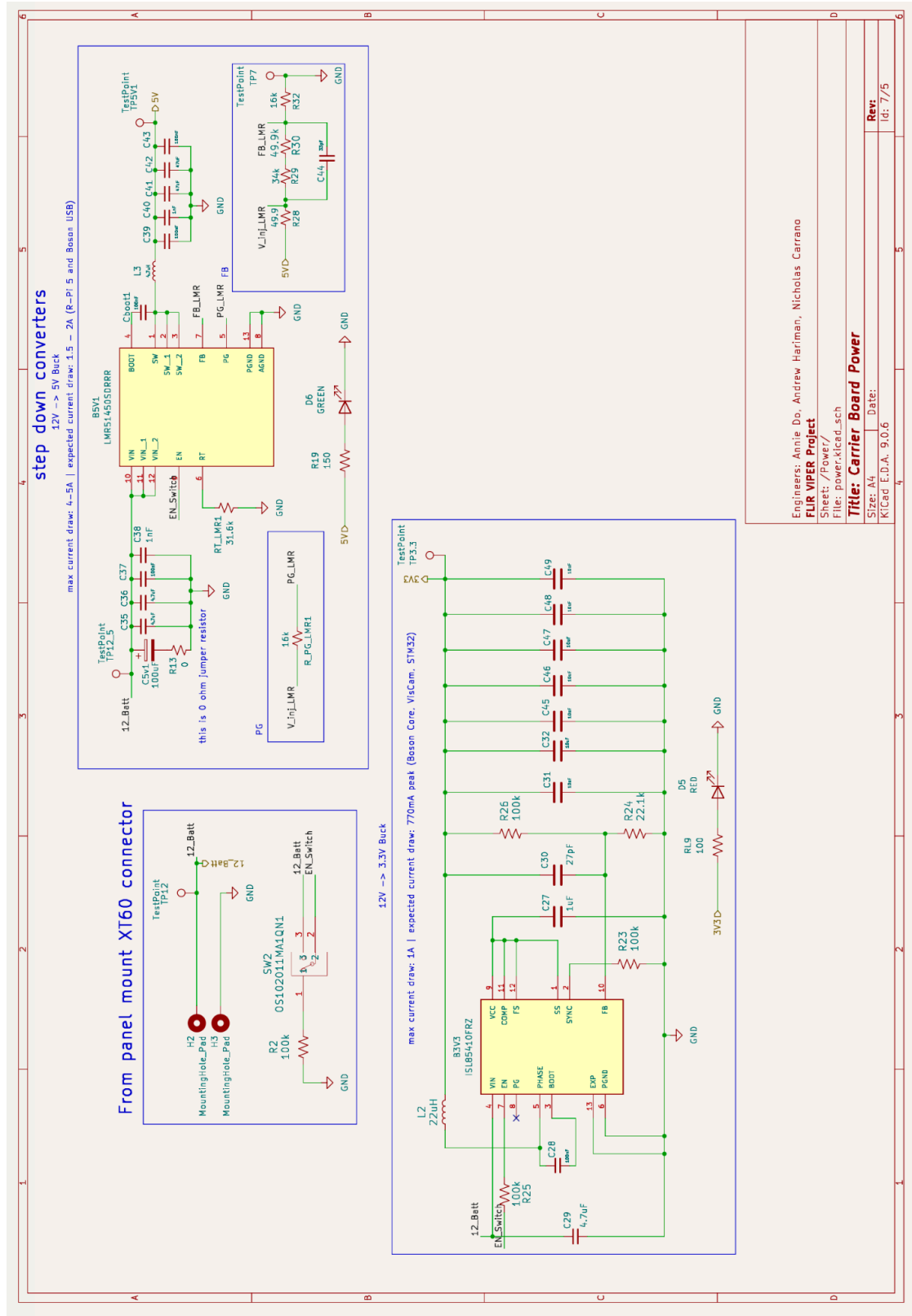


Figure 59.

11.2 - Sensor Board PCB Schematics

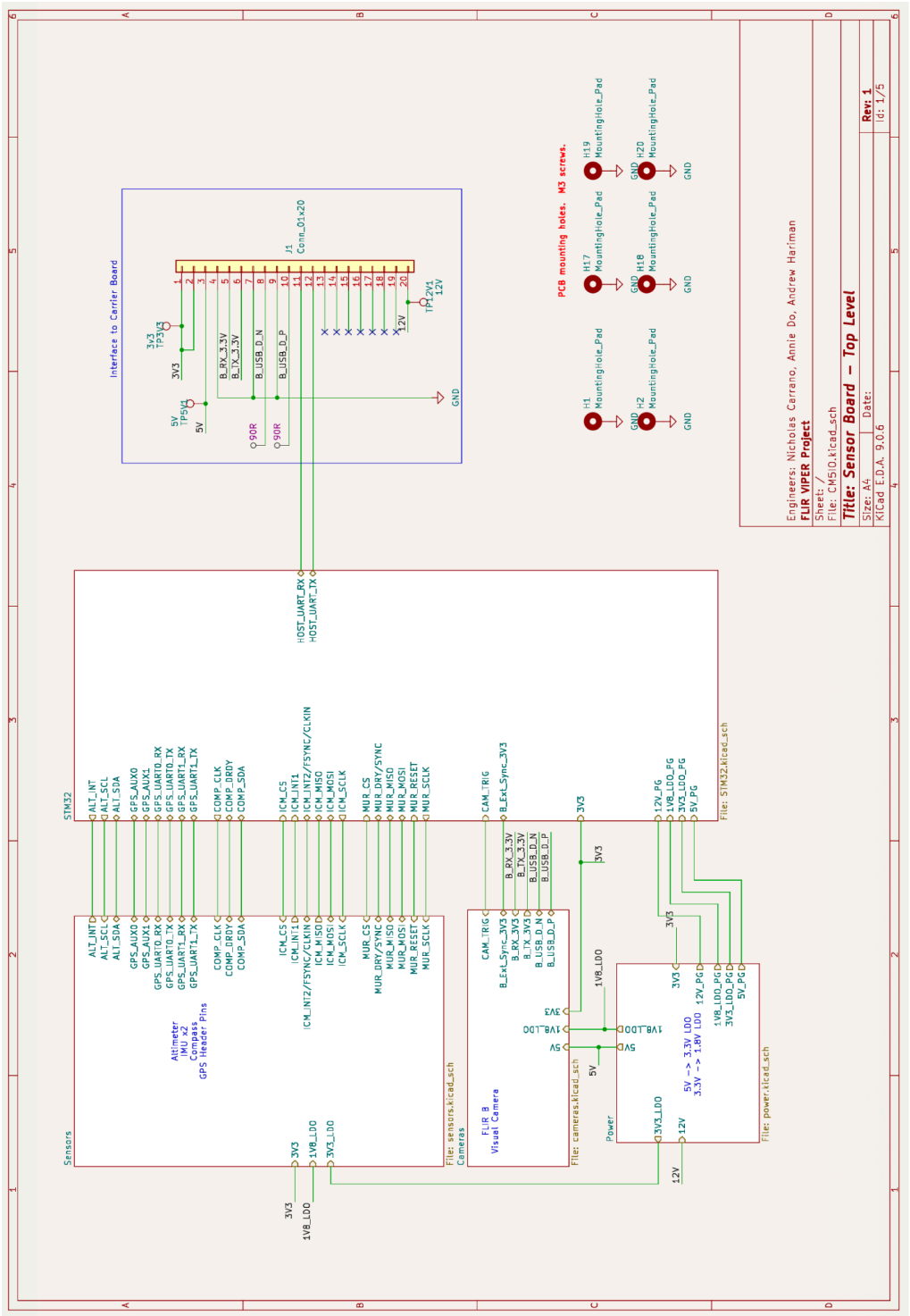


Figure 60.

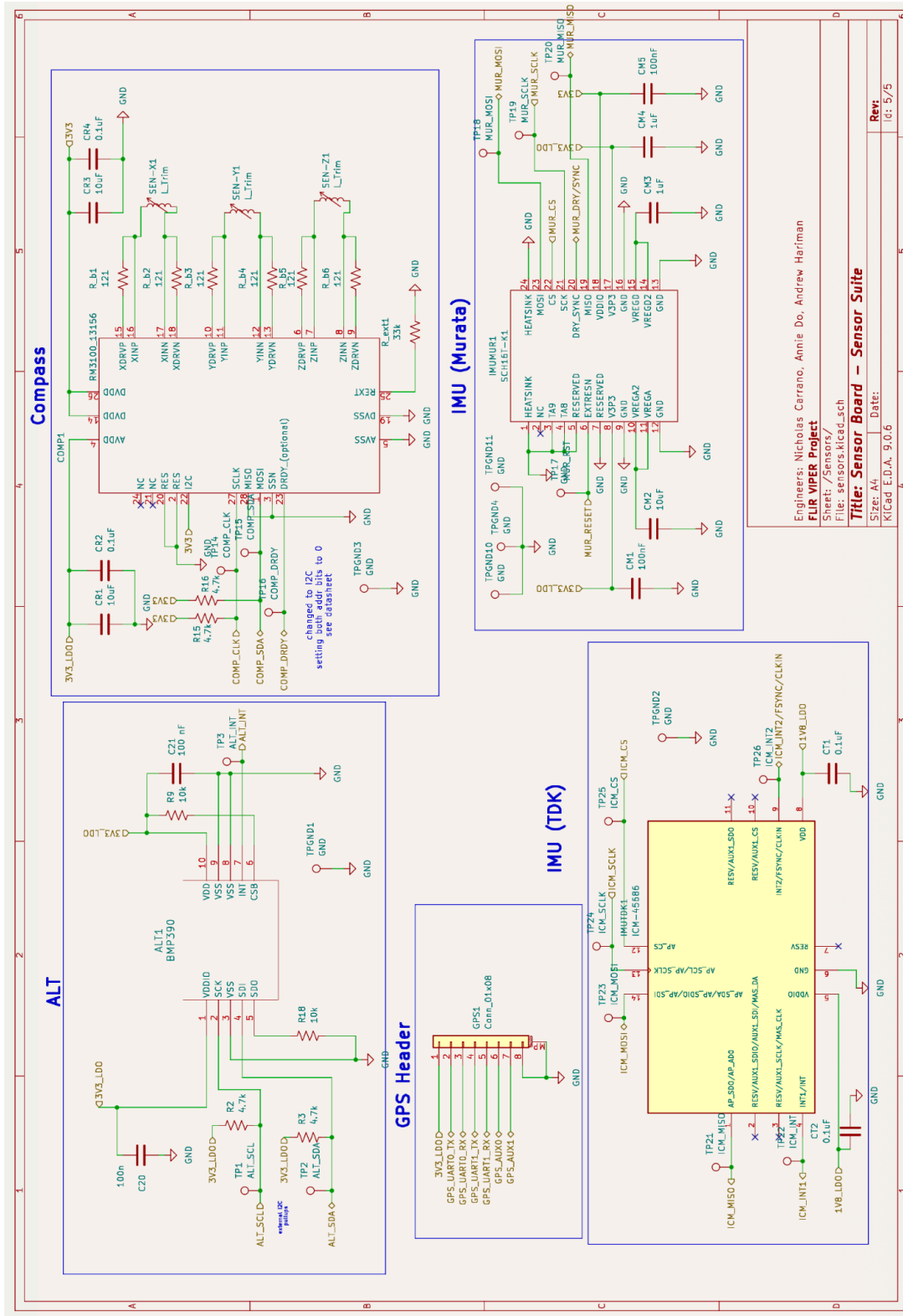
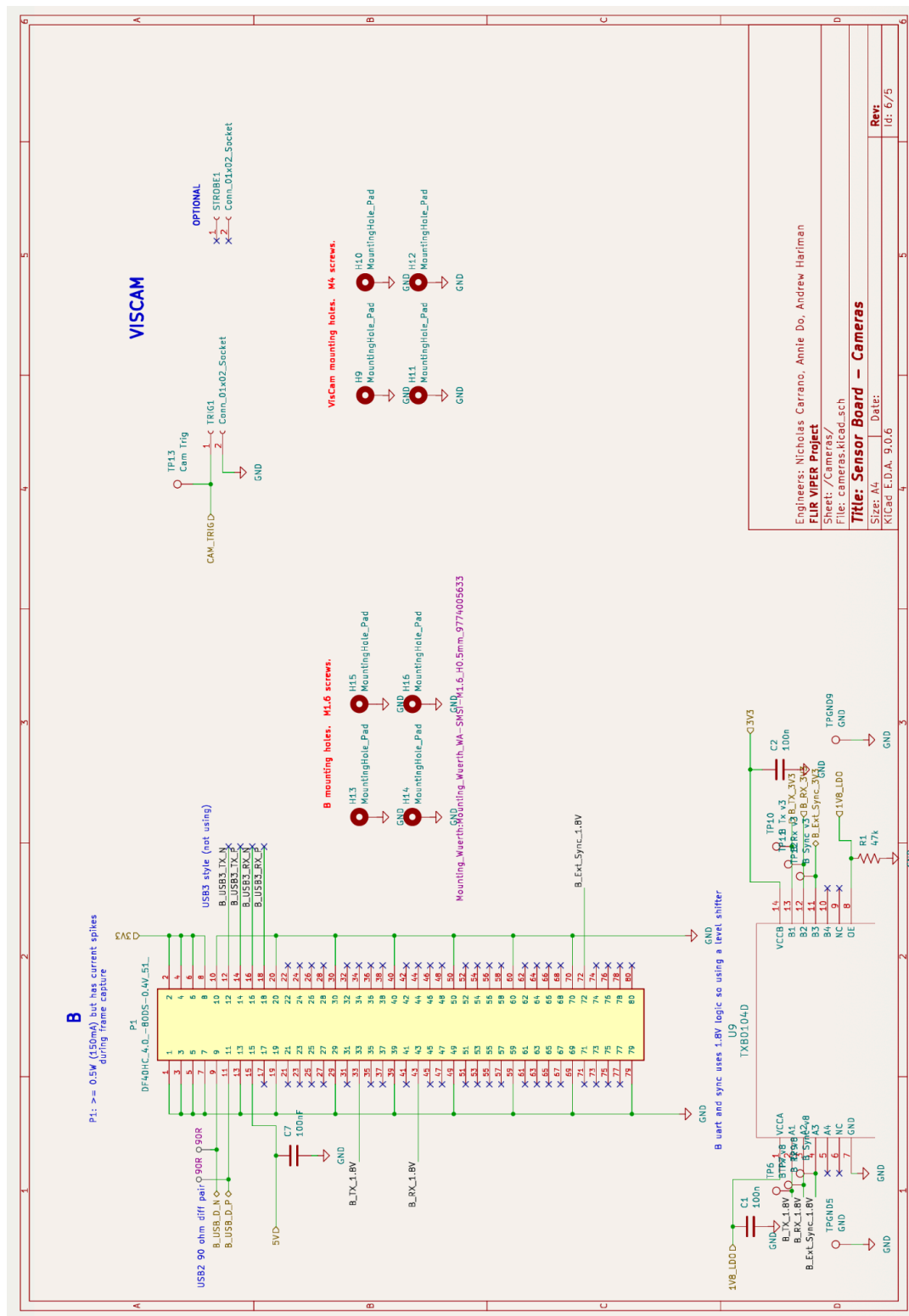
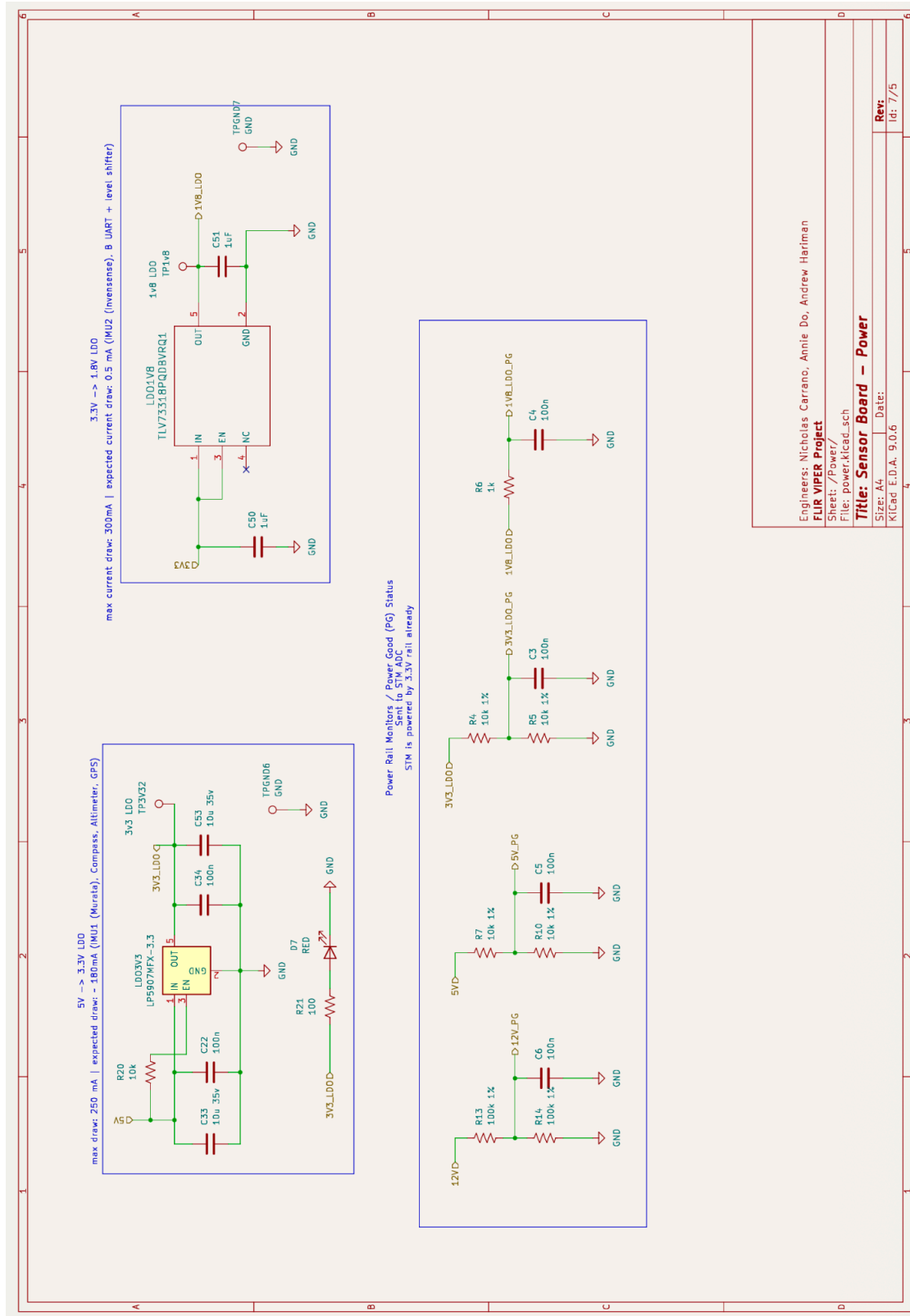


Figure 61.





ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	Capstone Boson Camera Core S95	Infrared Camera	1
2	Innomaker_OV9281	Visual Camera	1
3	Sensor Board 2-9	PCB #2	1
4	Carrier Board 2-9	PCB #1	1
5	98952A108	Aluminum Male-Female Threaded Hex Standoff	4
6	92000A116	Passivated 18-8 Stainless Steel Pan Head Phillips Screws	10
7	92000A002	Passivated 18-8 Stainless Steel Pan Head Phillips Screws	4
8	93657A204	Nylon Unthreaded Spacers	4
9	90695A031	Medium Strength Steel Thin Profile Hex Nut	4
10	95836A257	Black Oxide 18-8 Stainless Steel Pan Head Phillips Screw	2

PROPRIETARY AND CONFIDENTIAL

THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF <INSERT COMPANY NAME HERE>. ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT THE WRITTEN PERMISSION OF <INSERT COMPANY NAME HERE> IS PROHIBITED.

UNLESS OTHERWISE SPECIFIED:		NAME	DATE
DIMENSIONS ARE IN INCHES		DRAWN	S.F. 5/14/2026
TOLERANCES:		CHECKED	
FRACTIONAL ±		ENG APPR.	
ANGULAR: MACH ± BEND ±		MFG APPR.	
TWO PLACE DECIMAL ±		Q.A.	
THREE PLACE DECIMAL ±		COMMENTS:	
INTERPRET GEOMETRIC TOLERANCING PER:			
MATERIAL			
FINISH			

TITLE:

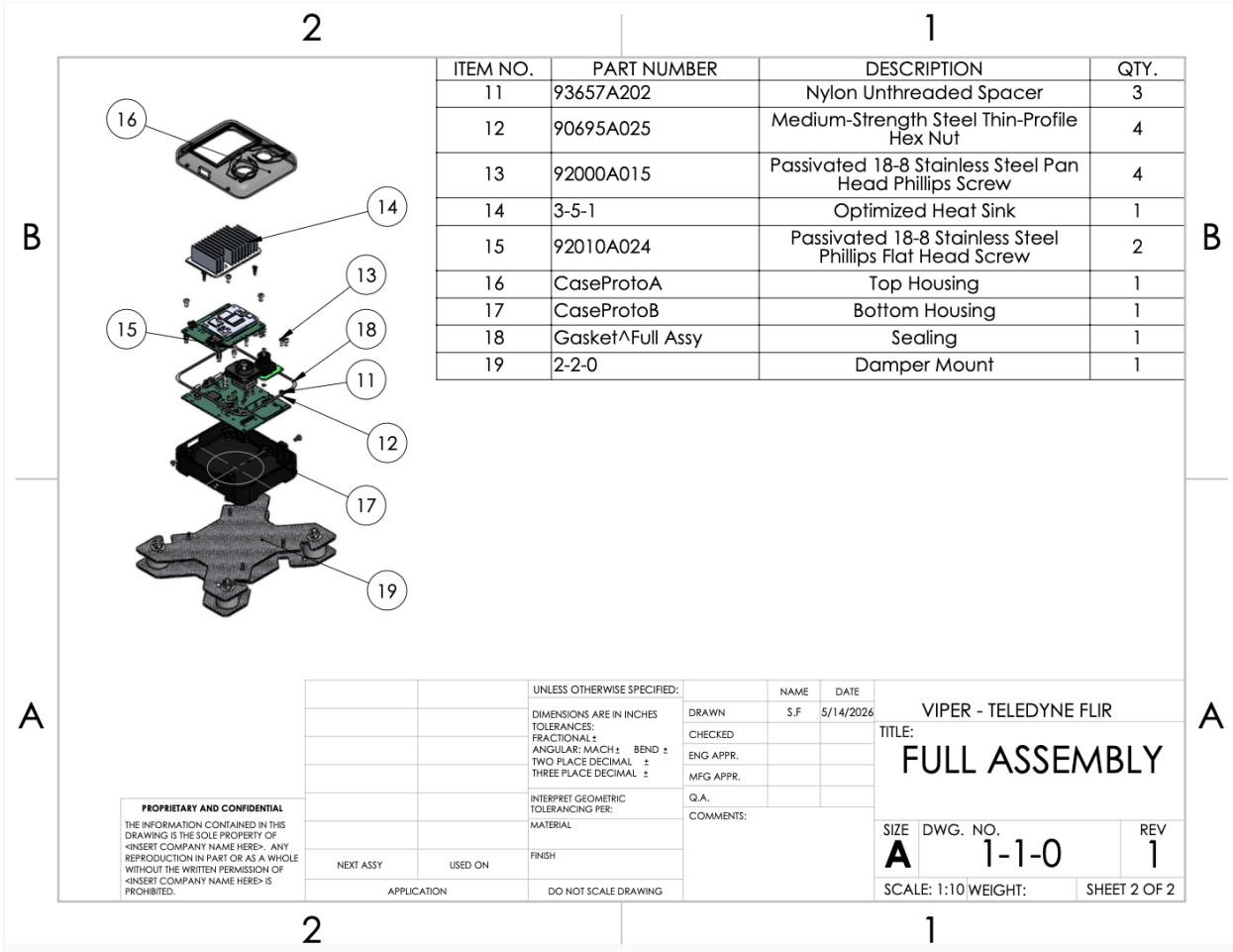
FULL ASSEMBLY

SIZE	DWG. NO.	REV
A	1-1-0	1

SCALE: 1:10 WEIGHT: SHEET 1 OF 2

NEXT ASSY	USED ON	APPLICATION	DO NOT SCALE DRAWING

Figure 65.



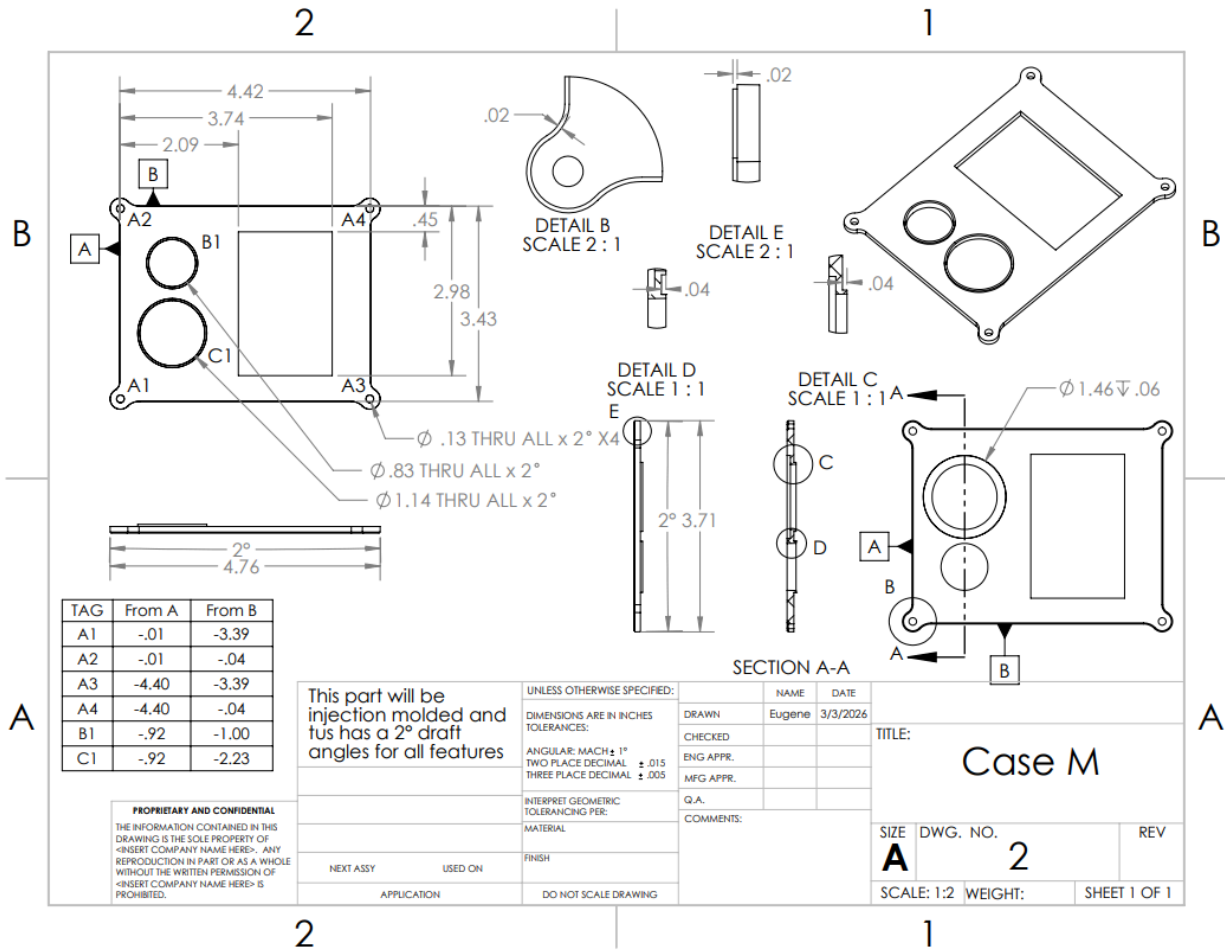


Figure 67.

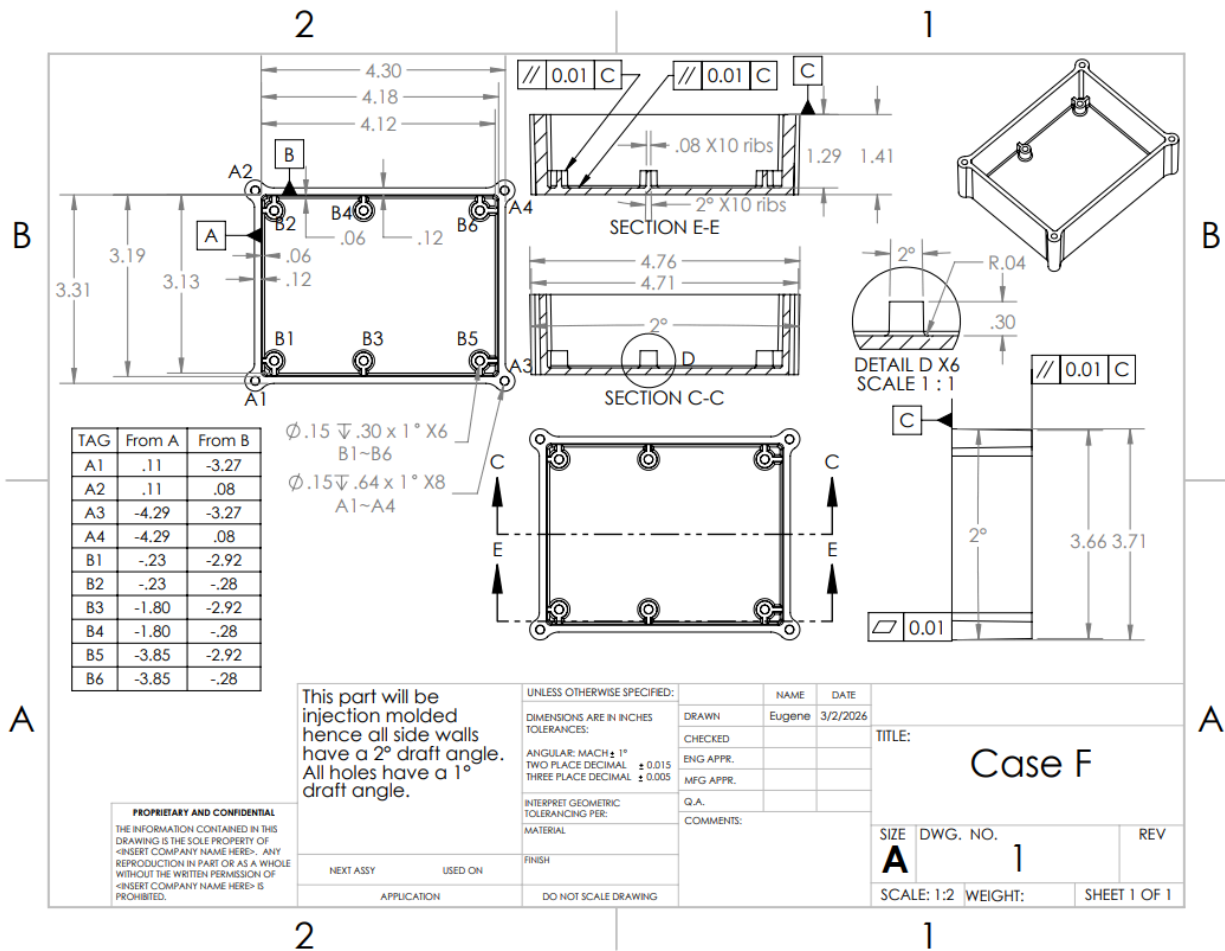


Figure 68.

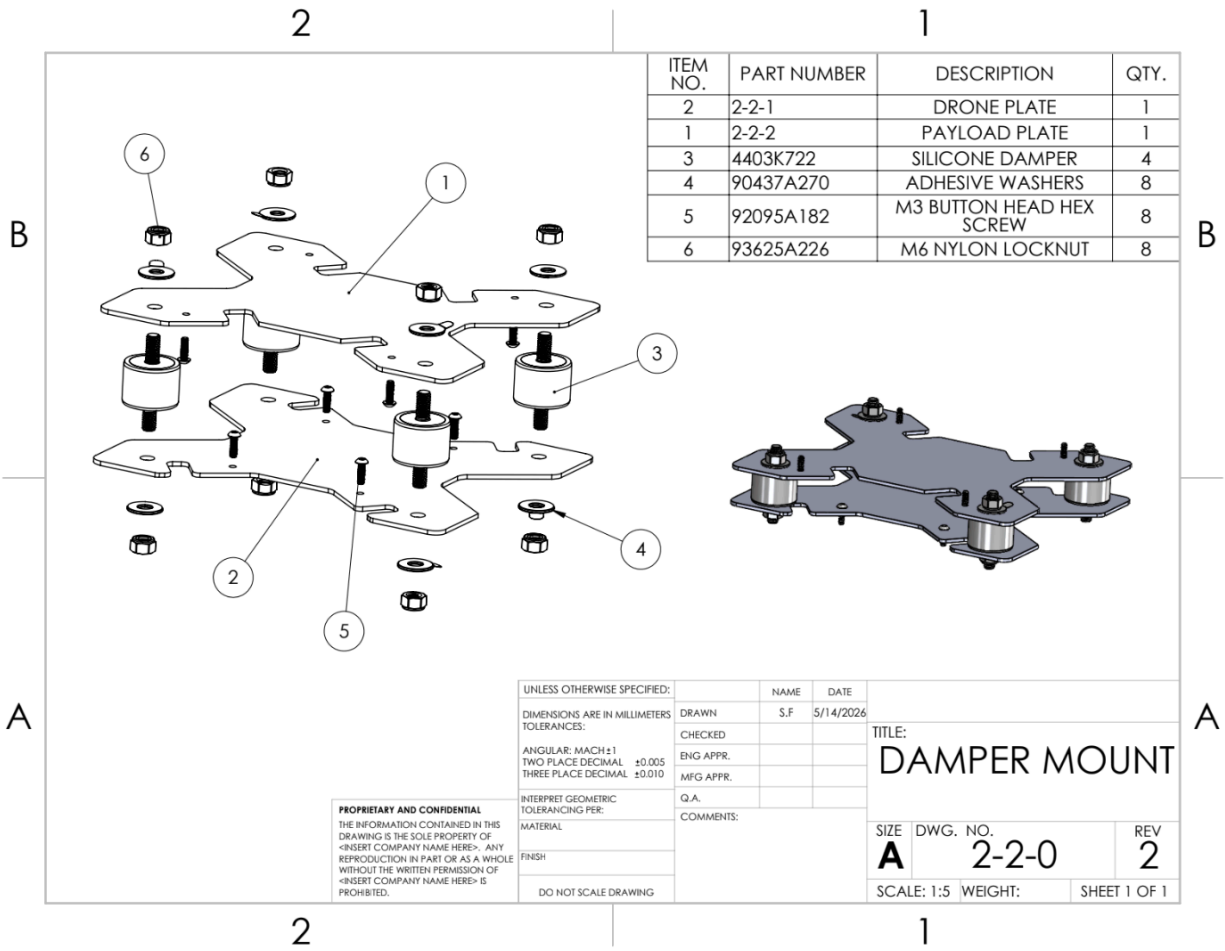


Figure 69.

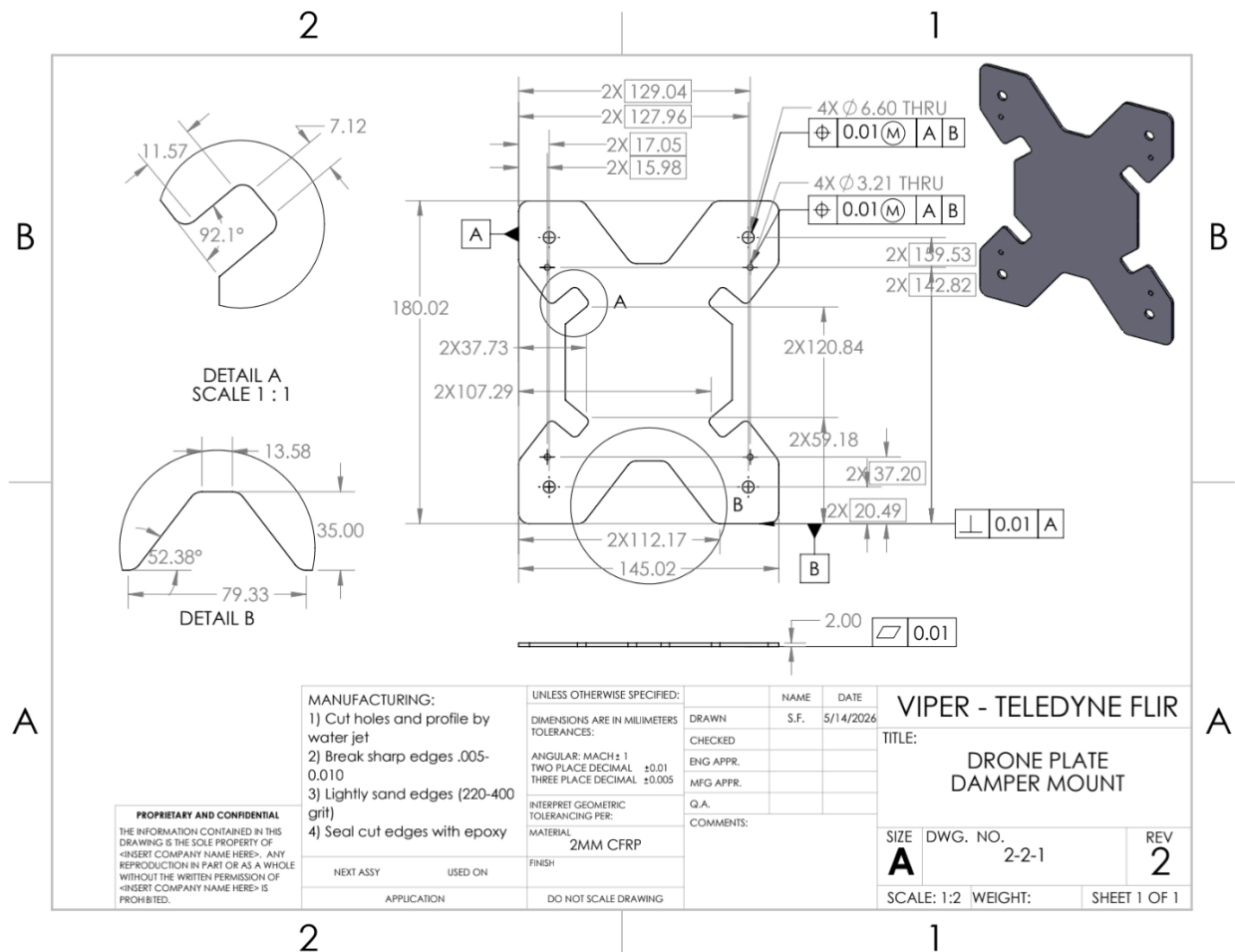


Figure 70.

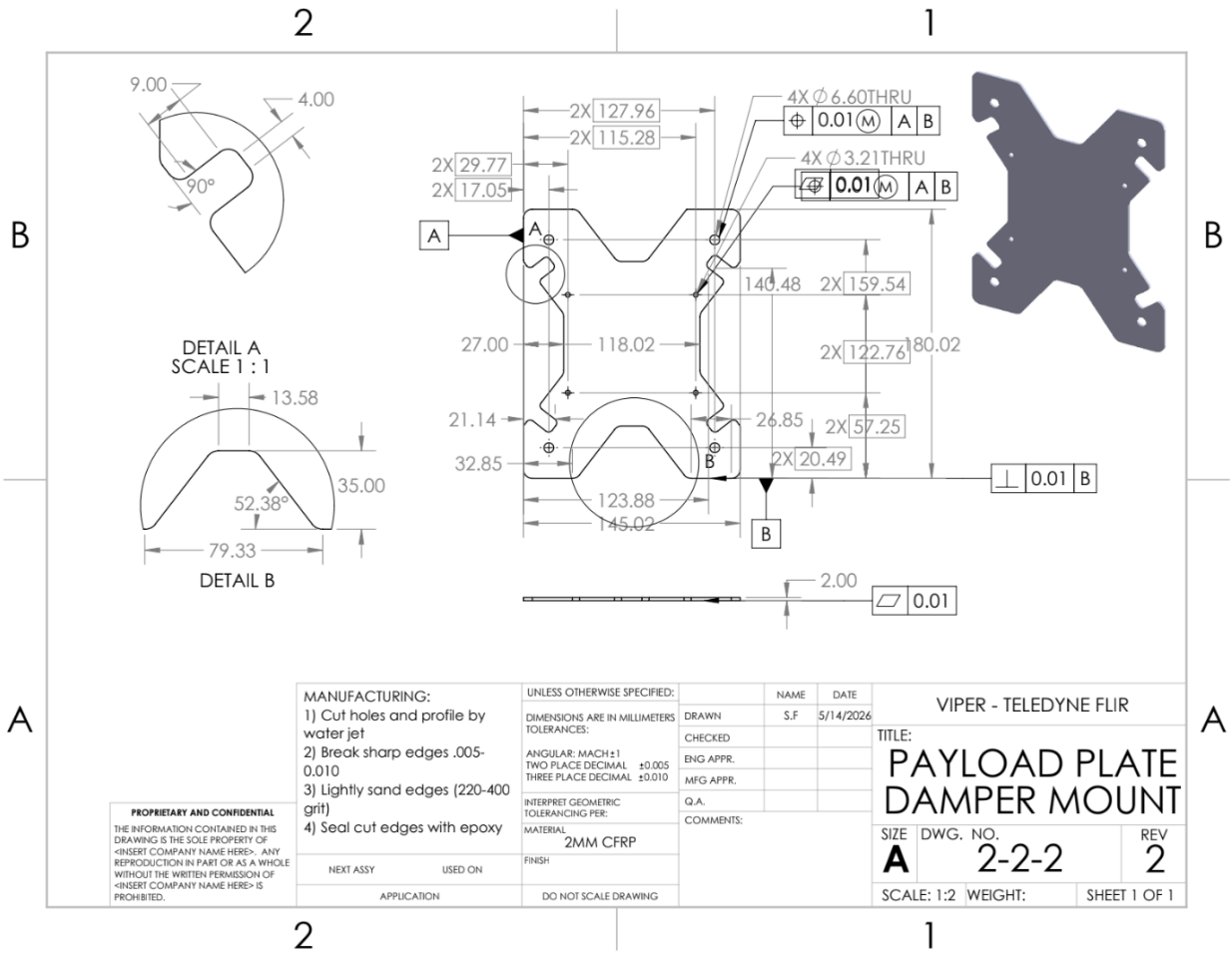


Figure 71.

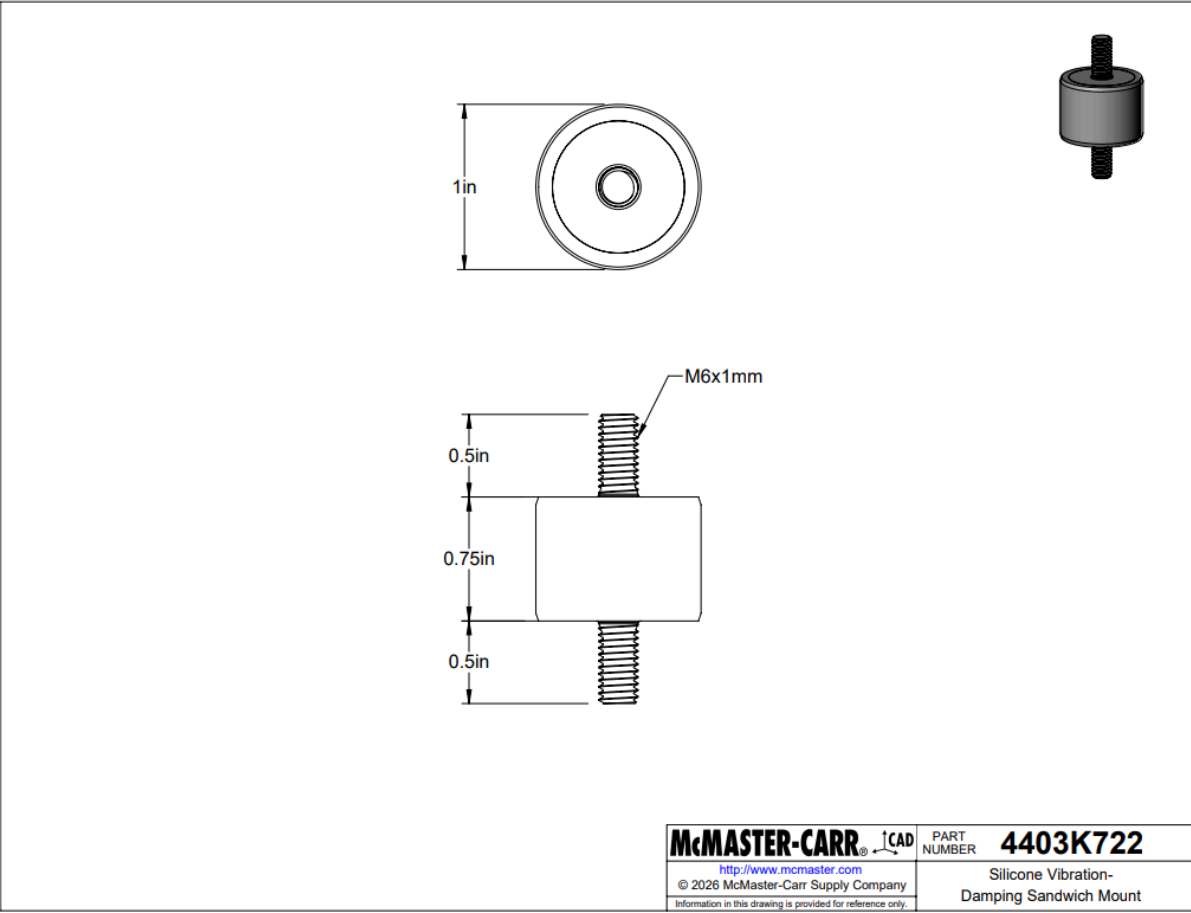


Figure 72.

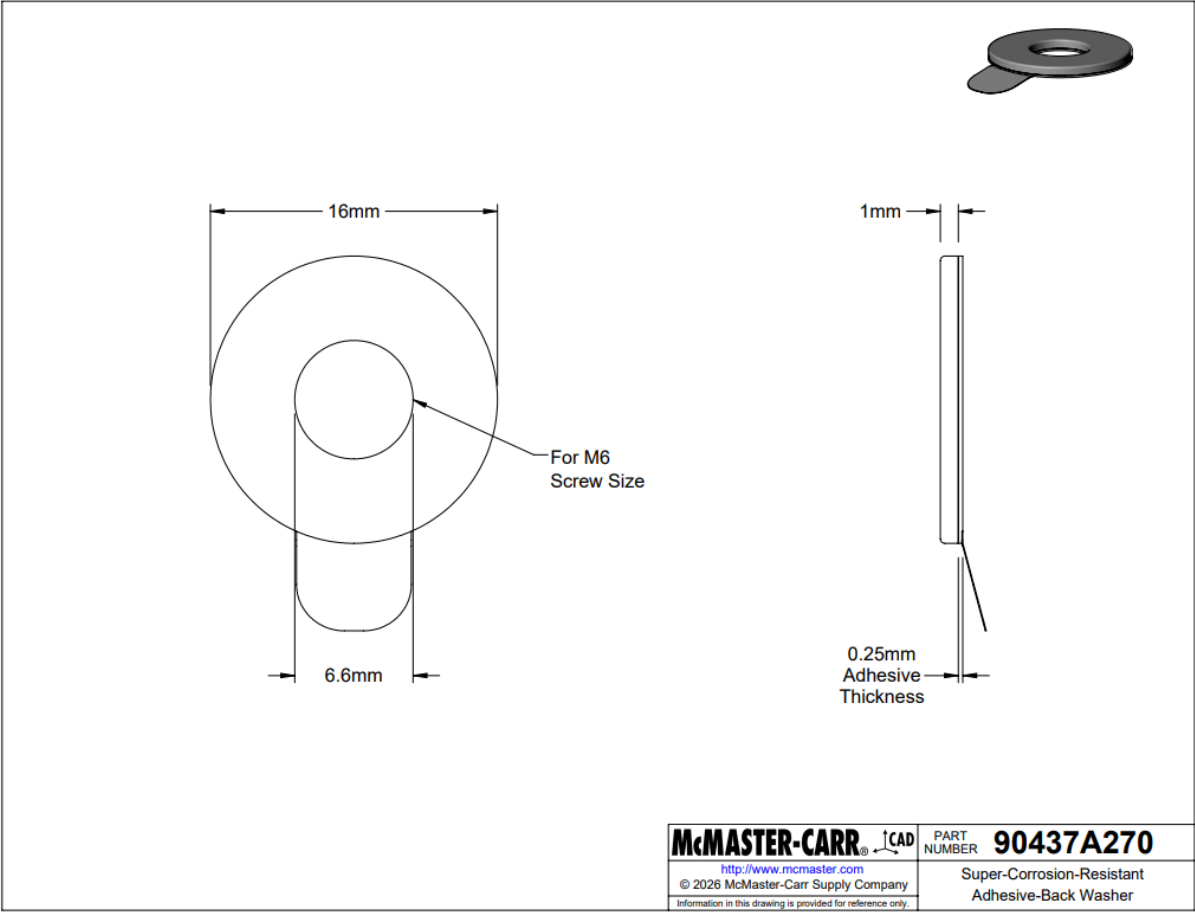


Figure 73.

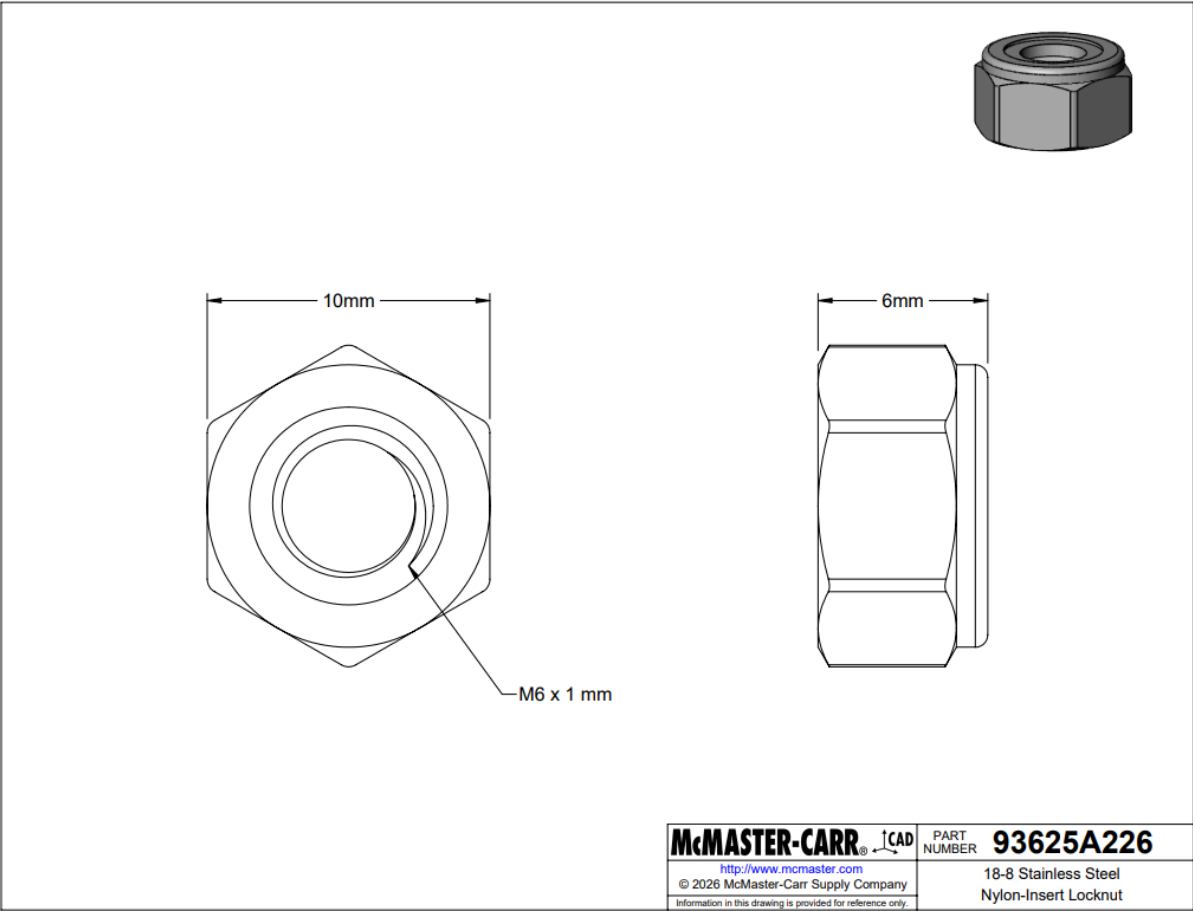


Figure 74.

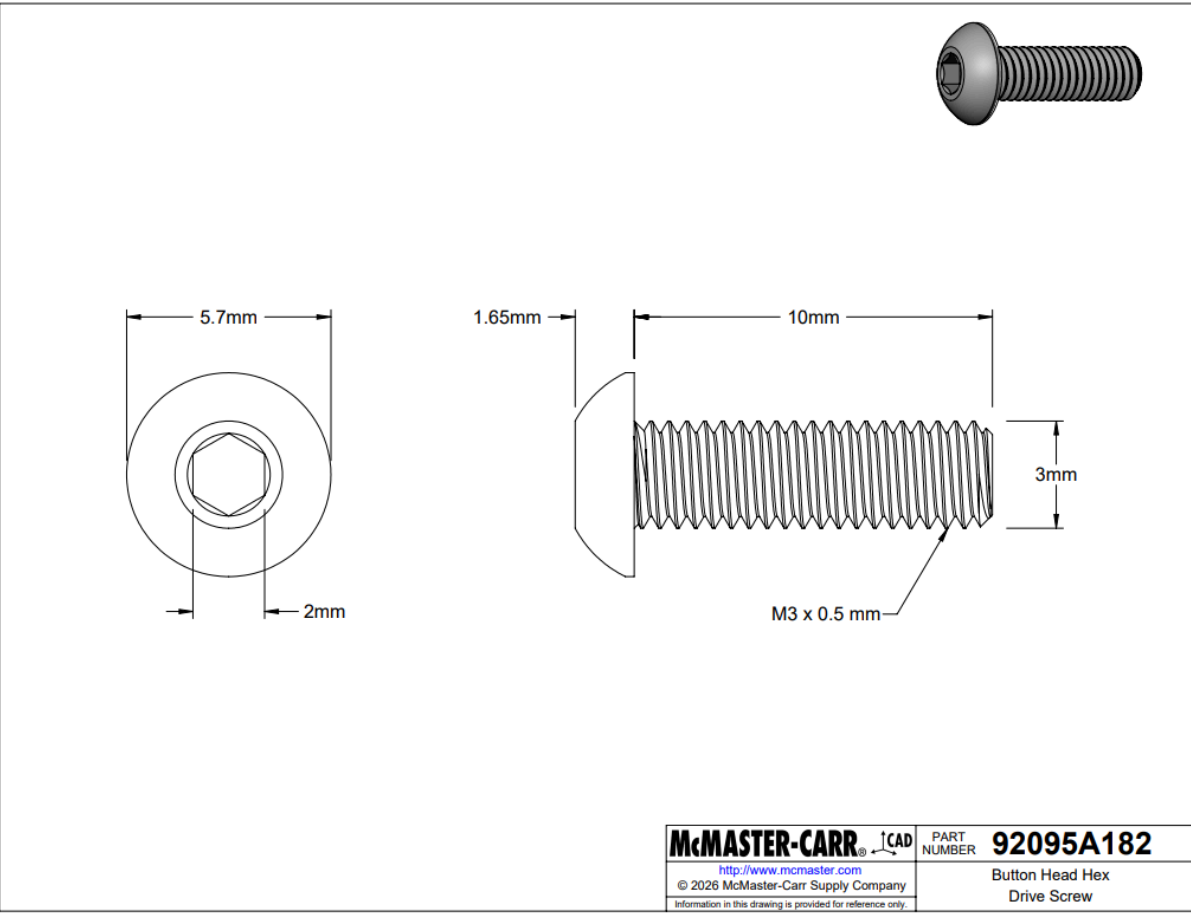


Figure 75.

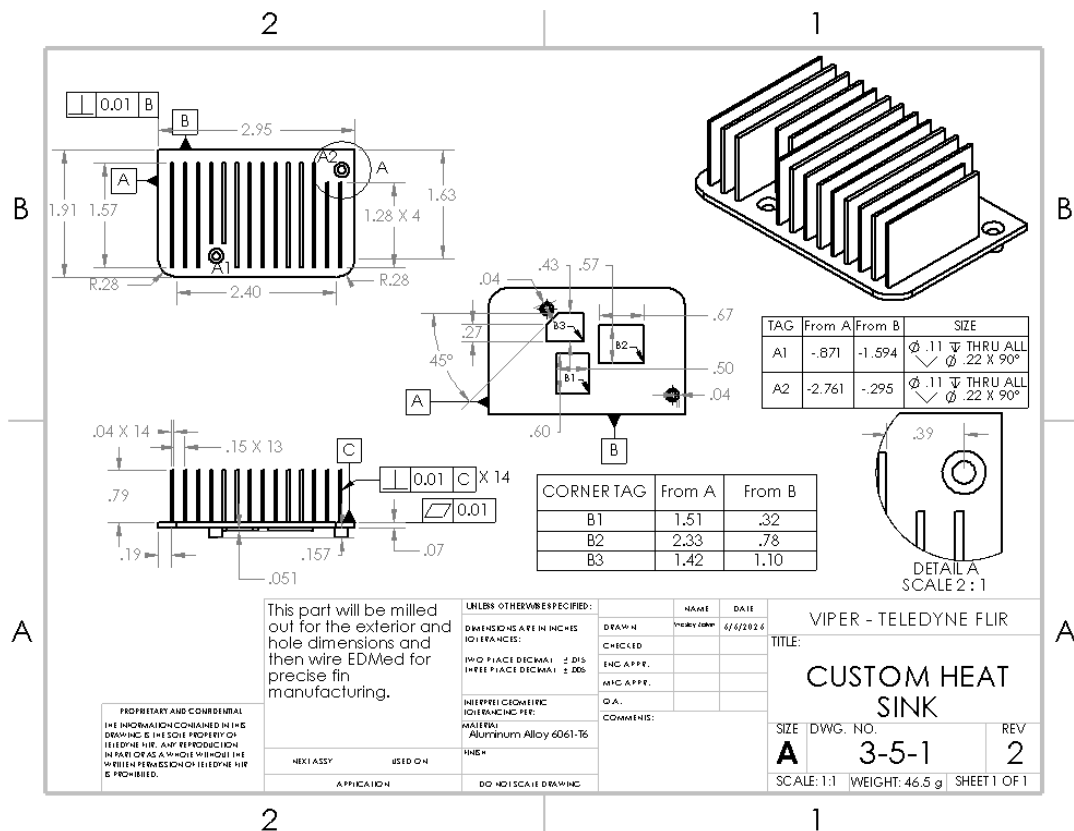


Figure 76.

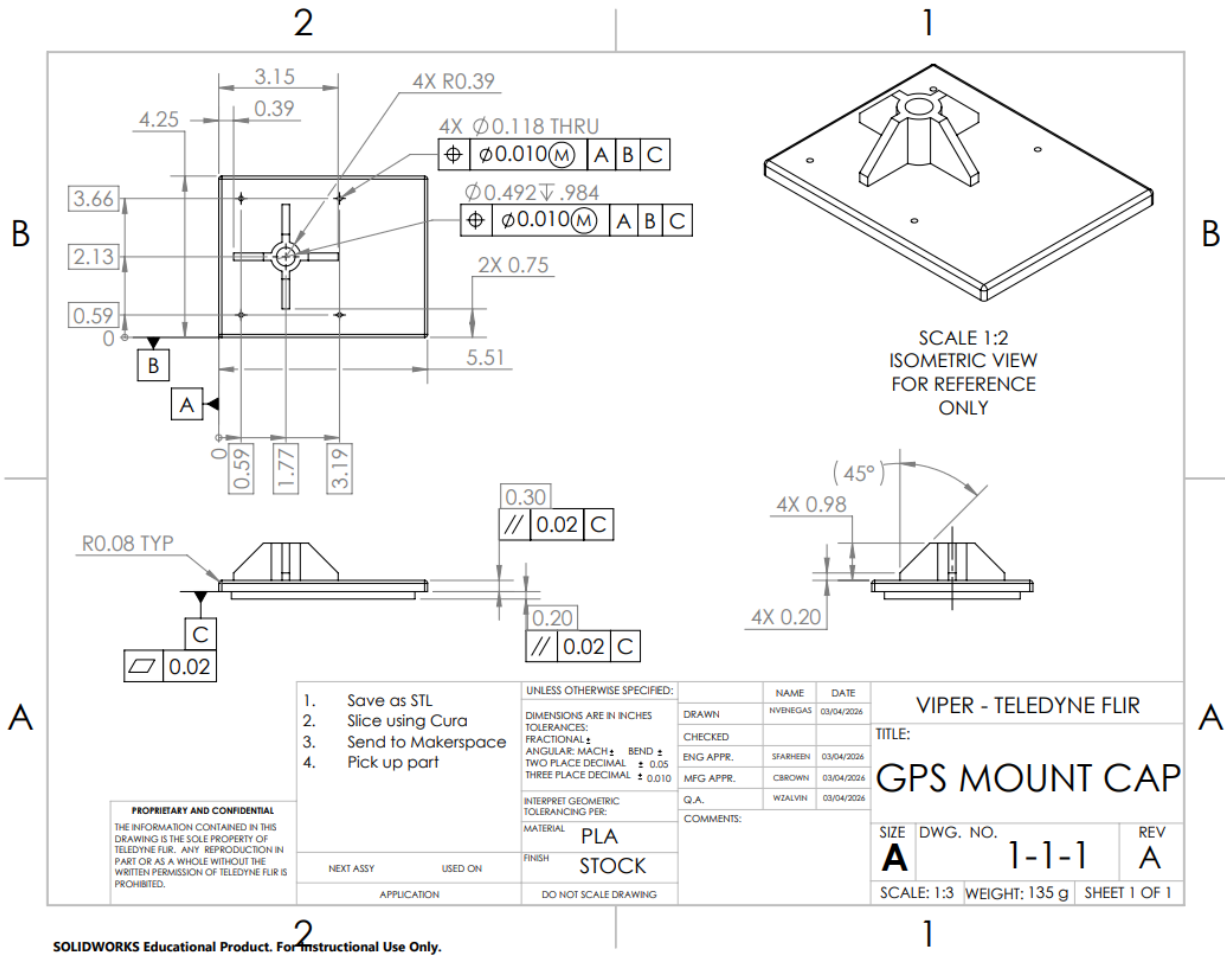
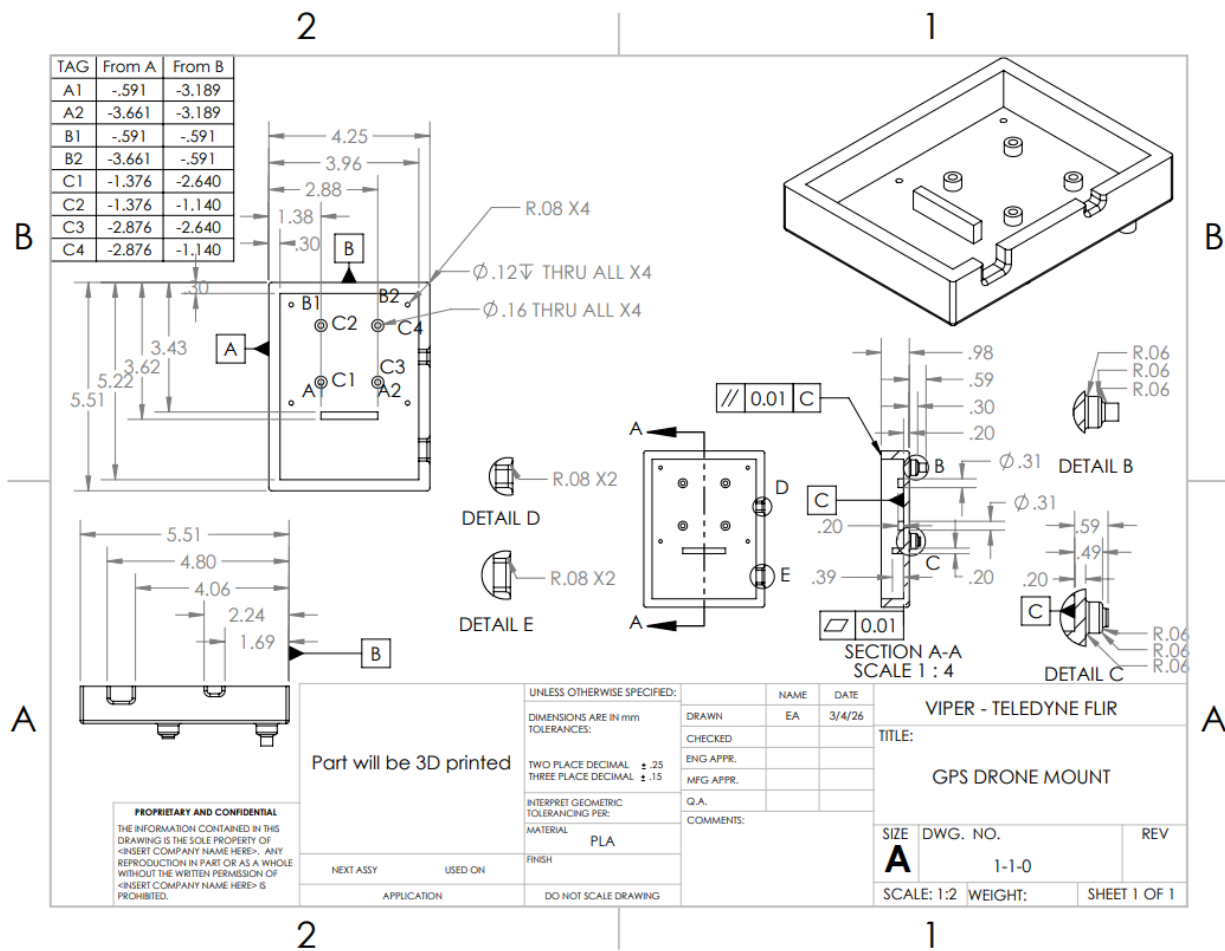


Figure 77.



11.4 - Thermal Benchmark Test SOP

Standard Operating Procedure: VIPER: Thermal Output Test

Author:	Wesley Zalvin, Carter Brown, Samina Farheen
Course:	Capstone, Teledyne FLIR
Building:	The Barn (UCSB Building #479)
Date:	Feb 4, 2026

Table 26.

1. Description

This SOP describes a test to characterize the thermal output and resulting internal temperatures of the VIPER payload's current design specifications. The thermal behavior of the system is characterized by the power dissipation of high-heat components (>1 Watt) within a sealed box, which will be modeled by load resistors that match the thermal output of those high-heat components. The transient temperature of six key points within the box is measured by an array of thermocouples. Data acquisition is accomplished with the Omega eight channel data acquisition module.

2. Hazards Overview

- Burn hazards: The electrical components heat up quickly and significantly.
 - Do not touch the components while the test is ongoing.
 - Wait at least 1 minute after testing has concluded to allow heat to dissipate from components.
 - If moving components before they are adequately cooled is necessary, use tongs or oven mitts.
- Shock: This test uses actively powered electrical components, so touching any of these during operation brings a risk of electrical shock.
 - Do not touch actively powered components.
 - Turn off power before making any changes to the testing set up.

3. Required Personal Protective Equipment (PPE)

- Safety Glasses
 - Should be worn during the whole experiment and after experiment when the device cools down.
- Tongs/oven mitts
 - Use in case something becomes displaced however touching hot components during the experiment should not be necessary.

4. Waste Disposal

This test is not expected to generate any waste. However, if any housing components were to melt, waste should be allowed to cool down before disposal into a normal trash can. Electronic waste should be disposed of properly.

5. Accident and Spill Procedure

In case of burn, flush skin under cold water for 15 minutes. Seek medical attention if discomfort lasts or skin is visibly burned.

In case of fire, use the fire extinguisher located near the exit of the barn.

1. Pull the pin.
2. Point nozzle at base of fire.
3. Squeeze trigger while sweeping back and forth over the fire.

6. Approvals Required

The experiment requires approval from one of the Mechanical Engineering Capstone Staff. Refer to Dr. Trevor Marks or Tyler Susko.

- tmarks@engineering.ucsb.edu
- susko@ucsb.edu

7. Procedure

This section contains **detailed**, step-by-step instructions for the procedure being performed.

The procedure section is typically written *concurrently* with the actual (physical) setup and preparation of the procedure. Note, approval of this SOP is required **prior** to performing the procedure.

The layout of this section is flexible and should be adapted to best suit your procedure. For example, some parts of your procedure might best be written as a check-list of steps or actions. If you have any figures, place them *at the end* of the document. Do not include any figure that is not referenced in the writeup.

Typically, the procedure section is split into four sections:

- I. **Assembly:** With the box assembled and holes drilled:
 1. Set a two channel power supply to 11 and 6.9 volts.
 2. With the power supply turned off, connect the load resistor circuits to the power supply
 3. Mount the load resistors in the box.
 4. Insert the thermocouples into their dedicated holes and secure them to their designated locations.
 5. Connect each thermocouple to the Omega eight channel data acquisition module, document which channel each thermocouple is connected to.
 6. Seal the holes of the box with tape or something similar.
 7. Close and secure the lid of the box.

II. Pre-run

1. Connect Omega data acquisition module to PC running the TC-08 software.
2. If testing forced convection, document where the fan is set up relative to the box.
3. Record room temperature and pressure.

4. Test wind speed at different points around the box with an anemometer.
5. Begin recording data from the Omega data acquisition module. This is the baseline.

III. Run

1. With the data recording, turn on the power supply.
2. Once steady state conditions have been reached, turn off the power supply.
 - a. Do not touch the components until the power supply is fully off.
3. Export data to a .csv file.
4. Repeat these steps for each trial.
 - a. If testing under different conditions, record the new conditions before proceeding.

IV. Post-Run

1. Disassemble all components, making sure to wipe off thermal paste with rubbing alcohol.